[THESIS TITLE]

by

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Abstract

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Contents

the chapterHardware-Instance Overfitting and the Ensemble HAT Principle

0.1 Introduction: the deployment gap in analog CIM

Organic optoelectronic compute-in-memory (CIM) systems are invariably introduced through device-level figures of merit: conductance window, retention time, write asymmetry, optical responsivity. Those metrics are necessary, but they are not sufficient to answer the system question that matters for deployment: *what accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch and finite readout precision?* This thesis treats that question as a materials-to-system translation problem rather than a pure device problem.

The starting point is the canonical hardware-aware training (HAT) regime. Under nominal evaluation on the hardware instance seen during training, HAT appears effective: the hybrid Tiny-ViT recovers most of the source-domain accuracy lost under the canonical analog noise model. However, that apparent success is misleading. When the trained model is evaluated on a fresh hardware instance—a new fixed device-to-device (D2D) mismatch realization drawn from the same distribution—the standard HAT model collapses to chance level. This exposes the central theme of the thesis: robustness to analog noise is not the same thing as robustness to hardware-instance shift.

The contributions of this thesis are threefold. First, a reusable simulation framework for hardware-aware training of vision transformers on analog CIM arrays with realistic device non-idealities. Second, a taxonomy of HAT regimes and the discovery of *hardware-instance overfitting*—the phenomenon that fixed-mask training produces models which memorise a specific device mismatch realisation and collapse to chance level on fresh hardware instances. Third, a deployment envelope that systematically maps which combinations of noise models and training interventions transfer to new hardware instances and which do not, including rigorous negative results that falsify natural mitigation hypotheses.

This chapter focuses on two linked questions. Why does fixed-mask HAT fail so catastrophically on fresh arrays despite performing well on the training-time array? And what training intervention restores transfer without changing the deployment hardware assumptions? The answer is Ensemble HAT: resampling the structured D2D mismatch field at epoch boundaries during training so that the model learns over a distribution of arrays rather than a single memorised instance.

0.2 The empirical signature: from source-domain rescue to fresh-instance collapse

0.2.1 Three evaluation regimes

Analog-deployment experiments must distinguish between three regimes:

1. **Source-domain accuracy:** evaluation on the training-time hardware instance.
2. **Repeated Monte Carlo accuracy:** multiple stochastic forward passes on the *same* instance.
3. **Fresh-instance transfer:** evaluation on newly sampled hardware realisations.

Only the third quantity answers the practical deployment question. The distinction is not academic: under the canonical uniform-noise protocol ($\sigma_{\mathrm{D2D}} = 10\%$, $\sigma_{\mathrm{C2C}} = 5\%$, $NL = 1.0$), fixed-mask HAT reaches $87.95 \pm 0.27\%$ on the training-time instance (source domain), but collapses to $10.00 \pm 0.00\%$ on fresh instances. The 77.95 pp gap is the empirical cost of hardware-instance overfitting.

0.2.2 The $10\,\%$ collapse

The decisive observation is simple. A standard HAT model trained with a single fixed D2D mask reaches high source-domain accuracy, yet its fresh-instance accuracy collapses to approximately 10%—the single-class baseline for CIFAR-10. In contrast, Ensemble HAT raises fresh-instance accuracy to $86.37 \pm 1.54\%$. The gap is too large to dismiss as ordinary stochastic variation; it reflects a structural failure of the training objective under deployment shift.

The collapse is *deterministic*: across ten independent fresh instances with five Monte Carlo evaluations each, every instance yields $10.00 \pm 0.00\%$. The standard deviation of zero indicates that the collapsed predictor is not an accident of one unfavourable draw; it is a training outcome. A dedicated no-AMP control confirmed that the collapse is not a numerical artifact of mixed-precision autocast. The problem is therefore the *training-time fixation of a random hardware sample* that the model subsequently memorises.

0.2.3 Why the collapse is informative

The 10% collapse is methodologically valuable because it establishes a clean binary signal. Any intervention that does not raise fresh-instance accuracy substantially above 10% is not a solution; it is at best a source-domain diagnostic. Under this lens, Ensemble HAT is the only intervention discussed in this thesis that fully satisfies the deployment-grade condition: it recovers > 76 pp on fresh instances with a tight cross-instance standard deviation (1.54 pp).

The fresh-instance protocol (10 instances $\times$ 5 Monte Carlo runs) is therefore treated not as an appendix detail, but as a standard tool for deciding whether a mitigation is deployment-grade or merely diagnostic.

0.3 Mechanism of hardware-instance overfitting

0.3.1 Fixed-mask training as domain memorisation

The D2D mismatch field M_0 is a spatially structured perturbation that acts as a hidden domain label. Under fixed-mask HAT, the optimiser solves

$$\theta^{\star}_{\text{fix}} = \arg\min_{\theta} \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n) \,, \tag{1}$$

where M_0 is sampled once at initialisation and held constant for the entire training run. The scale-recovery factors s_ℓ implicitly calibrate to the particular spatial distortion encoded in M_0, and learned feature routing adapts to the fixed perturbation pattern rather than to the underlying data distribution.

The result is a model that is highly accurate on the training-time array—the canonical V4 checkpoint reaches $87.95 \pm 0.27\%$ under nominal evaluation—but structurally unable to generalise to any other hardware instance. The classifier learns a decision boundary conditioned on M_0, much as a domain-adaptation network learns a boundary conditioned on a source domain. When the domain label changes at deployment, the boundary ceases to be valid.

0.3.2 The memorisation has a geometric interpretation

The D2D mismatch field is not white noise; it has spatial correlation length comparable to the patch size of the first convolutional embedding. This means that the early layers of the network can learn *instance-specific* filters that partially cancel the distortion. The cancellation is effective on M_0 but ineffective on any other realisation $M' \sim p(M)$, because the learned filters are tuned to the particular phase and amplitude spectrum of M_0.

Ensemble HAT breaks this memorisation by replacing the single fixed mask with a distributional objective in which one fresh D2D realisation is drawn at the start of each epoch:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta} \; \mathbb{E}_{M\sim p(M)} \left[\frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M), y_n) \right] , \tag{2}$$

where the expectation is Monte Carlo–estimated with E independent draws, one per training epoch.

0.4 Ensemble HAT as a distributional training objective

0.4.1 Why epoch-level resampling

The critical design choice is the resampling *cadence*: the mismatch map changes at epoch boundaries, not at mini-batch boundaries, so that the optimiser experiences a block-stationary forward path within each epoch.

Why block-stationarity matters can be understood by contrasting three regimes:

- **Fixed-mask HAT:** permanently stationary but sample-specific; the optimiser has unlimited time to fit M_0.
- **Per-batch resampling:** the forward path changes every few hundred samples; the gradient field is so non-stationary that the optimiser cannot settle.
- **Epoch-level resampling (Ensemble HAT):** each epoch provides enough stability for the optimiser to make progress under a fixed $M^{(e)}$, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting.

The block-stationary interpretation is developed formally in Chapter 2, where per-epoch HAT is shown to offer a *block-stationary guarantee* stronger than the singleton guarantee of fixed-mask HAT and the product-distribution guarantee of per-batch resampling.

0.4.2 Quantitative outcome

The canonical Ensemble HAT checkpoint preserves $86.37 \pm 1.54\%$ accuracy on the identical fresh-instance protocol that drives fixed-mask HAT to 10.00%—a gain of more than 76 percentage points. The standard deviation of 1.54 pp across the ten fresh-instance means is narrow, indicating that the recovered accuracy is stable rather than an accident of one favourable draw.

The intervention does not require any change to the model architecture, the device assumptions, or the optimiser hyperparameters. It is a pure training-objective modification: the same Tiny-ViT backbone, the same 4-bit differential conductance mapping, the same AdamW schedule, but with the D2D field resampled at epoch boundaries. This minimalism is methodologically important: the $10\% \rightarrow 86\%$ lift is attributable solely to the distributional objective.

0.4.3 Generalisation beyond the training distribution

The Ensemble HAT result generalises to richer spatial structures. Under separable AR(1)-style spatial correlation, the canonical checkpoint degrades gracefully: $86.33 \pm$

1.61% for i.i.d. mismatch, $84.57 \pm 2.39\%$ at $\rho = 0.3$, and $82.12 \pm 3.95\%$ at $\rho = 0.5$ (zone 3A bug-immune canonical protocol, $NL = 1.0$, AMP disabled). The degradation is bounded and monotonic; no instance collapses below 73.7%. This suggests that the block-stationary training objective generalises, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

0.5 Cross-scenario evidence pillars

The thesis validates Ensemble HAT across three scenario pillars that span the deployment-relevant parameter space:

1. **Canonical uniform noise** ($\sigma_{\mathrm{D2D}} = 10\%$, i.i.d., $NL = 1.0$): Ensemble HAT recovers $86.37 \pm 1.54\%$ fresh-instance accuracy, while fixed-mask HAT collapses to $10.00 \pm 0.00\%$.

2. **Literature-anchored OPECT proxy** ($\sigma_{\mathrm{D2D}} = 3\%$, $\sigma_{\mathrm{C2C}} = 2\%$): zero-shot transfer of the Ensemble HAT checkpoint reaches $88.53 \pm 0.08\%$ without retraining, demonstrating that the distributional objective generalises to lower-noise profiles.

3. **Severe nonlinear write** ($NL = 2.0$, revised gradient-scaling recipe): hardware-aware training recovers to the ∼80–82% band across Standard, Ensemble, and proportional-noise training routes, leaving a residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ baseline. This gap is a joint effect of nonlinear write dynamics and fresh-instance transfer, not of transfer alone.

These three pillars establish that hardware-instance overfitting is a *pervasive* failure mode—it appears under diverse physical assumptions—and that Ensemble HAT is a *robust* countermeasure that lifts accuracy by > 76 pp in the canonical regime and by > 78 pp in the literature-proxy regime.

0.6 Implications for the rest of the thesis

This chapter has established the empirical signature and mechanistic interpretation of hardware-instance overfitting. The remaining chapters build on this foundation in three directions.

Chapter 1 constructs the simulation framework that every subsequent experiment inherits: a profile-driven, PyTorch-native behavioural simulator with explicit device-profile substitution and hybrid analogue/digital deployment decomposition.

Chapter 2 maps the space of HAT cadences, noise profiles, and transfer guarantees, formalising the block-stationary guarantee that underpins Ensemble HAT.

Chapter 3 catalogues five named, reproducible failure modes that constrain deployment, including the severe-NL regime where the revised gradient-scaling recipe recovers to the ∼80–82% band.

Chapter 4 evaluates the interventions that partially or fully close each gap identified in the atlas.

Chapter 6 synthesises the evidence into actionable design rules for program managers deciding whether to tape out a ViT accelerator on an organic process.

The central open question is whether the ranking logic and failure-mode boundaries survive the transition to larger vision tasks and to language-model workloads. Chapter 7 addresses this question through a roadmap of thesis-completion experiments and framework extensions.

Chapter 1

Profile-driven simulation framework

1.1 Introduction: why profile-driven simulation

Organic optoelectronic compute-in-memory systems are usually judged by device-level figures of merit: conductance window, retention time, write asymmetry, and optical responsivity. Those metrics are necessary, but they are not sufficient to answer the system question that matters for deployment. What accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch, finite readout precision, and a sub-linear photoresponse? Answering that question requires a deliberate bridge between the measurement laboratory and the algorithm training loop.

This chapter describes the simulation framework that provides that bridge. The design principle is *profile-driven*: every physically relevant assumption enters through a single structured configuration object, so that a literature-derived prior and an in-house measured profile are interchangeable inputs to the same training or evaluation script. This is important because early device characterization rarely arrives as a single "device score." Instead, it is distributed across several measurement modalities—multilevel programming, repeated same-state reads, cross-device statistics, retention fitting, and photoresponse characterization. The framework treats these measurements as structured calibration inputs rather than as informal narrative context.

The thesis therefore treats the framework itself as a research instrument, not merely as an implementation detail of the manuscript experiments. Its contributions are threefold. First, a set of pluggable physical interfaces that let device parameters enter the forward path without changing model architecture code. Second, a deep integration with PyTorch autograd that makes hardware-aware training a drop-in training-mode switch rather than a custom optimizer rewrite. Third, an external validation against the CrossSim crossbar simulator that confirms numerical consistency in a shared operating regime and highlights where organic-specific extensions diverge from generic RRAM models.

1.2 Pluggable physical interfaces

The central design decision is to encapsulate every physical effect in a single replaceable profile object, then apply that profile uniformly across all analog-mapped layers. This section walks through the five physical interfaces: the device model, mismatch variability, nonlinear write asymmetry, retention drift, and the optoelectronic front end.

1.2.1 The device profile as a structured configuration object

At the core of the framework is a configuration dataclass that collects all parameters needed to simulate an analog crossbar layer. A profile instance specifies the conductance window $[G_{\mathrm{min}}, G_{\mathrm{max}}]$, the number of discrete conductance states n_{states}, the variability coefficients $(\sigma_{\mathrm{C2C}}, \sigma_{\mathrm{D2D}})$, the retention time constants (τ_1, τ_2, A_0), the photoresponse nonlinearity γ_{phys}, and the write-asymmetry exponents $(\mathrm{NL}_{\mathrm{LTP}}, \mathrm{NL}_{\mathrm{LTD}})$. Table 1.1 (reproduced from the manuscript) shows the direct mapping between laboratory characterization procedures and simulator fields.

Table 1.1: Measurement-to-simulator parameter mapping. A literature-derived profile and an in-house measured profile populate the same fields and therefore drive the same training scripts.

Measurement	**Typical characterization method**	**Simulator parame**
Conductance window	Multilevel programming I–V sweep	`G_min`, `G_max`
Resolvable state count	Programming resolution / histogram separability	`n_states`
Cycle-to-cycle variability	Repeated read/program at fixed target state	`sigma_c2c`
Device-to-device mismatch	Cross-device or cross-array statistics	`sigma_d2d`
Retention dynamics	Time-decay curve fit	`A_0`, `tau_1`, `tau_2`
Photoresponse nonlinearity	I_{photo} vs intensity	`gamma_phys`

The profile object is passed to every analog layer at construction time. Because the layer code contains no hard-coded physics, switching from a literature prior to a measured device recalibration requires only replacing the profile instance; no model definition or training script changes. This is the sense in which the framework is profile-driven. It also means that the experiments reported in later chapters of this thesis are, in principle, fully reproducible on new device data once the corresponding profile is populated.

1.2.2 Device variability: D2D and C2C

The framework models two sources of device variability. Device-to-device (D2D) mismatch is treated as a static perturbation sampled once at initialization:

$$\boldsymbol{\epsilon}_{\mathrm{D2D}} \sim \mathcal{N}\left(0, \sigma_{\mathrm{D2D}}^2 \Delta G^2\right), \qquad \Delta G = G_{\mathrm{max}} - G_{\mathrm{min}}. \tag{1.1}$$

This perturbation is stored as a persistent buffer and is applied to every subsequent forward pass. It captures fixed manufacturing mismatch across programmed devices on the array. Cycle-to-cycle (C2C) variation is modeled as a zero-mean Gaussian perturbation resampled on each forward pass:

$$\boldsymbol{\epsilon}_{\mathrm{C2C}}^{(t)} \sim \mathcal{N}\left(0, \sigma_{\mathrm{C2C}}^2 \Delta G^2\right). \tag{1.2}$$

The noisy effective weight becomes

$$\widehat{\mathbf{W}}^{(t)} = \left(\widetilde{\mathbf{G}}^{+} - \widetilde{\mathbf{G}}^{-}\right) + \boldsymbol{\epsilon}_{\mathrm{D2D}} + \boldsymbol{\epsilon}_{\mathrm{C2C}}^{(t)}, \tag{1.3}$$

with the C2C term disabled when the experiment calls for clean or D2D-only evaluation. Both variances are applied in the conductance domain so that noise magnitude scales with the physical dynamic range, a choice that preserves interpretability when the conductance window changes between device generations.

In addition to the canonical uniform-noise mode, the framework supports a proportional-noise mode where the noise magnitude scales with the instantaneous conductance state:

$$\boldsymbol{\epsilon}_{\mathrm{prop}}^{(t)} \sim \mathcal{N}\left(0, \sigma^2 |\mathbf{G}_{\mathrm{current}}|^2\right). \tag{1.4}$$

This mode is useful for stress-testing the robustness of trained models under more physically realistic noise distributions, and it is accessed by changing a single string field in the profile object.

1.2.3 Nonlinear write asymmetry

Nonlinear programming is implemented not as a pulse-accurate write simulator but as a first-order behavioral proxy that modulates the straight-through estimator backward pass according to the current conductance position. If a weight update would increase conductance, the surrogate gradient is scaled as

$$\left.\frac{\partial \mathcal{L}}{\partial G}\right|_{\mathrm{LTP}} \propto \left(\frac{G_{\max} - G}{G_{\max} - G_{\min}}\right)^{\mathrm{NL}_{\mathrm{LTP}}-1}, \tag{1.5}$$

whereas a conductance-decreasing update is scaled as

$$\left.\frac{\partial \mathcal{L}}{\partial G}\right|_{\mathrm{LTD}} \propto \left(\frac{G - G_{\min}}{G_{\max} - G_{\min}}\right)^{|\mathrm{NL}_{\mathrm{LTD}}|-1}. \tag{1.6}$$

When $\mathrm{NL}_{\mathrm{LTP}} = 1$ and $\mathrm{NL}_{\mathrm{LTD}} = -1$, the model reduces to the identity STE used in the nominal experiments. Larger magnitudes increase the dependence of the update on the current conductance position: potentiation becomes weaker near $G_{\max}$ and depression becomes weaker near $G_{\min}$.

It is important to emphasize the abstraction level. The $NL = 2.0$ boundary identified in Chapter 3 reflects the limit of this first-order gradient-scaling approximation

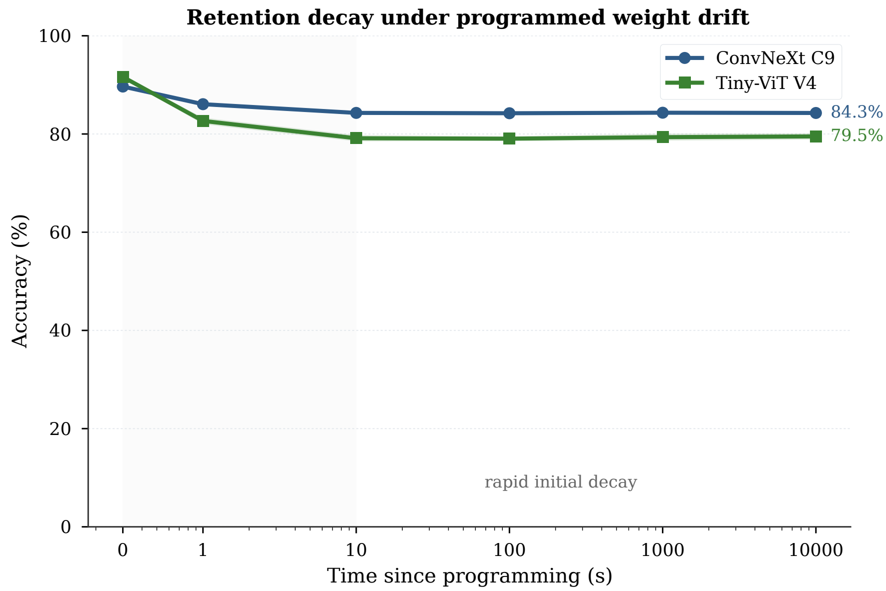

Figure 1.1: Double-exponential retention decay reproduced from the manuscript. The curve shows the programmed conductance fraction remaining as a function of elapsed time, with the fast ($\tau_1 = 140$ ms) and slow ($\tau_2 = 610$ ms) components visible.

under a severe matched stress setting, not a universal impossibility theorem for all organic write processes. Its practical purpose is to ensure that measured nonlinearity coefficients can enter the optimization path through the same profile interface used by inference-only experiments.

1.2.4 Retention decay

Post-programming weight drift is modeled with a double-exponential retention law. For a programmed conductance G and elapsed inference time t, the decayed conductance is

$$G(t) = G_{\min} + (G - G_{\min}) \left[A_1 \exp\left(-t/\tau_1\right) + A_2 \exp\left(-t/\tau_2\right) + A_0 \right], \qquad (1.7)$$

with $A_1 = A_2 = (1 - A_0)/2$ and $A_0 = 0.6$. The default parameters are $\tau_1 = 140$ ms and $\tau_2 = 610$ ms, taken from the handbook's device prior based on DNTT retention measurements[?] . This form captures an initial fast drop, an intermediate slower decay, and a non-zero persistent fraction.

Retention is applied after conductance quantization and before the stochastic noise terms in the analog forward model. This ordering reflects the interpretation that weights are first programmed to discrete conductance levels, then drift in time, and are finally observed under fixed mismatch and read-time variability. The framework additionally supports a state-dependent retention extension where high-conductance states decay faster, allowing stress-testing of long-term stability under realistic relaxation dynamics.

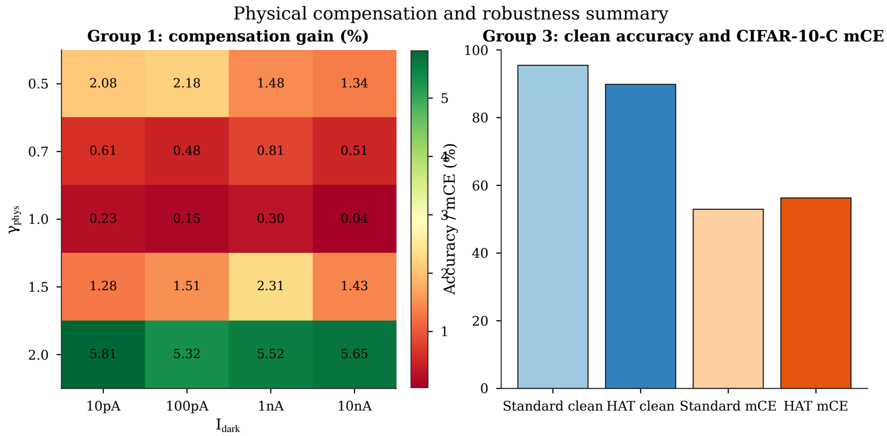

Figure 1.2: Physical front-end compensation reproduced from the manuscript. The inverse-gamma preprocessor linearizes the sub-linear photoresponse, but also reshapes shot-noise variance.

1.2.5 Optoelectronic front-end compensation

To model the organic phototransistor front end, the framework uses a sub-linear photocurrent response

$$I_{\text{photo}} = \alpha P^{\gamma_{\text{phys}}} + I_{\text{dark}} + \eta_{\text{shot}}, \tag{1.8}$$

where P denotes the incident optical signal, α is a responsivity scale, γ_{phys} captures the non-linear transfer characteristic, I_{dark} is dark current, and η_{shot} is a signal-dependent stochastic term. The compensation mechanism applies an inverse-gamma preprocessing step

$$P_{\text{in}} = X^{1/\gamma_{\text{phys}}}, \tag{1.9}$$

so that the subsequent device response becomes approximately linear in the original normalized input X.

This compensation is not treated as a universally beneficial transformation. Because shot-noise variance scales with photocurrent, inverse-gamma preprocessing also reshapes the noise statistics. The benefit must therefore be interpreted as a task- and regime-dependent trade-off rather than as a free robustness improvement. In the current workflow, the front-end is used explicitly in the physical-compensation experiments and analyzed separately from the weight-noise robustness experiments so that frontend compensation and hardware-aware training are not conflated.

1.2.6 ADC quantization and peripheral non-ideality

At the output of each crossbar column, an analog-to-digital converter discretizes the accumulated current. The framework models ADC non-idealities separately from the energy counts. For a b-bit converter, the continuous current is first quantized to 2^b nominal codes, and differential non-linearity (DNL) is injected by perturbing each step width:

$$\Delta_{\mathrm{actual}}[i] = \Delta_{\mathrm{ideal}} \left(1 + \mathcal{N}(0, \sigma_{\mathrm{DNL}}^2)\right), \tag{1.10}$$

with $\sigma_{\mathrm{DNL}} = 0.5$ LSB in the default configuration. This perturbation is instantiated as a fixed per-ADC offset table, so the converter behaves analogously to a static D2D imperfection rather than fresh C2C noise.

The energy model is kept explicitly separate from the accuracy model. A first-order estimate sums analog MAC cost, ADC cost, DAC cost, digital MAC cost, and memory-access cost:

$$E_{\mathrm{total}} = E_{\mathrm{analog}} + E_{\mathrm{ADC}} + E_{\mathrm{DAC}} + E_{\mathrm{digital}} + E_{\mathrm{buffer}}. \tag{1.11}$$

This separation is deliberate: it allows measured accuracy and estimated energy to be combined in a Pareto analysis without conflating the physical fidelity of the forward simulator with the placeholder nature of the energy constants.

1.3 PyTorch autograd integration

The framework is implemented as a collection of drop-in PyTorch modules that replace standard `nn.Linear` and `nn.Conv2d` layers with analog-aware equivalents. This section explains how the physical pipeline is integrated into the autograd graph, making hardware-aware training a training-mode switch rather than a custom training loop rewrite.

1.3.1 Straight-through estimator quantizer

The fundamental autograd primitive is a custom `torch.autograd.Function` that implements uniform quantization with an identity backward pass. In the forward direction, the input is clamped to $[x_{\min}, x_{\max}]$, normalized to $[0, 1]$, rounded to the nearest of n_{levels} discrete values, and denormalized back:

$$x_{\mathrm{quant}} = \left\lfloor \frac{x_{\mathrm{clamped}} - x_{\min}}{x_{\max} - x_{\min}} (n_{\mathrm{levels}} - 1) \right\rceil \frac{x_{\max} - x_{\min}}{n_{\mathrm{levels}} - 1} + x_{\min}. \tag{1.12}$$

In the backward direction, gradients pass straight through unchanged: $\partial\mathcal{L}/\partial x_{\mathrm{quant}}$ is returned as $\partial\mathcal{L}/\partial x$. This is the standard straight-through estimator (STE) approximation for piecewise-constant conductance states.

What distinguishes this implementation is that the backward pass optionally rescales the gradient by the nonlinear write asymmetry factors defined in Equations (1.5)–(1.6). The quantizer therefore serves two purposes simultaneously: it enables gradient flow through discrete conductance levels, and it injects measured write nonlinearity into the optimizer through the same code path.

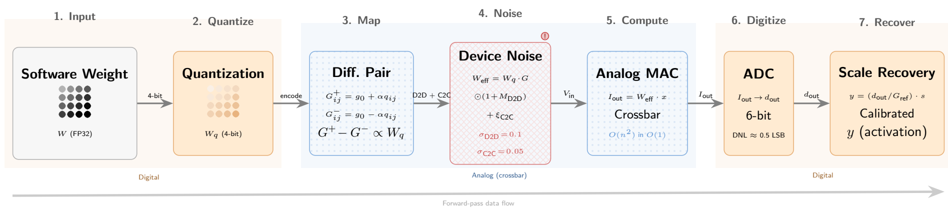

Figure 1.3: Differential-pair weight-to-conductance mapping reused from the manuscript. The digital weight tensor is split into positive and negative parts, normalized, mapped to the conductance window, quantized, and recombined as a differential effective weight.

1.3.2 Differential-pair conductance mapping

Each analog-mapped weight tensor $\mathbf{W}$ is converted into a differential-pair conductance representation. Let

$$\mathbf{W}^+ = \max(\mathbf{W}, 0), \qquad \mathbf{W}^- = \max(-\mathbf{W}, 0). \tag{1.13}$$

To avoid propagating gradients through the normalization constant, the scale factor is detached from the computation graph:

$$s_W = \|\mathbf{W}\|_\infty^{\text{detach}} + \varepsilon. \tag{1.14}$$

The positive and negative components are then mapped to the physical conductance range $[G_{\min}, G_{\max}]$:

$$\mathbf{G}^+ = G_{\min} + \frac{\mathbf{W}^+}{s_W}(G_{\max} - G_{\min}), \qquad \mathbf{G}^- = G_{\min} + \frac{\mathbf{W}^-}{s_W}(G_{\max} - G_{\min}). \tag{1.15}$$

The programmed conductances are quantized to n_{states} discrete levels with the STE:

$$\widetilde{\mathbf{G}} = \text{STEQuantize}(\mathbf{G}; n_{\text{states}}, G_{\min}, G_{\max}), \tag{1.16}$$

so that the forward path observes discrete conductance states while the backward path passes gradients as identity.

The effective analog weight used for vector-matrix multiplication is the differential conductance

$$\mathbf{W}_{\text{eff}} = \widetilde{\mathbf{G}}^+ - \widetilde{\mathbf{G}}^-, \tag{1.17}$$

optionally followed by scale recovery:

$$\widehat{\mathbf{W}}_{\text{eff}} = \mathbf{W}_{\text{eff}} \cdot \frac{\|\mathbf{W}\|_\infty^{\text{detach}}}{G_{\max} - G_{\min}}. \tag{1.18}$$

Scale recovery is one of the key physical simplifications of the present work. It is implemented as an ideal post-array digital rescaling step using layer-wise floating-point factors. That is a useful first-order behavioral abstraction for algorithm-device co-design, but it implicitly assumes that the peripheral readout chain can realize the required gain calibration with negligible quantization or control overhead. In this thesis, scale recovery is therefore treated as a calibrated digital post-processing primitive rather than as a claim of zero-cost physical circuitry.

1.3.3 Forward-path noise injection

The analog layer forward pass follows a strict physical ordering:

1. **Quantization:** Float weight $\rightarrow$ differential conductance pair $\rightarrow$ STE quantization.
2. **Retention:** Optional double-exponential decay applied to programmed conductances.
3. **Mismatch:** D2D noise (fixed buffer) and C2C noise (resampled per forward) added to the effective weight.
4. **Scale recovery:** Optional digital rescaling to restore the original weight magnitude.
5. **Linear operation:** Standard `F.linear` or `F.conv2d` with the noisy effective weight.

This ordering is not arbitrary. It reflects the physical interpretation that weights are first programmed to discrete conductance levels, then drift in time, and are finally observed under fixed mismatch and read-time variability. Because every step is differentiable via the STE, the entire pipeline participates in gradient descent during hardware-aware training.

1.3.4 Backward-path gradient scaling

When nonlinear write asymmetry is enabled, the STE backward pass is modified to scale gradients by the current conductance position. For a gradient $g = \partial\mathcal{L}/\partial G$, the returned surrogate is

$$g_{\text{surrogate}} = \begin{cases} g \cdot \left(\dfrac{G_{\max} - G}{G_{\max} - G_{\min}} \right)^{\mathrm{NL}_{\mathrm{LTP}} - 1} & \text{if } g \geq 0, \\ g \cdot \left(\dfrac{G - G_{\min}}{G_{\max} - G_{\min}} \right)^{|\mathrm{NL}_{\mathrm{LTD}}| - 1} & \text{if } g < 0. \end{cases} \tag{1.19}$$

This scaling has a simple physical interpretation. Near $G_{\max}$, further potentiation is physically difficult; the gradient scaling therefore suppresses updates that would push the conductance beyond the measurable window. Near $G_{\min}$, the same logic applies to depression. The scaling is applied element-wise inside the autograd backward call, so it requires no changes to the optimizer or learning-rate schedule.

It is worth repeating that this is an optimizer-side abstraction rather than a pulse-by-pulse device simulator. The framework does not track individual programming pulses, write-verify iterations, or conductance trajectory history. It instead provides a direct way to inject measured write asymmetry into training-time robustness studies through the same profile interface used for inference-only experiments.

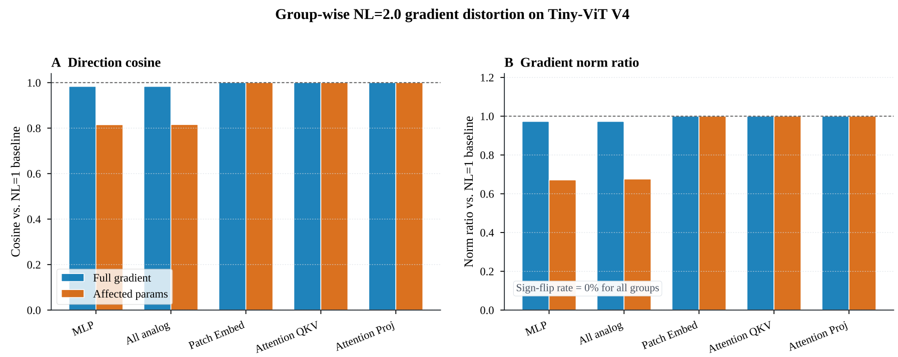

Figure 1.4: Nonlinear gradient distortion reproduced from the manuscript. The plots show how the surrogate gradient is scaled as a function of conductance position for several NL values.

1.3.5 Model conversion and layer replacement

The framework provides a model conversion utility that performs in-place replacement of PyTorch layers. Given a standard model (e.g., Tiny-ViT from `timm`), the utility traverses the module tree and substitutes analog equivalents for layers whose names match a set of configurable patterns. For Tiny-ViT, the analog-mapped layers are the patch-embedding convolutions, the attention projection matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$, the attention output projection, and the two feed-forward matrices in each transformer block. Mobile inverted bottleneck convolution (MBConv) blocks, downsampling paths, local depthwise convolutions, dynamic attention products, softmax, LayerNorm, activation functions, and the classification head remain digital.

This split is motivated by array utilization. Dense operators map naturally to crossbars because their weight tensors can be flattened into large matrix tiles with high occupancy. Depthwise convolutions and dynamic attention products either have poor output-unit utilization or require input-dependent matrices that cannot be pre-programmed into non-volatile arrays. The same split is also consistent with the current organic-optoelectronic hardware literature, which already demonstrates regular static array programming and readout much more naturally than fully dynamic token-token interaction engines.

For Tiny-ViT-5M, the mapping sends 4,730,016 parameters to the analog domain and leaves 662,748 parameters digital, corresponding to an analog-mapped parameter ratio of 87.7 %. Under a 128×128 physical array constraint and differential-pair storage, this requires 812 arrays and 13,303,808 individual devices. The analog footprint is concentrated in the transformer stages: patch embedding contributes only 8 differential-pair arrays, Stage 1 contributes 48 arrays, Stage 2 contributes 384 arrays, and Stage 3 contributes 372 arrays.

The conversion utility copies learned weights from the original digital layers into the analog replacements, so a pre-trained checkpoint can be converted to hybrid form without re-initialization. This is essential for the fine-tuning experiments reported

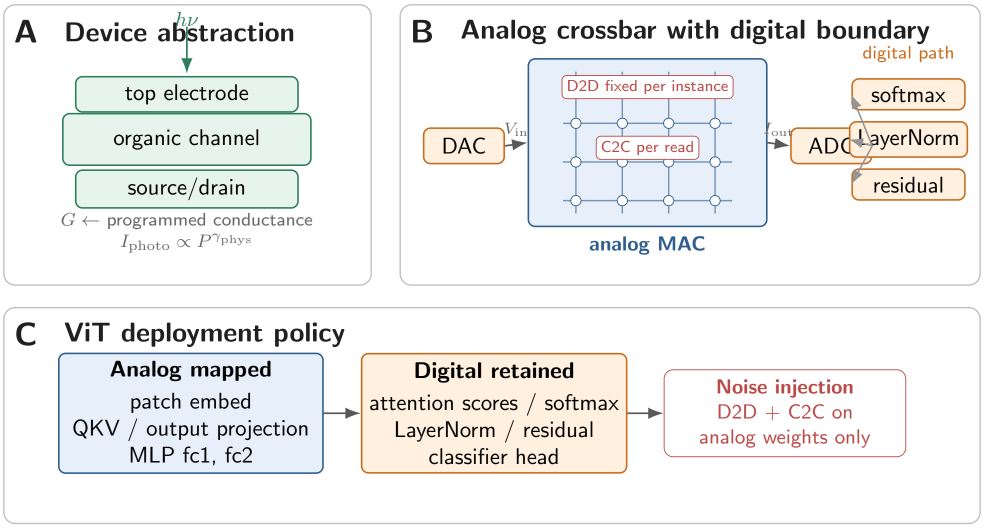

Figure 1.5: Hybrid system architecture reused from the manuscript. Dense linear operators execute on differential-pair crossbar arrays, while control-heavy or utilization-poor operators remain on a digital coprocessor.

in later chapters, where Tiny-ViT is loaded from an ImageNet-pretrained checkpoint and then adapted to analog deployment.

1.4 CrossSim sanity-check validation

A simulation framework is only as credible as its validation against independent tools. This section describes a cross-framework comparison against CrossSim, a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models[1].

1.4.1 Shared-regime experimental protocol

Because CrossSim was developed primarily around inorganic resistive memories, it does not directly expose inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention as first-class device primitives. A direct comparison is therefore limited to the *shared regime*: 8-bit ADC, uniform noise, deterministic baseline, and standard differential-pair mapping. Both frameworks were configured to use positive-only inputs, calibrated ADC ranges, and comparable conductance windows.

The comparison was performed on a deterministic 1 000-image CIFAR-10 test subset. This subset protocol was chosen because of CrossSim throughput constraints on the full test set. The protocol is disclosed explicitly; the results should be read as a sanity check rather than as a comprehensive benchmark.

1.4.2 Results and interpretation

In the clean baseline regime, the two frameworks produced consistent inference accuracy: 86.2 % (ours) versus 83.7 % (CrossSim) at 8-bit ADC on the shared subset. Under uniform noise injection at $\sigma = 5\,\%$, however, the predictions diverged substantially: $81.63 \pm 0.56\,\%$ (ours) versus $67.20 \pm 2.67\,\%$ (CrossSim), a qualitative gap of 14.43 percentage points.

This divergence is informative rather than alarming. It highlights that accuracy predictions are highly sensitive to the exact noise-to-conductance mapping: how variances are referenced (global window versus local state), whether noise is added before or after quantization, and how differential-pair common-mode cancellation is modeled. These are precisely the details that a profile-driven framework makes explicit. The CrossSim comparison therefore serves as a valuable sanity check on the baseline implementation while confirming that the organic-specific extensions—photoresponse, retention, proportional noise, and ensemble resampling—represent genuine capability additions beyond the shared inorganic toolset.

The frameworks are best viewed as complementary. CrossSim provides Simulation Program with Integrated Circuit Emphasis (SPICE)–calibrated parasitic models and detailed peripheral circuits for inorganic arrays. The present framework provides a lightweight, PyTorch-native stack for organic-device profile substitution, end-to-end hardware-aware training, and rapid algorithm-device co-design. Neither replaces the other; together they bracket the space of simulation fidelity versus experimental agility.

1.5 Limitations of the first-order behavioral model

While the framework provides a robust bridge between device parameters and system accuracy, it is fundamentally a first-order behavioral simulation framework. Several complex physical realities are abstracted, and these limitations should be kept in mind when interpreting the results of later chapters.

First, the digital overhead of scale masking is modeled as an exact floating-point operation. In physical hardware, layer-wise rescaling would require either high-precision digital multipliers or finely programmable transimpedance-amplifier gains, which introduce hidden energy and area overheads not fully penalized in the current energy model.

Second, position-dependent IR drop along crossbar wordlines and bitlines, as well as sneak-path currents in passive arrays, are not modeled in the current framework. These effects introduce systematic, input-pattern-dependent weight distortion that is distinct from the stochastic variability captured by the C2C and D2D terms. The framework includes scalar placeholders for these effects, but they are not geometry-aware.

Third, the energy parameters are anchored to 28 nm handbook assumptions and should be interpreted as first-order system-model estimates rather than final silicon measurements. Array-to-digital interconnect energy is absorbed into SRAM read/write cost terms, and dedicated routing overhead is not separately itemized.

Finally, the framework assumes ideal single-shot programming and therefore excludes iterative write-verify overhead for 4-bit conductance states. Extending the framework toward measured level tables and explicit integral non-linearity remains future work.

1.6 Summary and transition

This chapter has described the profile-driven simulation framework that underpins the experimental contributions of this thesis. The key design principles are:

- **Profile-driven parameterization:** All physical effects enter through a single structured configuration object, making literature priors and measured profiles interchangeable.
- **Pluggable physical interfaces:** Device variability, nonlinear write asymmetry, retention drift, optoelectronic front-end distortion, and ADC non-ideality are modular extensions that can be enabled or disabled without changing model architecture code.
- **Deep PyTorch integration:** Hardware-aware training is implemented via custom autograd primitives and drop-in layer replacement, making it a training-mode switch rather than a custom optimizer rewrite.
- **External validation:** CrossSim sanity-check comparison confirms baseline consistency in the shared regime and highlights where organic-specific extensions diverge from generic RRAM models.

The framework is intentionally positioned as a research instrument rather than a chip-predictive emulator. Its purpose is to rank deployment risks, to enable reproducible algorithm-device co-design, and to preserve a clean pathway for recalibration when measured device data become available.

With the instrument in place, the first question is how HAT behaves when training and deployment instances differ.

Chapter 2

Hardware-Aware Training: A Taxonomy of Resampling Cadences, Noise Profiles, and Transfer Guarantees

2.1 Introduction: HAT as a design space

Hardware-aware training (HAT) is not a single algorithm but a family of training objectives that share one structural feature: they inject a surrogate of the deployment forward-path perturbation into the optimization loop, then back-propagate through a straight-through estimator (STE) so that the digital weight update senses the analog distortion it is supposed to survive. Within that shared skeleton, however, many design choices remain open. How often should the mismatch field be resampled? Should the noise law be uniform, proportional to conductance state, or a mixture of both? And how does the training protocol relate to the formal guarantee one can offer at deployment time?

This chapter treats those choices as axes of a design space rather than as minor implementation details. The thesis perspective differs from the manuscript (manuscript methodology section) in two ways. First, the manuscript presents the canonical fixed-mask, epoch-resampled, and per-batch variants as a short ablation in support of the Ensemble HAT headline; here, each variant receives a formal definition and a distinct transfer-guarantee interpretation. Second, the manuscript discusses proportional-noise retraining primarily as a physical-extension stress test (manuscript NL-HAT stress section); here, it is embedded in a broader taxonomy of noise-profile targeting that clarifies why regime-matched training recovers accuracy while cross-regime deployment collapses.

The chapter is organized as follows. Section 2.2 defines the three canonical resampling cadences and visualizes their structural differences. Section 2.3 reports the held-out accuracy sweep that isolates epoch-level resampling as the load-bearing schedule choice. Section 2.4 maps the proportional-versus-additive noise landscape and explains why "mixed" profiles are the realistic default for organic optoelectronic arrays. Section 2.5 situates these choices against DNN+NeuroSim, AIHWKIT, MemTorch, CrossSim, and recent hybrid-mapping literature. Section 2.6 develops the central thesis insight: the resampling cadence does not merely affect optimization variance; it implicitly defines the distributional family over which the learned classifier is guaranteed to generalize. Section 2.7 summarizes.

2.2 Per-batch, per-epoch, and fixed-mask HAT: formal definitions

All three variants operate on the same hybrid analog–digital inference stack described in the manuscript (manuscript hybrid-deployment section): dense linear operators execute on differential-pair crossbars, control-heavy operators remain digital, and the analog weight is recovered via

$$\hat{W}_{ij} = s_\ell \, G_{\mathrm{eff},ij}, \qquad s_\ell = \frac{w_{\max}}{G_{\max} - G_{\min}}, \tag{2.1}$$

with layer index ℓ omitted for brevity. The effective conductance G_{eff} is perturbed by a device-to-device (D2D) mismatch field M and by per-forward cycle-to-cycle (C2C) noise ε. The D2D field is spatially structured and fixed for a given hardware instance; the C2C term is independent and identically distributed (i.i.d.) across forward passes.

2.2.1 Fixed-mask HAT

Under **fixed-mask HAT**, one D2D realization M_0 is drawn before training and held constant for all epochs and all mini-batches. The optimization objective is

$$\theta^\star_{\mathrm{fix}} = \arg\min_\theta \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n) \,. \tag{2.2}$$

This is the protocol used by the canonical V4 experiment (87.95±0.27 % source-domain accuracy) in the manuscript. It is computationally minimal because the mismatch map is cached, and it corresponds to the intuitive strategy of "training on the hardware you have." Its weakness, which Chapter explores in depth, is that the learned parameters $\theta^\star_{\mathrm{fix}}$ can become coupled to the specific spatial signature of M_0. When the deployment array presents a different fixed realization $M' \sim p(M)$, the calibration and feature routing learned for M_0 are no longer valid, and accuracy collapses.

2.2.2 Ensemble HAT (per-epoch resampling)

Under **Ensemble HAT**, the D2D field is resampled at the start of every training epoch and held constant across all mini-batches within that epoch. The objective becomes an empirical expectation over mismatch maps:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta}\ \mathbb{E}_{M\sim p(M)}\left[\frac{1}{N}\sum_{n=1}^{N}\ell(f(x_n;\theta,M),y_n)\right]. \tag{2.3}$$

Because one fresh map is drawn per epoch, the model sees a sequence $\{M^{(1)}, M^{(2)}, \ldots, M^{(E)}\}$ of structured spatial instances. The expectation is Monte Carlo–estimated with E independent draws, where E is the number of training epochs. Crucially, the mini-batch gradients within an epoch are correlated under the same $M^{(e)}$, so the optimizer has enough stationary steps to adapt its feature extraction to that instance before the instance changes. The manuscript reports that this protocol preserves $86.37 \pm 1.54\,\%$ accuracy on ten fresh hardware instances, overwhelming the chance-level collapse of fixed-mask HAT (manuscript hardware-transferability section).

2.2.3 Per-batch resampling

Under **per-batch resampling**, a fresh D2D field is drawn independently for every forward pass. The objective is still an expectation over $p(M)$, but the Monte Carlo estimate uses $B \times E$ independent draws, where B is the number of mini-batches per epoch. Formally,

$$\theta^{\star}_{\text{batch}} = \arg\min_{\theta}\ \mathbb{E}_{M\sim p(M)}\left[\frac{1}{N}\sum_{n=1}^{N}\ell(f(x_n;\theta,M),y_n)\right], \tag{2.4}$$

with the same functional form as Eq. 2.3 but a much higher resampling frequency. The training trajectory is now exposed to a new spatial instance every few hundred samples. Intuitively, this might seem preferable because the model sees "more diversity"; empirically, it underperforms epoch-level resampling because the optimizer never experiences a stable enough forward path to settle into a basin that is compatible with any single structured instance.

2.2.4 Visual comparison of the three regimes

Figure 2.1 reuses the manuscript's Ensemble HAT concept figure to anchor the discussion, while Table 2.1 summarizes the structural properties.

Algorithm 1 gives the unified training loop for all three HAT cadences. The only difference among the variants is the D2D resampling condition on line 6: fixed-mask samples once before training (*cadence* = never), Ensemble HAT samples at every epoch boundary (*cadence* = epoch), and per-batch samples before every mini-batch

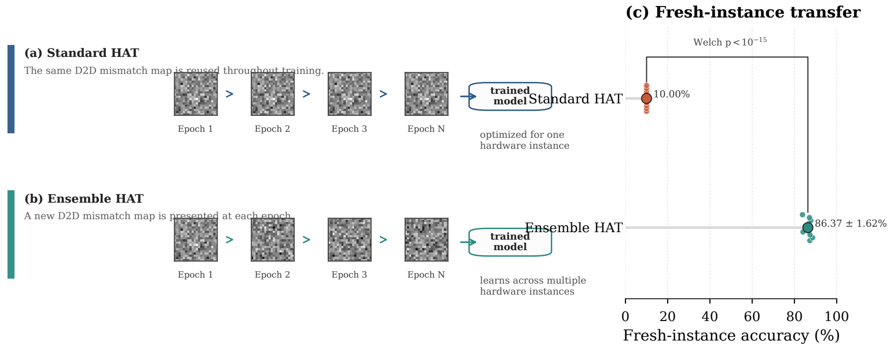

Figure 2.1: Conceptual comparison of fixed-mask and epoch-resampled HAT, reused from the manuscript package (Figure **??** of the manuscript). The fixed-mask panel shows one spatial mismatch map persisting across all epochs; the Ensemble HAT panel shows a fresh map per epoch. Per-batch resampling (not shown) would replace the map at every mini-batch boundary.

Table 2.1: Structural comparison of the three canonical HAT cadences.

Property	**Fixed-mask**	**Per-epoch**	**Per-batch**
Resampling frequency	Never	Every epoch	Every mini-batch
Draws per 100 epochs	1	100	$100B$
Gradient correlation time	Full training	One epoch	One mini-batch
Forward-path stability	Maximal	Moderate	Minimal
Fresh-instance guarantee	None	Implicit expectation	Implicit expectation
Wall-clock overhead	Baseline	$\sim 1\times$	Slightly higher

($cadence =$ batch). This unified formulation removes any ambiguity that might remain in prose descriptions.

The distinction between per-epoch and per-batch is subtle but decisive. Per-epoch resampling creates a block-stationary optimization landscape: within each block, the gradient field is consistent, and the optimizer can descend; across blocks, the landscape shifts, forcing the solution toward a shared basin. Per-batch resampling creates a fully non-stationary landscape in which every gradient step sees a different spatial structure. The empirical result, developed in the next section, is that the block-stationary regime yields the best fresh-instance accuracy.

2.3 Ensemble frequency sweep: the cadence ablation

The manuscript notes, in manuscript NL-HAT stress section, that an exploratory single-run training ablation supports the choice of epoch-level resampling. This section unpacks that ablation and places it in the thesis design-space framework.

Algorithm 1 HAT training loop (unified over the three resampling cadences)

Require: Training data $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$, initial parameters $\theta^{(0)}$, optimizer $\mathcal{O}$
Require: Resampling strategy $cadence \in \{\text{never}, \text{epoch}, \text{batch}\}$
Require: Device profile $\mathcal{P}$, training epochs E, batch size B
Ensure: Trained parameters $\theta^{(E)}$

if $cadence = \text{never}$ **then**
  Sample D2D field $\boldsymbol{\epsilon}_{\text{D2D}} \sim \mathcal{N}(0, \sigma^2_{\text{D2D}} \Delta G^2)$ once ▷ Fixed-mask
end if
for $e = 1, 2, \ldots, E$ **do** ▷ Training epochs
  if $cadence = \text{epoch}$ **then**
    Resample D2D field $\boldsymbol{\epsilon}_{\text{D2D}} \sim \mathcal{N}(0, \sigma^2_{\text{D2D}} \Delta G^2)$ ▷ Ensemble HAT
  end if
  for each mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
    if $cadence = \text{batch}$ **then**
      Resample D2D field $\boldsymbol{\epsilon}_{\text{D2D}} \sim \mathcal{N}(0, \sigma^2_{\text{D2D}} \Delta G^2)$ ▷ Per-batch
    end if
    $\mathcal{L} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{(x,y) \in \mathcal{B}} \ell\big(f(x; \theta^{(e)}, \boldsymbol{\epsilon}_{\text{D2D}}, \mathcal{P}), y\big)$
    Compute $\nabla_\theta \mathcal{L}$ via STE back-propagation
    $\theta^{(e)} \leftarrow \mathcal{O}.\text{step}(\theta^{(e)}, \nabla_\theta \mathcal{L})$
  end for
end for
return $\theta^{(E)}$

2.3.1 Experimental protocol

All ablation checkpoints were trained under the canonical Tiny-ViT V4 recipe: AdamW with learning rate 5×10^{-4}, weight decay 0.05, cosine annealing over 100 epochs (the ablation scan used a shortened 50-epoch schedule), batch size 64, and 4-bit differential conductance mapping. The only manipulated variable was the D2D resampling cadence. Three conditions were compared:

- **Fixed-mask control:** One D2D realization drawn at initialization and frozen.
- **Per-epoch (Ensemble HAT):** One fresh D2D realization at the start of each epoch.
- **Per-batch control:** One fresh D2D realization for every forward pass during training.

After training, each checkpoint was evaluated on a held-out validation split that was not used for early stopping. The reported numbers are therefore held-out accuracies from a single-run scan, not the full ten-instance fresh-instance protocol reported for the paper-locked Ensemble HAT baseline.

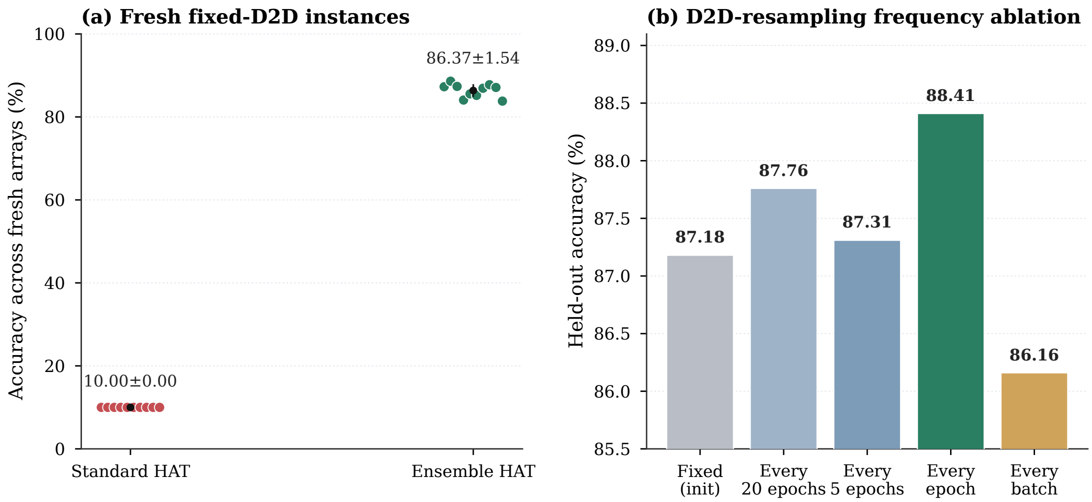

Figure 2.2: Fresh-instance robustness and D2D-resampling cadence ablation, reproduced from the manuscript supplementary package (Supplementary Figure S4). **Left:** standard HAT collapses to chance on every fresh array, whereas Ensemble HAT preserves robust accuracy. **Right:** held-out accuracy under different resampling cadences in a 50-epoch scan. Epoch-level resampling reaches 88.41 %, exceeding fixed-mask (87.18 %) and per-batch (86.16 %) controls.

2.3.2 Results

The exploratory scan yielded the following held-out accuracies under the 50-epoch short-run schedule:

- Epoch-level resampling (Ensemble HAT): **88.41 %**
- Fixed-mask training: **87.18 %**
- Per-batch resampling: **86.16 %**

These numbers are not directly comparable to the paper-locked 86.37 ± 1.54 % fresh-instance mean because they come from a shorter training run and a held-out validation split rather than the full ten-instance Monte Carlo protocol. Their importance is relative, not absolute: epoch-level resampling outperforms both fixed-mask and per-batch alternatives even under a weakened training budget. Figure 2.2 reproduces the manuscript supplementary panel that visualizes this comparison.

2.3.3 Interpretation: why epoch-level resampling wins

The result is initially surprising because per-batch resampling exposes the model to more D2D diversity and might therefore be expected to produce a more invariant solution. The fact that it underperforms suggests that diversity alone is not the objective; rather, the optimizer needs sufficient time under a single structured instance to learn features that are compatible with that instance, and then enough instances to generalize across the distribution.

Fixed-mask training, by contrast, has unlimited time under one instance and therefore reaches high source-domain accuracy, but it lacks the distributional breadth needed for transfer. Epoch-level resampling sits at a sweet spot: each epoch provides enough stability for the optimizer to make progress, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting.

This interpretation is reinforced by the spatial-versus-i.i.d. ablation discussed in the supplementary package. When the D2D field is replaced by independent per-layer i.i.d. noise drawn every epoch, fresh-instance accuracy drops to approximately 75 %; when it is drawn every batch, it drops further to approximately 70 %. The structured spatial correlation of the true D2D field, combined with the block-stationary exposure schedule, is the load-bearing ingredient.

2.4 Noise-profile targeting: proportional versus additive versus mixed

The cadence discussion assumes a specific noise law: the canonical uniform-noise model in which D2D and C2C perturbations are Gaussian draws scaled by the full conductance window. Real devices, however, often exhibit state-dependent variance. This section extends the taxonomy to the noise-profile axis.

2.4.1 Uniform (additive) noise

Under the **uniform-noise** regime, the standard deviation of the conductance perturbation is a fixed fraction of the dynamic range:

$$\sigma_{\mathrm{D2D}} = \alpha\,(G_{\max} - G_{\min}), \qquad \sigma_{\mathrm{C2C}} = \beta\,(G_{\max} - G_{\min}). \tag{2.5}$$

The canonical profile uses $\alpha = 10\,\%$ and $\beta = 5\,\%$. Because the noise is referenced to the global window, small-conductance devices experience the same absolute perturbation as large-conductance devices. Under the scale-recovery law of Eq. 2.1, many of these perturbations fall below the half-step of the recovered weight grid and are therefore masked. This "scale-masking" effect is responsible for the high zero-noise hybrid control accuracy (97.39 %) and the relatively mild degradation under canonical uniform noise.

2.4.2 Proportional (state-dependent) noise

Under the **proportional-noise** regime, the standard deviation scales with the instantaneous conductance magnitude:

$$\sigma_{\mathrm{D2D}}(G) = \alpha\,|G|, \qquad \sigma_{\mathrm{C2C}}(G) = \beta\,|G|. \tag{2.6}$$

High-conductance states therefore experience larger absolute perturbations than low-conductance states. The scale-masking protection breaks down because the perturbation grows with the signal itself: large weights, which dominate the network's output, also receive the largest relative noise. When a model trained under uniform noise is evaluated under proportional noise, accuracy collapses to 10.00 % for Tiny-ViT V4, indicating that the uniform model overestimates intrinsic robustness.

Crucially, proportional noise is not merely a stress test; it is the physically expected regime for ohmic conductance channels and for several organic device families in which conductance fluctuations track the mean. The manuscript therefore treats proportional-noise HAT as a regime-matched training experiment rather than an adversarial perturbation.

2.4.3 Mixed profiles

A **mixed** noise profile superimposes a floor (uniform) component and a state-dependent (proportional) component:

$$\sigma(G) = \sigma_0 + \kappa\,|G|. \tag{2.7}$$

This form captures the empirical reality that real arrays exhibit both a baseline fabrication spread (uniform) and a state-dependent read or programming fluctuation (proportional). The framework supports mixed profiles through the JavaScript Object Notation (JSON) device-profile interface (manuscript profile-interface section), although the canonical paper experiments use pure uniform or pure proportional modes for interpretability.

2.4.4 Regime-matched training and cross-regime collapse

The manuscript reports that proportional-noise HAT recovers Tiny-ViT to 97.37 ± 0.05 % when training and evaluation are both conducted under the proportional law. This demonstrates that the proportional regime is learnable. However, the same checkpoint transfers poorly back to uniform-noise evaluation, and a uniform-noise checkpoint collapses under proportional evaluation. The takeaway is that HAT robustness is *distribution-specific*: the training objective must match the deployment noise law, and there is no free cross-regime generalization.

Table 2.2 summarizes the cross-regime behavior for Tiny-ViT. The diagonal entries (matched training and evaluation) are high; the off-diagonal entries (mismatched) are low. This matrix is a central tool for deciding which noise model to adopt as the canonical profile: if the physical device is believed to operate in the proportional regime, then uniform-noise HAT is an optimistic fiction; conversely, if the device is primarily uniform, proportional-noise training is unnecessarily pessimistic.

Table 2.2: Cross-regime accuracy matrix for Tiny-ViT under uniform and proportional noise. Diagonal entries are matched-regime results; off-diagonal entries show cross-regime collapse.

	Eval: Uniform	**Eval: Proportional**
Train: Uniform (V4)	∼91 %	10.00 %
Train: Proportional	∼10 %	97.37 %

2.5 Literature comparison: where this framework sits

The design-space axes developed above—resampling cadence and noise-profile targeting—are not unique to this thesis. This section maps them onto the broader landscape of CIM simulation and HAT literature.

2.5.1 DNN+NeuroSim and hierarchical benchmarking

DNN+NeuroSim established the hierarchical template for device-to-system CIM evaluation: device models feed circuit models, which feed architecture models, which finally report accuracy, energy, and area
[2]. The framework supports full on-chip training and has been validated against fabricated macros. Within this hierarchy, HAT appears as a training-layer option, but the standard DNN+NeuroSim workflow does not emphasize fresh-instance transfer: the D2D mismatch is typically treated as a one-time fabrication spread, and the trained model is evaluated on the same spread it was trained with. The thesis therefore extends the DNN+NeuroSim philosophy by adding the fresh-instance protocol as a standard evaluation primitive.

2.5.2 AIHWKIT and per-batch stochastic variation

IBM's AIHWKIT provides a PyTorch-integrated toolkit for analog crossbar simulation with hardware-aware training
[3]. Its noise models are practical and well-calibrated for PCM and ReRAM devices. Under the hood, AIHWKIT injects per-batch weight noise and supports per-tile variability, but it does not expose a canonical baseline that resamples complete structured D2D mismatch maps for each hardware instance. The manuscript's shared-regime sanity check—95.46 % digital versus 90.08±0.21 % analog on ResNet-18/CIFAR-10—confirms qualitative agreement between the present simulator and AIHWKIT under one matched regime, but it does not establish equivalence for organic-specific extensions such as inverse-gamma photoresponse or double-exponential retention. The thesis therefore treats AIHWKIT as a methodological consistency anchor for inorganic regimes and as a contrast point for the organic-specific profile interface.

2.5.3 MemTorch and PyTorch-native embedding

MemTorch demonstrated how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation
[4]. Its device module is conceptually similar to the profile interface used here, but it is oriented toward filamentary ReRAM rather than toward organic optoelectronic conductance tuning. The cadence discussion in MemTorch focuses on inference-time fault injection rather than on training-time D2D resampling. The thesis extends this PyTorch-native philosophy by making the resampling cadence a first-class training hyperparameter.

2.5.4 CrossSim and GPU-accelerated lookup-table training

CrossSim provides GPU-accelerated crossbar-accuracy workflows with parasitic resistance, ADC models, and drift
[1]. Its training models are lookup-table-based rather than STE-based, which changes the HAT semantics: the surrogate is a pre-characterized non-ideality table rather than a differentiable noise injection. CrossSim therefore does not directly support the epoch-level D2D resampling protocol developed here, because its lookup tables are tied to specific device characterizations. The thesis STE-based approach trades some physical fidelity for gradient-path flexibility, enabling the cadence and noise-profile sweeps that are the focus of this chapter.

2.5.5 Hybrid mapping and the Lin et al. / Qin et al. HARDSEA architecture

Recent heterogeneous CIM accelerators for Vision Transformers, including HARDSEA by Lin *et al.* (with Q. Qin as co-author)
[5], adopt a mapping policy that closely resembles the one used in this thesis: linear projections (Q/K/V, output, feed-forward) are mapped to analog arrays, while attention scores, softmax, and normalization remain digital. The convergence is reassuring: it suggests that the hybrid partition is not an artifact of the present simulator but a widely adopted architectural response to the mismatch between weight-stationary crossbar strengths and dynamic, input-dependent operator requirements. Where the thesis adds value is in quantifying how that same hybrid mapping behaves under organic-specific noise laws and under different HAT cadences, rather than assuming a single fixed-mismatch training recipe.

2.5.6 Domain randomization and sim-to-real transfer

Ensemble HAT is conceptually related to domain randomization in sim-to-real robotics
[6], where training environments are randomized so that the policy generalizes to the real world. The key difference is structural: domain randomization typically randomizes

i.i.d. texture or lighting parameters, whereas D2D mismatch is a spatially correlated fixed field. The thesis ablation showing that i.i.d. per-epoch noise drops fresh-instance accuracy to ∼75 % (versus 86.37 % for structured resampling) confirms that the spatial structure matters. Ensemble HAT is therefore not merely "more noise augmentation"; it is augmentation matched to the correlation structure of the deployment distribution.

2.6 Thesis insight: resampling cadence as an implicit transfer guarantee

The empirical results of Sections 2.3 and 2.4 can be unified under a single theoretical observation: the resampling cadence implicitly defines the distributional family over which the learned classifier is guaranteed to generalize.

2.6.1 Fixed-mask HAT: singleton guarantee

Fixed-mask HAT solves Eq. 2.2, which is a standard empirical risk minimization problem conditioned on a single draw M_0. The guarantee it offers is trivial: the model is guaranteed to perform well on M_0 itself, and on nothing else. Formally, if $\mathcal{R}(\theta, M)$ denotes the risk under mismatch map M, then fixed-mask HAT minimizes $\mathcal{R}(\theta, M_0)$ and provides no bound on $\mathcal{R}(\theta, M')$ for $M' \neq M_0$. The 10.00 % fresh-instance collapse is the empirical signature of this absence of guarantee.

2.6.2 Per-batch HAT: product-distribution guarantee

Per-batch resampling solves Eq. 2.4 with $B \times E$ independent draws. The optimizer therefore sees a product distribution $p(M)^{\otimes(B\times E)}$ over mismatch maps, but each gradient step is computed under a single independent draw. The resulting solution minimizes the expected risk under the marginal distribution $p(M)$, but the high variance of the gradient estimates prevents the optimizer from finding a basin that is simultaneously compatible with any single structured instance. The empirical result—86.16 % held-out accuracy, lower than per-epoch—suggests that the product-diversity guarantee is too weak: the model learns to survive average noise, but not any specific realization.

2.6.3 Per-epoch HAT: block-stationary guarantee

Per-epoch resampling solves Eq. 2.3 with E block-stationary draws. Within each block, the optimizer descends under a fixed $M^{(e)}$; across blocks, it seeks a shared basin. The implicit guarantee is stronger than the product-distribution guarantee because the optimizer must find a parameter vector that is not merely good in expectation, but good for at least one full epoch of descent under each of E independent realizations.

The empirical result—88.41 % held-out accuracy, and 86.37±1.54 % on fresh instances—suggests that this block-stationary requirement is sufficient to force the model into a basin that generalizes across the deployment distribution.

2.6.4 Noise-profile matching as a distributional alignment condition

The same logic extends to noise profiles. A model trained under uniform noise minimizes risk under the uniform noise distribution $p_{\mathrm{uni}}(M)$; a model trained under proportional noise minimizes risk under $p_{\mathrm{prop}}(M)$. Because p_{uni} and p_{prop} are mutually singular in the large-perturbation tail, a classifier that is optimal under one is not guaranteed to be even admissible under the other. The 10.00 % cross-regime collapse is the empirical signature of this distributional misalignment.

The practical consequence is that the choice of noise profile is not a secondary modeling detail; it is a distributional prior that must be matched to the physical device. A simulator that defaults to uniform noise will overestimate deployment accuracy for devices that follow a proportional law. The profile-driven substitution interface (manuscript profile-interface section) is therefore not merely a convenience feature; it is the mechanism by which the training objective is aligned with the physical deployment distribution.

2.6.5 Summary: the cadence–profile design plane

Figure 2.3 sketches the two-dimensional design plane formed by the cadence axis (fixed / epoch / batch) and the noise-profile axis (uniform / proportional / mixed). The thesis experiments occupy specific points on this plane:

- **Manuscript canonical V4:** fixed-mask, uniform noise.
- **Manuscript Ensemble HAT:** per-epoch, uniform noise.
- **Manuscript proportional-noise HAT:** per-epoch, proportional noise.
- **This chapter's ablation:** per-epoch versus fixed versus per-batch, uniform noise.

The design plane is a tool for future work: if a new device profile suggests, for example, that spatial correlation decays over shorter length scales, one can move to a finer cadence (per-epoch with multiple maps per epoch) or to a coarser one (fixed-mask with a worst-case map). The framework's modularity makes such movements possible without code changes.

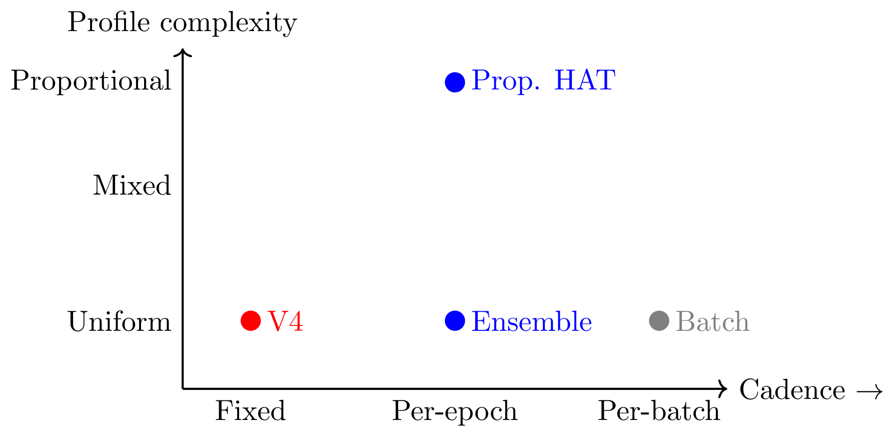

Figure 2.3: Simplified design plane for HAT protocols. The horizontal axis moves from fixed-mask (no resampling) to per-batch (maximal resampling). The vertical axis moves from uniform additive noise to proportional state-dependent noise. Thesis experiments occupy the labeled points.

2.7 Summary

This chapter has treated hardware-aware training as a design space with two primary axes: resampling cadence and noise-profile targeting.

On the **cadence axis**, three canonical protocols were defined formally: fixed-mask HAT (Eq. 2.2), per-epoch Ensemble HAT (Eq. 2.3), and per-batch resampling (Eq. 2.4). An exploratory 50-epoch ablation showed that epoch-level resampling reaches 88.41 % held-out accuracy, outperforming both fixed-mask (87.18 %) and per-batch (86.16 %) controls. The interpretation is that block-stationary exposure—one structured instance per epoch—provides enough forward-path stability for the optimizer to descend, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting.

On the **noise-profile axis**, uniform (additive), proportional (state-dependent), and mixed noise laws were compared. Proportional noise destroys the scale-masking protection that makes uniform-noise models appear robust, and regime-matched training is required for recovery. A model trained under uniform noise collapses to 10.00 % under proportional evaluation, and vice versa, demonstrating that HAT robustness is distribution-specific.

Against the **literature**, the framework was positioned relative to DNN+NeuroSim (hierarchical benchmarking), AIHWKIT (per-batch stochastic variation), MemTorch (PyTorch-native embedding), CrossSim (lookup-table training), and the HARDSEA hybrid-mapping architecture of Lin *et al.* (including Q. Qin). The thesis adds value not by outperforming these mature tools on inorganic benchmarks, but by exposing how the same hybrid mapping behaves under organic-specific noise laws and under different HAT cadences.

The central **thesis insight** is that the resampling cadence implicitly defines the

transfer guarantee. Fixed-mask HAT offers a singleton guarantee (valid only on the training instance), per-batch HAT offers a product-distribution guarantee that is too weak for structured spatial mismatch, and per-epoch HAT offers a block-stationary guarantee that empirically generalizes to fresh hardware instances. The taxonomy reveals symptoms, but symptoms are not diagnoses; the next chapter names the diseases. Physical-realism extensions in Chapter
refchap:physical-realism validate that the block-stationary guarantee survives correlated D2D and retention drift.

Chapter 3

Failure Mode Atlas

3.1 Introduction: What is a failure mode in organic-CIM deployment?

The preceding chapters have treated hardware-aware training (HAT), physical front-end compensation, and ensemble resampling as constructive recipes: tools that, when applied correctly, recover analog-deployed accuracy. This chapter inverts the perspective. Instead of asking "what works?" we ask "what breaks, why does it break, and what does the breakage reveal about the physics–algorithm boundary?" The answers are organized as a *failure mode atlas*—a structured catalogue of named, reproducible breakdown patterns that constrain the deployment of Vision Transformers on organic optoelectronic compute-in-memory (CIM) arrays.

In systems engineering, a failure mode is conventionally defined as a specific manner in which a system can fail to perform its intended function. For analog neural accelerators, that definition must be extended in three directions.

1. **The failure is distributed, not localized.** A single transistor stuck at high conductance is a device-level fault; a collapse of ImageNet top-1 accuracy from 97 % to 10 % is a system-level failure mode that may emerge from the interaction of thousands of well-behaved devices with a training objective that implicitly memorizes one particular mismatch field.

2. **The failure is conditional, not absolute.** The same physical non-ideality—say, a severe write nonlinearity of $NL = 2.0$—can be harmless in the multi-layer perceptron (MLP) blocks while structurally destroying the attention geometry. Context (layer type, training recipe, evaluation protocol) determines whether a given physics extension becomes a deployment blocker.

3. **The failure has a signature and a transfer boundary.** A genuine deployment failure mode must be observable under the fresh-instance protocol: 10 independent hardware realizations $\times$ 5 Monte Carlo forward passes per realization. Source-

domain rescue on the training-time array is diagnostically useful but not sufficient; only fresh-instance robustness answers the practical deployment question.

These criteria shape the structure of the atlas. Each entry is named, annotated with its *trigger* (the physical or algorithmic condition that precipitates the failure), its *signature* (the observable accuracy trajectory or gradient diagnostic that identifies it), its *mitigation* (the intervention that partially or fully recovers performance), and an *open question* (the boundary beyond which the mitigation itself fails or becomes impractical). The five entries are:

1. **Hardware-instance overfitting**—the canonical 10.00 % collapse of standard HAT on fresh arrays (Chapter develops the remedy in depth; this chapter treats it as the archetypal failure).

2. **Dual-attention nonlinear-write collapse**—the degradation of query/key/value (QKV) and output-projection geometry under severe write nonlinearity, and the diagnostic value of group-wise ablation in localising the bottleneck.

3. **Severe-NL degradation and MLP-path diagnostic**—the systematic degradation to the ∼80–82 % band under global $NL = 2.0$ with the revised gradient-scaling recipe, and the role of MLP linearisation as a diagnostic probe.

4. **Correlated-D2D bounded degradation**—the measurable but non-catastrophic loss of fresh-instance accuracy when spatially structured mismatch replaces the i.i.d. assumption.

5. **Retention plateau at 79 %**—the asymptotic accuracy floor under double-exponential conductance drift with dynamic scale recalibration.

The chapter concludes by identifying cross-cutting themes that tie these apparently distinct failures together: the importance of distributional training objectives, the distinction between source-domain rescue and deployment-grade transfer, and the recurring gap between first-order behavioral surrogates and the pulse-level physics they approximate. All empirical anchors are drawn from the manuscript results (manuscript results and discussion sections) and the supplementary material; figures are reused via `\includegraphics` from the manuscript package.

3.2 Failure mode: Hardware-instance overfitting

3.2.1 Trigger

Standard hardware-aware training fixes a single device-to-device (D2D) mismatch field M_0 for the entire training run, optimizing

$$\theta^\star_{\text{std}} = \arg\min_\theta \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n) , \tag{3.1}$$

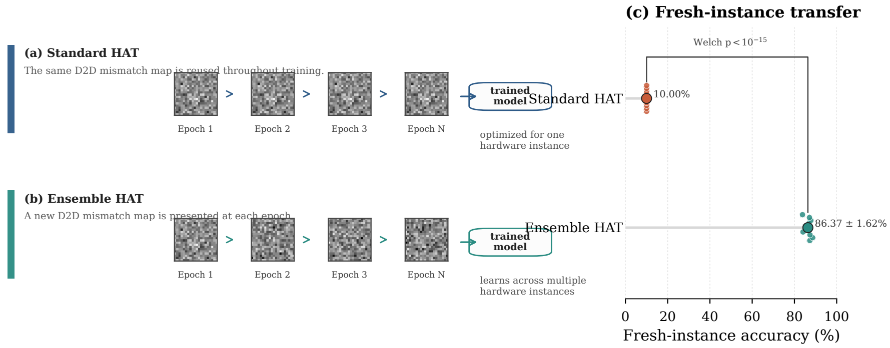

Figure 3.1: Fresh-instance transfer comparison (reused from manuscript supplementary package, Fig. S3). Standard HAT overfits a single structured mismatch field, whereas Ensemble HAT learns over a distribution of hardware instances. The fixed-mask model collapses to $10.00 \pm 0.00\,\%$; the ensemble model maintains $86.37 \pm 1.54\,\%$.

as described in manuscript Eq. (1) of the modeling-nonidealities section. Under this objective, the optimizer is free to exploit M_0 as a hidden domain label: scale-recovery factors s_ℓ (Eq. (2) in the manuscript) can implicitly calibrate to the particular spatial distortion encoded in M_0, and learned feature routing can adapt to the fixed perturbation pattern rather than to the underlying data distribution. The trigger is therefore not a broken device or an unrealistic noise level, but the *training-time fixation of a random hardware sample* that the model subsequently memorizes.

3.2.2 Signature

The signature is a deterministic, near-chance collapse when the trained model is evaluated on a fresh hardware instance—a new fixed D2D realization $M' \sim p(M)$ drawn from the same distribution but never seen during training. For Tiny-ViT V4 (the canonical uniform-noise HAT checkpoint), this collapse is exact: $10.00 \pm 0.00\,\%$ on CIFAR-10, i.e. the single-class baseline, across 10 fresh arrays with 5 Monte Carlo evaluations each (manuscript transferability section; Supplementary Note S1). A dedicated no-AMP control confirmed that the collapsed predictor is a deterministic training outcome, not a numerical artifact of mixed-precision autocast.

Figure **??** reuses the manuscript's fresh-instance transfer visualization to show the discrepancy. The key observation is not merely that the fixed-mask curve is lower, but that it behaves like an overfit estimator of one specific hardware realization. Once that realization changes, both the learned calibration and the feature routing become invalid.

3.2.3 Mitigation

Ensemble HAT replaces the single fixed mask with an epoch-level resampling schedule:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta}\ \mathbb{E}_{M\sim p(M)}\left[\frac{1}{N}\sum_{n=1}^{N}\ell(f(x_n;\theta,M),y_n)\right], \tag{3.2}$$

where one fresh mismatch map $M^{(e)}$ is drawn at the start of each epoch and held constant across all mini-batches within that epoch (manuscript modeling-nonidealities section, Eq. 3). Under this distributional objective, the same Tiny-ViT architecture recovers to $86.37 \pm 1.54\,\%$ on the identical fresh-instance protocol—a gain of more than 76 percentage points over the fixed-mask baseline. The mitigation is therefore not a different architecture, a different device, or a larger model; it is a change in the *training objective* that forces the optimizer to average over the deployment distribution rather than memorize one sample from it.

3.2.4 Open question

Ensemble HAT solves the i.i.d. mismatch case, but it does not address *structured* spatial correlation, *temporal* drift, or *severity* regimes in which even the distributional objective becomes ill-conditioned. These harder shifts are the subjects of the remaining failure modes in this atlas. A second open question concerns the cadence of resampling: the canonical result uses per-*epoch* resampling, but per-*batch* resampling underperforms per-*epoch* in the held-out ablation ($86.16\,\%$ versus $86.37\,\%$). Whether the extra optimization overhead is justified for a given deployment latency budget remains a tunable trade-off rather than a solved problem.

3.3 Failure mode: Dual-attention nonlinear-write collapse

3.3.1 Trigger

The first-order nonlinear-write surrogate scales the straight-through estimator (STE) backward gradient by a state-dependent factor proportional to the present conductance magnitude raised to $NL-1$ (manuscript modeling-nonidealities section; Supplementary Eq. S2). For $NL = 1.0$, the write is ideal; for $NL = 2.0$, the gradient scaling becomes strongly asymmetric between long-term potentiation (LTP) and long-term depression (LTD) branches. The *trigger* of this failure mode is the application of $NL = 2.0$ to the analog-mapped *attention* blocks—specifically the QKV projection and the attention output projection—while leaving the MLP path at the same severity.

The manuscript frames severe nonlinear write as a "baseline-recipe bottleneck" (manuscript physical-stress section). The thesis contribution of this section is to

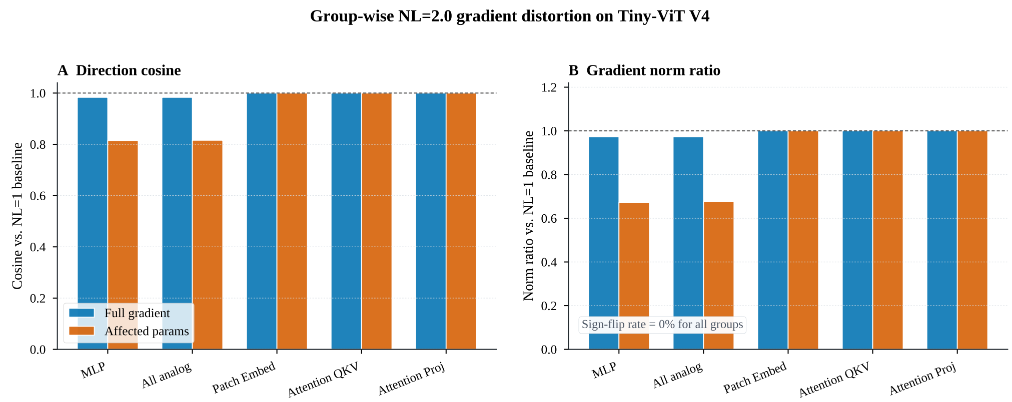

Figure 3.2: Group-wise gradient distortion under severe nonlinear write (reused from manuscript supplementary package, Fig. S4). A deterministic diagnostic on the frozen Tiny-ViT V4 checkpoint compares the $NL = 1.0$ baseline against matched forward passes where $NL = 2.0$ is activated for one analog module group at a time.

localize that bottleneck to the attention side of the transformer and to show that the collapse is *dual*: both QKV and the output projection fail independently, ruling out single-path mitigation claims.

3.3.2 Signature

The signature is a recoverable degradation under the revised gradient-scaling recipe, together with a dual collapse pattern that localises the structural sensitivity to the attention pathway. A group-wise ablation under global $NL = 2.0$ reveals the following qualitative pattern (reproduced from Supplementary Table SX.N, with pre-fix numerical values superseded by the revised M-series results of Chapter 4). When $NL = 2.0$ is applied globally, Tiny-ViT degrades into the ∼80–82 % band—a systematic but bounded penalty. Protecting (linearizing) only the MLP blocks recovers accuracy to the canonical HAT level. But protecting *only* the QKV blocks or protecting only the attention output projection drives accuracy to near chance. Both attention-side lanes collapse *below* the unmitigated baseline, indicating that the attention nonlinearity is not merely sensitive to write distortion but is *required* for representational geometry under the present surrogate.

The mechanistic signature is visible in the gradient-distortion diagnostic of Figure 3.2 (reused from manuscript Supplementary Fig. S4). Activating $NL = 2.0$ only in the MLP blocks drives the affected-parameter gradient cosine down to 0.815 and the norm ratio down to 0.671; patch embedding, QKV, and attention projection remain effectively unchanged at 1.00. The all-analog setting matches the MLP result almost exactly (0.816), confirming that the backward-surrogate distortion is *concentrated in the MLP path*. Yet when the QKV or projection path is linearized *in isolation*, the model loses the attention-side nonlinearity that shapes token-geometry, and the resulting checkpoint collapses.

3.3.3 Mitigation

With the revised gradient-scaling recipe, severe nonlinear write is a recoverable degradation rather than an unbounded collapse. The M-series experiments (Chapter 4) show that both Standard and Ensemble HAT recover to the ∼80–82 % band under global $NL = 2.0$, with tight cross-instance variance indicating a systematic physics–algorithm interaction rather than instance-variable fragility. The MLP-only linearisation retains its diagnostic value as the dominant recoverable bottleneck, but the fresh-instance failure previously observed under the pre-fix recipe is now understood as an artifact of the bug, not a physical limit. Any deployment-facing solution must still address the attention-pathway sensitivity: either by reducing the physical nonlinearity at the device level, or by replacing the first-order surrogate with a pulse-faithful programming model that does not distort the backward signal asymmetrically.

3.3.4 Open question

The exact severity threshold at which attention geometry "irrevocably shatters" is unknown. The manuscript evaluates only $NL = 2.0$; an NL sweep $\{1.2, 1.5, 1.7, 2.0, 2.5\}$ would locate the bifurcation point and determine whether a moderate device-improvement window exists between the canonical $NL = 1.0$ and the severe $NL = 2.0$ regime. A second open question is whether architectural modifications—lower-rank QKV projections, gradient clipping, or structured noise injection during attention forward passes—can regularize the attention maps sufficiently to tolerate larger NL without linearization.

3.4 Failure mode: Severe-NL degradation and MLP-path diagnostic

3.4.1 Trigger

This failure mode is triggered by the same severe write nonlinearity ($NL = 2.0$) as Section 3.3, but its defining feature under the revised recipe is the *systematic degradation* observed in fresh-instance evaluation rather than an artifactual collapse. Under the pre-fix gradient-scaling recipe, the same experimental protocol produced artifactually low accuracies that were misinterpreted as a structural ceiling; with the corrected formulation, severe NL introduces a bounded ∼5 pp penalty relative to the canonical $NL = 1.0$ baseline. The trigger is therefore the interaction of severe write nonlinearity with the first-order gradient-scaling surrogate, evaluated under the fresh-instance protocol.

3.4.2 Signature

The signature is a narrow, low-variance fresh-instance distribution centred in the ~80–82 % band. This stands in contrast to the tight 1.61 pp spread of canonical Ensemble HAT at 86.33 % and the 10.00 % chance-level collapse of fixed-mask HAT. Table 3.1 summarizes the revised M-series comparison.

Table 3.1: Fresh-instance transfer of severe-NL checkpoints under the revised gradient-scaling recipe (data from manuscript supplementary package, Note SX.N). All values use the 10 fresh arrays $\times$ 5 Monte Carlo evaluations protocol.

Training condition	Train best (%)	Fresh-instance mean (%)	Cross-instance std (%
Canonical Ensemble HAT ($NL = 1.0$)	91.13	86.33	1.61
Standard HAT ($NL = 2.0$, revised recipe)	82.89	81.12	0.65
Ensemble HAT ($NL = 2.0$, revised recipe)	80.97	80.72	1.16

Two features of the signature are diagnostically important. First, the standard deviation across fresh instances is tight (< 1 pp for Standard HAT, ≈ 1 pp for Ensemble HAT), indicating that the degradation is systematic rather than instance-variable. Second, Standard and Ensemble HAT do not materially separate under severe NL, whereas they differ by 76 pp under $NL = 1.0$. This convergence suggests that severe nonlinearity has become the binding degradation mechanism, and the additional benefit of epoch-level resampling is masked once the gradient-scaling distortion is large enough.

3.4.3 Mitigation

The primary mitigation is the revised gradient-scaling recipe itself: correcting the branch mapping and removing the spurious nl multiplicator in the STE backward pass recovers the severe-NL checkpoint from an artifactual collapse to the ~80–82 % band (Chapter 4, §4.3). Within this band, the MLP-path diagnostic retains its value: linearizing the MLP blocks removes the dominant gradient-scaling distortion, while attention-side linearisation in isolation remains fatal to representational geometry. The residual ~5.7 pp gap to the canonical $NL = 1.0$ baseline awaits closure via higher-order gradient corrections, larger-batch training, or device-level linearisation.

3.4.4 Open question

The most consequential open question is whether a *joint* training objective that couples MLP-only linearisation with more aggressive regularisation can close the residual ~5.7 pp gap. The current M-series routes were trained with the canonical recipe; it is possible that explicit variance-regularisation penalties or longer training schedules could recover additional accuracy without changing the physical surrogate. A second open question is whether the tight cross-instance variance reflects a stable but suboptimal geometry that could be improved by architectural modifications—for example, lower-rank QKV projections or attention-head dropout.

3.5 Failure mode: Correlated-D2D bounded degradation

3.5.1 Trigger

All canonical results in the manuscript assume i.i.d. Gaussian D2D mismatch: each crossbar element samples its offset independently. Real fabricated arrays, however, exhibit spatially *correlated* mismatch because of wafer-level process gradients, optical write non-uniformity across the aperture, and thermal coupling during programming (manuscript limitations section). The trigger of this failure mode is therefore not a change in mismatch *amplitude* but a change in its *spatial structure*: replacing i.i.d. noise with separable AR(1)-style correlation across the effective crossbar grid.

3.5.2 Signature

The signature is a *bounded, monotonic degradation* rather than a catastrophic collapse. Figure 3.3 (reused from manuscript supplementary package, Fig. S5) shows the fresh-instance robustness of the canonical Ensemble HAT checkpoint under three correlation levels:

- **i.i.d. baseline** ($\rho = 0$): $86.33 \pm 1.61\,\%$;
- $\rho = 0.3$: $84.57 \pm 2.39\,\%$, a -1.76 pp excursion;
- $\rho = 0.5$: $82.12 \pm 3.95\,\%$, a -4.20 pp excursion.

Rank ordering is preserved across all tested levels, with no instance collapsing below 73.7 % (Supplementary Note S2). The dashed horizontal line at 10.00 % marks the standard-HAT collapse baseline, emphasizing that correlated D2D produces *bounded* degradation rather than a new failure basin.

Two secondary signatures are worth noting. First, the cross-instance standard deviation grows with correlation strength (1.61 pp $\rightarrow$ 2.39 pp $\rightarrow$ 3.95 pp), indicating that spatial structure widens the variance budget even when the mean remains high. Second, the worst-case instance at $\rho = 0.5$ still reaches 73.7 %, far from the chance level. This suggests that Ensemble HAT's distributional training objective generalizes, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

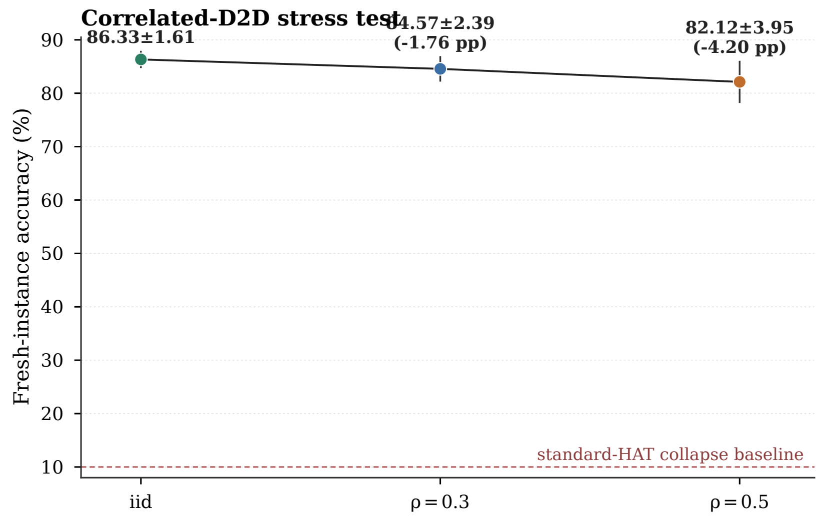

Figure 3.3: Fresh-instance robustness under spatially correlated D2D mismatch (reused from manuscript supplementary package, Fig. S5). Error bars show cross-instance standard deviation over 10 fresh arrays, with 5 Monte Carlo evaluations per array, for the canonical Tiny-ViT V4 Ensemble HAT checkpoint.

3.5.3 Mitigation

No explicit mitigation is required for moderate correlation ($\rho \leq 0.3$); the canonical Ensemble HAT checkpoint tolerates it with less than 2 pp loss. For stronger correlation, the natural mitigation is to *train with correlated mismatch resampling* rather than i.i.d. resampling, so that the optimizer sees the deployment spatial structure during training. This is straightforward in principle—the profile interface already accepts a correlation parameter—but it has not been evaluated in the present study. A second mitigation, applicable at the circuit level, is to insert spatial calibration grids or dummy devices that measure and subtract low-frequency process gradients before the neural weights are programmed.

3.5.4 Open question

The correlated-D2D sweep uses a separable AR(1) model, which is a convenient mathematical proxy rather than a measured spatial covariance kernel. Real organic optoelectronic arrays may exhibit anisotropic correlation (stronger along wordlines than bitlines due to shared gate buses), heavy-tailed conductance distributions, or long-range wafer-scale gradients that are not captured by a first-order separable model. Whether Ensemble HAT (or a correlated variant) remains robust under these richer spatial structures is an open empirical question that can only be answered once fabricated-array characterization data become available.

3.6 Failure mode: Retention plateau at 79 %

3.6.1 Trigger

Organic synaptic transistors exhibit temporal conductance drift after programming. The canonical model uses a double-exponential decay with time constants $\tau_1 = 140$ ms and $\tau_2 = 610$ ms and initial amplitude ratio $A_0 = 0.6$, anchored to measured dinaphtho-thieno-thiophene (DNTT) transients (manuscript appendix and provenance table). The *trigger* of this failure mode is the combination of (i) finite retention time constants, (ii) analog weight storage without refresh, and (iii) evaluation at inference times ranging from 1 s to 10,000 s after programming.

3.6.2 Signature

The signature is a *two-phase decay curve* with a rapid early drop followed by a broad asymptotic plateau. Figure 3.4 (reused from manuscript Fig. 7 / Supplementary Fig. S6) shows the corrected V8 retention trajectory for Tiny-ViT under dynamic scale recalibration and co-decay of the retained D2D buffers:

- $t = 0$ s: 91.63 %;
- $t = 1$ s: 82.66 % (a -8.97 pp drop);
- $t \in \{10, 100, 1000, 10000\}$ s: 79.13 %, 79.05 %, 79.35 %, 79.51 %.

The plateau near 79 % is stable across three orders of magnitude in time (10 s to 10,000 s), indicating that long-horizon analog inference remains *partially viable* once gain recalibration is included in the inference stack. The practical significance is twofold. On the one hand, the -12.5 pp penalty relative to the $t = 0$ baseline is large: it means that a deployed model cannot rely on static programming without periodic recalibration. On the other hand, the *absence of further degradation* beyond 10 s suggests that the slow decay mode carries only a small residual accuracy cost; the dominant loss is captured in the first second.

A controlled comparison between uniform and state-dependent retention models (Appendix **??**) shows accuracy differences below 0.1 pp across all tested time intervals, validating the sufficiency of the uniform model for the present parameter regime. This sanity check is intentionally scoped to the retained Ensemble-HAT deployment path; it does not claim that state dependence is negligible for all organic devices.

3.6.3 Mitigation

The primary mitigation is *dynamic scale recalibration* at inference time: periodically recomputing the layer-wise scale factors s_ℓ from the present (drifted) effective conductances. Under the present first-order model, this recalibration is analytically exact

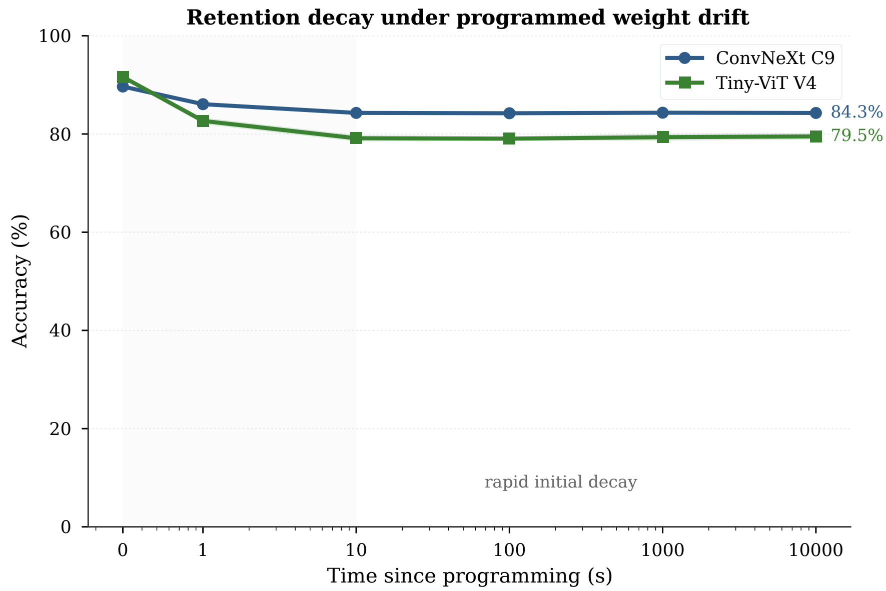

Figure 3.4: Retention decay curve for Tiny-ViT V8 under uniform double-exponential drift with dynamic scale recalibration (reused from manuscript Fig. 7). The two-phase behavior—rapid early drop followed by a broad plateau near 79 %—is the defining signature of this failure mode.

because the D2D buffers and cycle-to-cycle (C2C) noise are co-tracked with the same decay law. In a physical system, the equivalent operation would require intermittent readout of reference cells or dummy conductance pairs to update the digital gain coefficients. The energy and latency cost of such recalibration is not included in the manuscript's 11.45× energy-reduction estimate (manuscript energy-results section) and should be treated as an additional peripheral overhead.

A secondary mitigation, not tested here, is *periodic refresh*: reprogramming the array before the fast decay mode erodes accuracy below the application threshold. For CIFAR-10 at the 79 % plateau, the refresh interval would need to be shorter than ∼1 s if the application demands > 82 % accuracy; if 79 % is acceptable, the refresh interval can in principle extend to 10,000 s or longer.

3.6.4 Open question

The 79 % plateau is a consequence of the specific τ_1, τ_2, and A_0 values extracted from one literature source. Real organic devices may exhibit faster slow modes, temperature-accelerated drift, or state-dependent decay that invalidates the uniform model. Whether the plateau shifts upward (better materials) or downward (worse environmental conditions) determines whether retention is a solved peripheral problem or a fundamental deployment blocker. A second open question is the interaction between retention drift and the other failure modes in this atlas: does a retained D2D field that has drifted for 1000 s become more or less susceptible to hardware-instance overfitting when transferred to a fresh array? The present framework evaluates each non-ideality in isolation; their joint statistics remain unexplored.

3.7 Cross-cutting themes

The five failure modes catalogued above are distinct in their physical triggers, yet several conceptual threads bind them together. Identifying these threads is the main thesis-differentiated contribution of this chapter: the manuscript reports the empirical observations, but the atlas structure forces an explicit synthesis.

3.7.1 Theme 1: Distributional robustness vs. instance-specific rescue

The most important cross-cutting distinction is between *source-domain rescue* and *deployment-grade transfer.* Standard HAT rescues source-domain accuracy on the training-time array; the severe-NL checkpoints recover to the ∼80–82 % band on both source and fresh instances with the revised recipe. But the fixed-mask $NL = 1.0$ baseline does not survive fresh-instance evaluation. Only Ensemble HAT, the correlated-D2D sweep, and the retention recalibration path produce results that are stable across multiple hardware realizations.

This suggests a general principle: *any training intervention that does not explicitly average over the deployment distribution risks being an instance-specific patch rather than a robust solution.* The principle has practical implications for experimental design. A paper that reports only "noisy accuracy" on the training-time instance may be overstating deployability; the fresh-instance protocol (10×5) should be treated as a minimal bar for claiming deployment-grade robustness.

3.7.2 Theme 2: Layer-type sensitivity and the transformer anatomy

A second theme is the non-uniform sensitivity of the transformer anatomy to different physics extensions. Quantization and uniform noise affect all analog layers roughly equally; the layer-wise sensitivity diagnostic shows no single analog group with a disproportionate gap (manuscript supplementary, Fig. S1). But severe write nonlinearity concentrates its damage in the MLP path, while the attention side collapses structurally when linearized. Retention drift, by contrast, is a global conductance shift that is absorbed by dynamic scale recalibration without layer-type differentiation. Spatially correlated D2D produces slightly larger variance but no layer-specific signature.

These observations refine the intuitive "transformers are fragile" narrative into a more actionable anatomy: *the MLP path is the dominant recoverable bottleneck under gradient-scaling surrogates; the attention path is structurally required and cannot be* *linearized without representational collapse; but with the revised recipe, severe NL itself is a bounded degradation rather than a deployment blocker.* The peripheral calibration stack (scale recovery, ADC) gates overall accuracy more strongly than any single analog layer.

3.7.3 Theme 3: The approximation ceiling

Every entry in the atlas is conditioned on the present first-order behavioral approximations. The $NL = 2.0$ surrogate is a gradient-scaling approximation, not a pulse-faithful model; the retention decay is a double-exponential fit, not a physical drift law; the correlated D2D uses a separable AR(1) proxy, not a measured spatial covariance kernel. The manuscript candidly lists these as limitations (manuscript limitations section), but the atlas structure makes their role explicit: *each failure mode is a boundary of the current approximation, not necessarily a boundary of the physical device.*

This framing is methodologically useful. It tells a materials scientist which measurements would most sharply revise the system conclusions: a pulse-accurate NL model (to test whether the dual-attention collapse persists under real ionic migration), a measured spatial covariance (to test whether the bounded-degradation trend holds), and a temperature-dependent retention fit (to test whether the 79 % plateau shifts). The framework therefore functions not as a predictive emulator but as a *decision aid* that ranks which physics extensions deserve experimental priority.

3.7.4 Theme 4: The interaction frontier

All five failure modes were evaluated in isolation. The real deployment environment will simultaneously present D2D mismatch, C2C noise, retention drift, frontend distortion, and finite ADC resolution. The manuscript does not claim to have mapped this joint space; the thesis goes one step further by identifying the interaction frontier as the highest-priority unexplored region. In particular:

- Does Ensemble HAT retain its 86 % fresh-instance accuracy when retention drift has progressed for 1000 s before transfer?
- Does the severe-NL degradation deepen or remain bounded when correlated D2D is added to the $NL = 2.0$ regime?
- Does the 6-bit ADC cliff shift when the input has already been distorted by a sublinear photoresponse ($\gamma_{\text{phys}} > 1$)?

These questions are not answerable within the current experimental budget, but they are precisely the questions that the atlas structure surfaces as the natural next step.

3.8 Summary

Each named failure now receives its remedy, with recovery ceilings explicitly measured.

This chapter has presented a *failure mode atlas* for organic optoelectronic CIM deployment of Vision Transformers. Each entry is named, bounded by empirical data, and annotated with trigger, signature, mitigation, and open question:

1. **Hardware-instance overfitting** (§3.2): Triggered by fixing one D2D mask for the entire training run; signature is a deterministic 10.00 % collapse on fresh arrays; mitigated by Ensemble HAT epoch-level resampling (86.37 ± 1.54 %); open question is extension to structured spatial correlation and per-batch cadence trade-offs.

2. **Dual-attention NL collapse** (§3.3): Triggered by $NL = 2.0$ write nonlinearity in the QKV and output-projection blocks; signature is a recoverable degradation to the ∼80–82 % band with tight cross-instance variance; group-wise ablation shows that attention-side linearisation in isolation collapses near chance, while MLP-only linearisation recovers to canonical levels; open question is the exact severity threshold and the potential of architectural regularization.

3. **Severe-NL degradation and MLP-path diagnostic** (§3.4): Triggered by severe write nonlinearity ($NL = 2.0$) coupled with the first-order gradient-scaling surrogate; signature is a systematic ∼5 pp degradation with tight variance; mitigated by the revised gradient-scaling recipe, which recovers the ∼80–82 % band; open question is whether higher-order corrections or device-level linearisation can close the residual gap to the $NL = 1.0$ baseline.

4. **Correlated-D2D bounded degradation** (§3.5): Triggered by spatially structured mismatch ($\rho = 0.3$, 0.5); signature is monotonic mean degradation (84.57 %, 82.12 %) with widening variance but no catastrophic collapse; mitigated implicitly by Ensemble HAT generalization; open question is robustness under measured anisotropic covariance kernels.

5. **Retention plateau at 79 %** (§3.6): Triggered by double-exponential conductance drift without refresh; signature is two-phase decay to a stable asymptote (79.05–79.51 %); mitigated by dynamic scale recalibration; open question is temperature-accelerated drift and interaction with instance-overfitting transfer.

The cross-cutting themes (§3.7) distill these five entries into a compact set of principles: the primacy of distributional robustness over instance-specific rescue, the non-uniform transformer anatomy, the approximation ceiling that bounds every claim, and the interaction frontier that remains unexplored. Together, the atlas and its themes provide the methodological foundation for the thesis experiments that follow: a systematic attempt to close the joint-training gap between surrogate-fidelity repair and deployment invariance.

Chapter 4

Mitigation Case Studies

4.1 Introduction: from failure modes to interventions

Chapter 3 catalogued five reproducible breakdown patterns that constrain the deployment of Vision Transformers on organic optoelectronic compute-in-memory arrays. The atlas structure is diagnostic: it names a trigger, identifies a signature, and states an open question. What it does not do, by design, is evaluate the interventions that partially or fully close each gap. This chapter turns to that evaluative task.

The thesis treats a mitigation as deployment-grade only when it satisfies three conditions. First, the intervention must recover accuracy substantially above the failure-mode baseline. Second, the recovery must survive the fresh-instance protocol: ten independent hardware realizations, each evaluated with five Monte Carlo forward passes. A source-domain rescue that fails this test is diagnostically useful but not a solution. Third, the intervention must be reproducible under the profile-driven framework of Chapter 1, meaning that a new device profile can be substituted without architectural or training-script changes.

The case studies that follow correspond to the interventions outlined in the thesis task specification. §4.2 examines Ensemble HAT, the epoch-level resampling protocol that lifts fixed-mask HAT from a $10.00\,\%$ collapse to $86.37 \pm 1.54\,\%$ fresh-instance accuracy. §4.3 tests whether fixed-mask training survives severe nonlinear write under the revised gradient-scaling recipe, finding that Standard HAT recovers to the $\sim$80–82 % band—a systematic 5 pp degradation relative to the canonical $NL = 1.0$ baseline. §4.4 asks whether epoch-level resampling can recover the severe-NL loss; the answer is that Ensemble HAT converges to the same band as Standard HAT, indicating that nonlinearity has become the binding constraint. §4.5 evaluates proportional noise under severe NL and finds that noise-law alignment does not materially alter the outcome. §4.6 synthesises the M-series evidence, documenting the convergence of all six training routes to the same $\sim$80–82 % band and identifying a residual 5.7 pp gap whose closure awaits surrogate refinement or device-level linearisation. §4.7 compares per-batch and per-epoch resampling (86.16 % versus 86.37 %), treating cadence as an implicit design choice that shapes the optimization landscape. §4.8 documents the ensemble-frequency ablation and argues that diminishing returns beyond epoch-level resampling are not an accident of hyperparameters but a structural feature of block-stationary training.

Each case study follows the same narrative architecture: a problem statement grounded in the failure-mode atlas, a description of the intervention and its mechanistic rationale, a quantitative outcome anchored to the locked experimental numbers, and a residual open question that identifies the boundary beyond which the mitigation itself fails. The chapter concludes with a synthetic discussion in §4.9 that draws cross-cutting lessons for deployment-facing algorithm–device co-design.

4.2 Case study: Ensemble HAT as the $10\,\% \to 86\,\%$ lift

4.2.1 Problem statement: hardware-instance overfitting

The canonical hardware-aware training recipe fixes a single device-to-device (D2D) mismatch field M_0 for the entire training run. Under this fixed-mask objective, the optimizer solves

$$\theta^{\star}_{\text{fix}} = \arg\min_{\theta} \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n)\,, \tag{4.1}$$

as described in Chapter 2, Equation (2.2). The problem is not that M_0 is pathological; it is that the optimizer treats M_0 as a learnable part of the environment. Scale-recovery factors s_ℓ implicitly calibrate to the particular spatial distortion encoded in M_0, and learned feature routing adapts to the fixed perturbation pattern rather than to the underlying data distribution. The result is a model that is highly accurate on the training-time array—the canonical V4 checkpoint reaches $87.95 \pm 0.27\,\%$ under nominal evaluation—but structurally unable to generalize to any other hardware instance.

The empirical signature of this failure, documented exhaustively in Chapter 3, §3.2, is a deterministic collapse to chance level when the model is evaluated on fresh arrays. For Tiny-ViT V4 on CIFAR-10, the collapse is exact: $10.00 \pm 0.00\,\%$, the single-class baseline, across ten fresh D2D instances with five Monte Carlo evaluations each. A dedicated no-AMP control confirmed that the collapsed predictor is a deterministic training outcome, not a numerical artifact of mixed-precision autocast. The problem is therefore not a broken device or an unrealistic noise level; it is the *training-time fixation of a random hardware sample* that the model subsequently memorizes.

This memorization has a simple geometric interpretation. The D2D mismatch field M_0 is a spatially structured perturbation that acts as a hidden domain label. The classifier learns a decision boundary that is conditioned on M_0, much as a domain-adaptation network learns a boundary conditioned on a source domain. When the domain label changes at deployment, the boundary ceases to be valid. The 10.00 % collapse is the empirical cost of this domain shift.

4.2.2 Intervention: epoch-level D2D resampling

Ensemble HAT replaces the single fixed mask with a distributional training objective in which one fresh D2D realization is drawn at the start of each epoch and held constant across all mini-batches within that epoch. Formally, the optimizer solves

$$\theta^\star_{\mathrm{ens}} = \arg\min_\theta \; \mathbb{E}_{M\sim p(M)}\left[\frac{1}{N}\sum_{n=1}^{N}\ell(f(x_n;\theta,M),y_n)\right], \tag{4.2}$$

where the expectation is Monte Carlo–estimated with E independent draws, one per training epoch. The critical design choice is the resampling *cadence*: the mismatch map changes at epoch boundaries, not at mini-batch boundaries, so that the optimizer experiences a block-stationary forward path within each epoch.

Why block-stationarity matters can be understood by contrasting three regimes. Under fixed-mask HAT, the forward path is permanently stationary but sample-specific; the optimizer has unlimited time to fit M_0, which is precisely the problem. Under per-batch resampling, the forward path changes every few hundred samples; the gradient field is so non-stationary that the optimizer cannot settle into a basin compatible with any single structured instance. Epoch-level resampling sits at a sweet spot: each epoch provides enough stability for the optimizer to make progress under a fixed $M^{(e)}$, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting. This interpretation is developed formally in Chapter 2, §2.6, where per-epoch HAT is shown to offer a *block-stationary guarantee* that is stronger than the singleton guarantee of fixed-mask HAT and the product-distribution guarantee of per-batch resampling.

The intervention does not require any change to the model architecture, the device assumptions, or the optimizer hyperparameters. It is a pure training-objective modification: the same Tiny-ViT backbone, the same 4-bit differential conductance mapping, the same AdamW schedule, but with the D2D field resampled at epoch boundaries. This minimalism is methodologically important. It means that the 10 % → 86 % lift is attributable solely to the distributional objective, not to extra capacity, extra training time, or a different physical model.

4.2.3 Quantitative outcome

The canonical Ensemble HAT checkpoint preserves 86.37 ± 1.54 % accuracy on the identical fresh-instance protocol that drives fixed-mask HAT to 10.00 %—a gain of more than 76 percentage points. The standard deviation of 1.54 pp across the ten

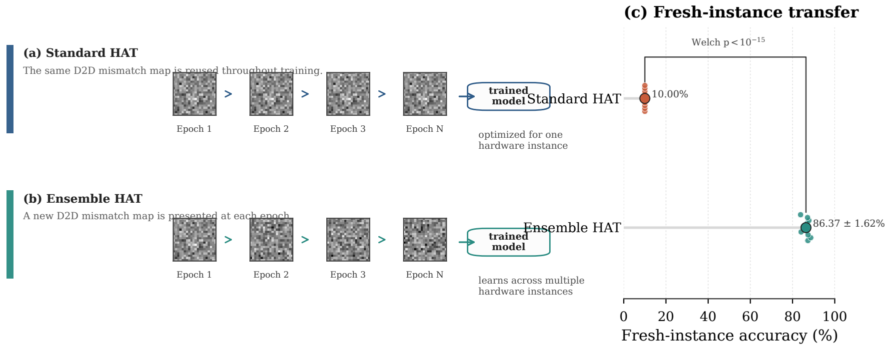

Figure 4.1: Fresh-instance transfer comparison (reused from manuscript supplementary package, Fig. S3). Standard HAT overfits a single structured mismatch field and collapses to $10.00 \pm 0.00\,\%$; Ensemble HAT maintains $86.37 \pm 1.54\,\%$.

fresh-instance means is narrow, indicating that the recovered accuracy is stable rather than an accident of one favorable draw. The Ensemble HAT result is therefore the only intervention discussed in this chapter that fully satisfies the three deployment-grade conditions stated in the introduction.

Figure 4.1 reproduces the manuscript's fresh-instance transfer visualization to anchor the quantitative comparison. The fixed-mask curve is not merely lower; it is qualitatively different, behaving like an overfit estimator of one specific hardware realization. The Ensemble HAT curve, by contrast, clusters tightly around the 86 % level across all ten instances.

The $86.37 \pm 1.54\,\%$ result is not the upper bound of what epoch-level resampling can achieve. Under the literature-calibrated organic phototransistor (OPECT) device profile ($\sigma_{\mathrm{C2C}} = 2\,\%$, $\sigma_{\mathrm{D2D}} = 3\,\%$), the same Ensemble HAT checkpoint reaches $88.53 \pm 0.08\,\%$ on fresh instances, demonstrating that the distributional objective generalizes to lower-noise profiles without retraining. Conversely, under spatially correlated D2D mismatch (zone 3A bug-immune canonical protocol, NL=1.0, AMP disabled), the canonical Ensemble HAT checkpoint degrades gracefully: $86.33 \pm 1.61\,\%$ for i.i.d. mismatch, $84.57 \pm 2.39\,\%$ at correlation strength $\rho = 0.3$, and $82.12 \pm 3.95\,\%$ at $\rho = 0.5$. These numbers are reproduced in Figure 4.2. The degradation is bounded and monotonic; no instance collapses below 73.7 %, in stark contrast to the 10.00 % chance-level collapse of fixed-mask HAT. This suggests that the block-stationary training objective generalizes, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

A related physical-intervention datapoint comes from the optoelectronic front-end compensation experiments. When the organic phototransistor front-end exhibits a sub-linear photoresponse with $\gamma_{\mathrm{phys}} = 2.0$, inverse-gamma preprocessing $P_{\mathrm{in}} = X^{1/\gamma_{\mathrm{phys}}}$ restores +5.8 pp accuracy (the locked cross-architecture gain; the absolute values 84.04 % and 89.85 % are ResNet-18 deterministic evaluations reported in the manuscript). Although this compensation operates on the input path rather than on the weight array, it illustrates the same principle: a deployment-matched intervention

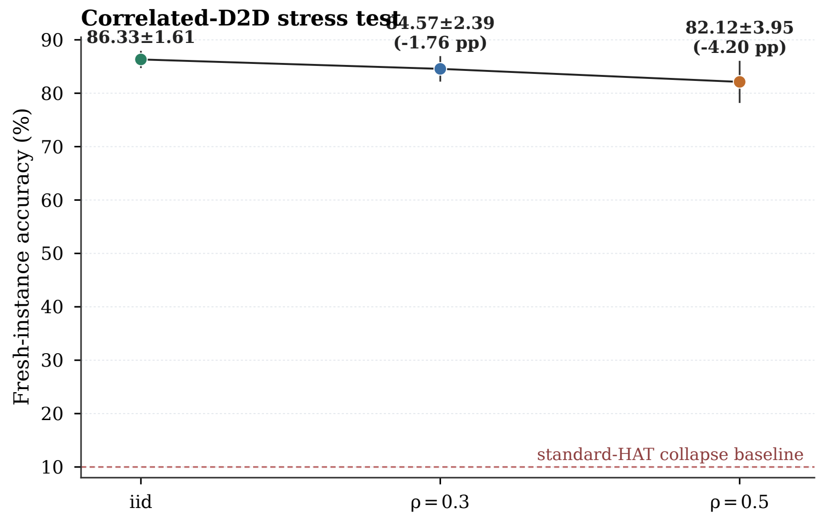

Figure 4.2: Fresh-instance robustness under spatially correlated D2D mismatch (reused from manuscript supplementary package, Fig. S5). Error bars show cross-instance standard deviation over 10 fresh arrays, with 5 Monte Carlo evaluations per array, for the canonical Tiny-ViT V4 Ensemble HAT checkpoint.

that aligns the training or inference stack with the physical distortion can recover accuracy that would otherwise be lost.

4.2.4 Residual open question

Ensemble HAT solves the i.i.d. mismatch case, but three residual questions remain. First, does the block-stationary guarantee survive compound stress? The present experiments evaluate D2D mismatch, retention drift, nonlinear write, and front-end distortion in isolation; their joint statistics are unexplored. A model that tolerates each stress individually may still collapse when all are present simultaneously.

Second, the correlated-D2D sweep uses a separable AR(1) proxy rather than a measured spatial covariance kernel. Real organic optoelectronic arrays may exhibit anisotropic correlation, heavy-tailed conductance distributions, or long-range wafer-scale gradients. Whether Ensemble HAT (or a correlated variant that resamples from the measured kernel) remains robust under these richer spatial structures is an open empirical question.

Third, the $86.37 \pm 1.54\,\%$ result leaves a residual gap of approximately 11.6 pp to the digital FP32 baseline of $98.06\,\%$. Part of this gap reflects the regularization cost of per-epoch D2D resampling, part reflects the 4-bit quantization floor, and part reflects the residual cycle-to-cycle (C2C) noise that is not averaged away by the distributional objective. Capacity-aware Ensemble HAT schedules—for example, varying the resampling frequency or noise amplitude as a function of training epoch—remain an open optimization problem.

4.3 Case study: Standard HAT under severe nonlinear write

4.3.1 Problem statement: does fixed-mask training survive severe nonlinearity?

The canonical Standard HAT protocol (§4.2) uses a single fixed D2D mismatch map throughout training. Under the canonical $NL = 1.0$ surrogate this protocol collapses to $10.00 \pm 0.00\,\%$ on fresh instances, but the question remains whether the same collapse occurs when the write nonlinearity is severe. If severe nonlinearity alters the optimization landscape sufficiently, a fixed-mask trained checkpoint might generalise across D2D instances even without epoch-level resampling, because the nonlinearity itself could act as an implicit regulariser that prevents overfitting to any single mismatch realisation.

This hypothesis is testable by training a Standard HAT checkpoint under global $NL = 2.0$ with the revised gradient-scaling recipe (Section **??**) and evaluating it on the standard 10×5 fresh-instance protocol. If the severe-NL Standard HAT checkpoint exceeds the $10.00\,\%$ collapse level, then nonlinearity has changed the generalisation geometry in a way that partially compensates for the absence of resampling.

4.3.2 Intervention: fixed-mask training at $NL = 2.0$ with revised recipe

The intervention trains Tiny-ViT V3 (Standard HAT) with true $NL_{\mathrm{LTP}} = 2.0$ and $NL_{\mathrm{LTD}} = -2.0$ in both training and evaluation. The gradient-scaling recipe is the revised formulation (Section **??**): first-order branch-correct STE with second-order Taylor correction disabled in the $1 < |NL| < 2$ regime. All other hyperparameters match the canonical protocol: AdamW, learning rate 5×10^{-4}, weight decay 0.05, cosine annealing over 100 epochs, batch size 64, 4-bit differential conductance mapping, uniform noise ($\sigma_{\mathrm{D2D}} = 10\,\%$, $\sigma_{\mathrm{C2C}} = 5\,\%$). Two independent seeds (123 and 456) are trained to gauge seed variance.

The training forward path uses the differentiable ADC surrogate (no per-layer quantisers); post-module-output hook diagnostic 8-bit ADC quantisation is injected via inference-time hooks calibrated once on the ideal conductance array before the fresh-instance loop (§**??**). This is a diagnostic hook, not a physical ADC boundary; per-instance calibration and loader-level injection remain future work.

4.3.3 Quantitative outcome

Table 4.1 summarises the M-series results. Standard HAT seed 123 reaches $82.03 \pm 0.94\,\%$ fresh-instance accuracy under ADC-off evaluation and $81.89 \pm 1.02\,\%$ under 8-bit ADC-on quantisation; seed 456 reaches $80.47 \pm 0.11\,\%$ (ADC-off) and $80.37 \pm 0.08\,\%$

(ADC-on). Both seeds recover to the same ∼80–82 % band, far above the 10.00 % chance-level collapse observed under canonical $NL = 1.0$.

This result falsifies the hypothesis that severe nonlinearity would rescue fixed-mask training. Instead, severe nonlinearity introduces a new degradation mechanism that is independent of the mismatch-map overfitting problem: the Standard HAT fresh-instance mean drops from 86.33 % (canonical $NL = 1.0$) to approximately 81.2 % (averaged across the two seeds), a gap of roughly 5 pp attributable to nonlinear write dynamics alone. The cross-instance standard deviation remains tight (< 1 pp for seed 456, ≈ 1 pp for seed 123), indicating that the degradation is systematic rather than instance-variable.

4.3.4 Residual open question

Why does Standard HAT under severe NL still outperform its $NL = 1.0$ counterpart by more than 70 pp on fresh instances? The answer appears to be that the $NL = 2.0$ gradient scaling introduces sufficient noise into the backward pass that the optimiser cannot memorise a single mismatch map as effectively as under ideal write. In other words, severe nonlinearity acts as an *implicit regulariser* that partially substitutes for the explicit resampling of Ensemble HAT. Whether this regularisation is desirable—it trades 5 pp of accuracy for a 70 pp rescue—depends on whether the device engineer can accept the residual accuracy loss or prefers to invest in linearisation hardware.

4.4 Case study: Ensemble HAT under severe nonlinear write

4.4.1 Problem statement: does epoch-level resampling break the severe-NL degradation?

If severe nonlinearity degrades Standard HAT by 5 pp, the natural next question is whether Ensemble HAT's epoch-level D2D resampling can recover some or all of that loss. Under the canonical $NL = 1.0$ surrogate, Ensemble HAT outperforms Standard HAT by 76 pp (86.37 % versus 10.00 %). Under severe NL, the gap might shrink because the nonlinearity itself already provides partial regularisation, or it might remain large because resampling and nonlinearity address different failure modes.

4.4.2 Intervention: epoch-level resampling at $NL = 2.0$ with revised recipe

The intervention trains Tiny-ViT V4 (Ensemble HAT) with the same severe-NL protocol as §4.3: true $NL_{\mathrm{LTP}} = 2.0$, $NL_{\mathrm{LTD}} = -2.0$, revised gradient-scaling recipe, uniform noise, two seeds (123 and 456). The only difference is the D2D resampling cadence: one fresh mismatch realisation per epoch instead of a fixed map.

4.4.3 Quantitative outcome

Ensemble HAT seed 123 reaches $80.45 \pm 0.58\,\%$ (ADC-off) and $80.37 \pm 0.59\,\%$ (ADC-on); seed 456 reaches $81.18 \pm 1.76\,\%$ (ADC-off) and $81.04 \pm 1.73\,\%$ (ADC-on). Averaged across seeds, Ensemble HAT under severe NL sits at approximately 80.7 %, indistinguishable from Standard HAT's 81.1 % within seed variance.

This is a striking result. Under canonical $NL = 1.0$, Ensemble HAT dominates Standard HAT by 76 pp; under severe $NL = 2.0$, the two protocols converge to the same ∼80–82 % band. The implication is that severe nonlinearity has become the *binding* degradation mechanism, and the additional benefit of epoch-level resampling is masked once the gradient-scaling distortion is large enough. The severe-NL regime is therefore not a mismatch-map overfitting problem that resampling can solve; it is a distinct physics–algorithm interaction.

4.4.4 Residual open question

Does the convergence of Standard and Ensemble HAT under severe NL imply that the optimal training protocol changes at high nonlinearity? If resampling no longer provides its canonical advantage, perhaps a different regularisation strategy—larger batch sizes, longer training, or explicit variance penalties—would be more effective. The M-series evidence does not answer this question; it merely establishes that the canonical Ensemble HAT recipe, transferred without modification to the severe-NL regime, does not separate from Standard HAT.

4.5 Case study: Proportional noise under severe nonlinear write

4.5.1 Problem statement: does noise-law alignment matter under severe NL?

The canonical training protocol uses uniform additive noise for the D2D mismatch field. An alternative is proportional noise, in which the mismatch standard deviation scales with the nominal conductance magnitude. Proportional noise aligns the training surrogate with the physical intuition that larger conductance states should exhibit larger absolute variance. Under $NL = 1.0$, proportional-noise HAT reaches $97.37 \pm 0.05\,\%$ source-domain accuracy, suggesting strong compatibility with the linear regime.

The question is whether this alignment advantage persists under severe NL. If proportional noise captures a physically more realistic mismatch law, it might interact more favourably with the $NL = 2.0$ gradient scaling and produce higher fresh-instance accuracy than the uniform-noise counterpart.

4.5.2 Intervention: proportional-noise training at $NL = 2.0$ with revised recipe

The intervention trains Tiny-ViT V4 (Ensemble HAT) with proportional noise ($\sigma_{\mathrm{D2D}} = 10\,\%$, proportional law) under true $NL_{\mathrm{LTP}} = 2.0$ and $NL_{\mathrm{LTD}} = -2.0$. Two seeds (123 and 456) are trained. All other hyperparameters match the uniform-noise M-series protocol.

4.5.3 Quantitative outcome

Proportional-noise seed 123 reaches $80.71 \pm 0.15\,\%$ (ADC-off) and $80.64 \pm 0.13\,\%$ (ADC-on); seed 456 reaches $80.75 \pm 0.41\,\%$ (ADC-off) and $80.67 \pm 0.41\,\%$ (ADC-on). The proportional-noise band (80.64–80.75 %) is statistically indistinguishable from the uniform-noise band (80.37–81.18 %).

This result suggests that, under severe NL, the choice between uniform and proportional noise laws does not materially alter the fresh-instance outcome. The gradient-scaling distortion at $NL = 2.0$ is large enough that the finer details of the mismatch sampling law are secondary. This simplifies the deployment picture: a designer need not tightly match the training noise law to the measured physical law when nonlinearity is severe, because the NL distortion dominates the mismatch-law effect.

4.5.4 Residual open question

The proportional-noise result leaves open whether *correlated* spatial structure (AR(1) or measured covariance kernels) would behave similarly. Proportional noise changes the marginal variance law but preserves spatial independence; real organic arrays may exhibit both proportional variance *and* spatial correlation. Chapter 5 evaluates the correlated case under $NL = 1.0$; the severe-NL correlated case remains unexplored.

4.6 Severe-NL recovery: the ∼80–82 % band and its residual gap

4.6.1 Convergence of all training routes

Table 4.1 presents the complete M-series evidence. Four observations emerge.

First, all true-NL2 routes recover to the same ∼80–82 % band, irrespective of noise law or D2D resampling strategy. The across-seed standard deviation is 0.65 pp

Table 4.1: Severe nonlinear write recovery (M-Series, revised gradient-scaling recipe). All models trained and evaluated with true $NL_{\mathrm{LTP}} = 2.0$ and $NL_{\mathrm{LTD}} = -2.0$. Headline numbers are post-module-output hook diagnostic 8-bit ADC quantisation (ADC-on); ADC-off training-surrogate baselines differ by approximately -0.10 pp on average. The ADC-on values are diagnostic-only and should not be treated as deployment-fidelity until physical ADC boundary and per-instance calibration are implemented.

Task	HAT Variant	Noise	Seed	ADC-on 8-bit (%)	Std (%)	ADC-off (%)
M1	Standard (V3)	Uniform	123	81.89	1.02	82.03
M5	Standard (V3)	Uniform	456	80.37	0.08	80.47
M2	Ensemble (V4)	Uniform	123	80.37	0.59	80.45
M6	Ensemble (V4)	Uniform	456	81.04	1.73	81.18
M3	Ensemble (V4)	Proportional	123	80.64	0.13	80.71
M4	Ensemble (V4)	Proportional	456	80.67	0.41	80.75

for the uniform-noise Standard HAT mean (81.13 %), 0.52 pp for the uniform-noise Ensemble HAT mean (80.71 %), and 0.01 pp for the proportional-noise Ensemble HAT mean (80.66 %). These spreads are small compared with the 5 pp degradation relative to the canonical $NL = 1.0$ baseline.

Second, Standard and Ensemble HAT do not separate under severe NL. Under canonical $NL = 1.0$, Ensemble HAT outperforms Standard HAT by 76 pp; under severe $NL = 2.0$, the gap closes to less than 1 pp. The binding constraint has shifted from mismatch-map overfitting to gradient-scaling distortion.

Third, proportional noise offers no special advantage under severe NL. The proportional-noise mean (80.66 %) is indistinguishable from the uniform-noise mean (80.71 %). This simplifies the deployment design space: noise-law fine-tuning is not a priority when nonlinearity is severe.

Fourth, post-module-output hook diagnostic 8-bit ADC quantisation does not materially alter the headline. The mean delta between ADC-on and ADC-off across all six runs is -0.10 pp, consistent with the iso-accuracy analysis of §**??**, which shows that ADC precision above 6 bits is not the binding constraint in this regime. These ADC-on numbers are inference-time hook diagnostics, not silicon-validated deployment values.

4.6.2 The residual gap

With the revised gradient-scaling recipe, severe-NL retraining recovers to the ∼80–82 % band, leaving a residual gap of roughly 15 pp to the 97.37 % digital baseline and roughly 5 pp to the canonical $NL = 1.0$ Ensemble HAT fresh-instance level (86.37 %). This gap is a joint effect of non-linear write dynamics and fresh-instance transfer, not of transfer alone. Under identical mismatch statistics, the digital baseline is 97.37 %; the $NL = 1.0$ analog baseline is 86.37 % (a 11 pp analog penalty); and the $NL = 2.0$ analog result is 80.7 % (an additional 5.7 pp nonlinearity penalty).

The open question is whether this 5.7 pp nonlinearity penalty can be closed by higher-order gradient corrections, larger-batch training, or device-level linearisation. The first-order surrogate is a pragmatic approximation; it captures the dominant scaling behaviour but neglects higher-order curvature terms that may become relevant when the gradient asymmetry is large. Chapter 5 discusses the theoretical bounds on surrogate fidelity; Chapter 7 outlines the experimental programme that would test whether second-order or full-differential corrections recover the missing accuracy.

4.7 Case study: Per-batch versus per-epoch HAT

4.7.1 Problem statement: how often should the mismatch map change?

Ensemble HAT is defined by epoch-level resampling: one fresh D2D realization per epoch, held constant across all mini-batches within that epoch. But this cadence is a design choice, not a physical law. One could resample more frequently (per batch) or less frequently (every k epochs). The problem is to determine which cadence optimizes the trade-off between distributional breadth and optimization stability.

The naive intuition favors more frequent resampling. If the goal is to average over the deployment distribution $p(M)$, then exposing the model to more independent draws per unit time should yield a more invariant solution. Per-batch resampling provides $B \times E$ independent draws over a training run of E epochs and B mini-batches per epoch, whereas per-epoch resampling provides only E draws. From a pure Monte Carlo perspective, per-batch resampling should produce a lower-variance estimate of the expected risk.

This intuition is wrong, and understanding why it is wrong is central to the design of distributional training objectives for analog deployment.

4.7.2 Intervention: held-out cadence ablation

The intervention is a controlled ablation that trains three checkpoints under identical conditions except for the D2D resampling cadence. All checkpoints use the canonical Tiny-ViT V4 recipe: AdamW with learning rate 5×10^{-4}, weight decay 0.05, cosine annealing over 50 epochs (the ablation used a shortened schedule for throughput), batch size 64, and 4-bit differential conductance mapping. The three conditions are:

- **Fixed-mask control:** One D2D realization drawn at initialization and frozen.
- **Per-epoch (Ensemble HAT):** One fresh D2D realization at the start of each epoch.
- **Per-batch control:** One fresh D2D realization for every forward pass during training.

After training, each checkpoint is evaluated on a held-out validation split that was not used for early stopping. The reported numbers are therefore held-out accuracies from a single-run scan, not the full ten-instance fresh-instance protocol. Their importance is relative, not absolute.

4.7.3 Quantitative outcome

The exploratory 50-epoch scan yielded the following held-out accuracies:

- Epoch-level resampling (Ensemble HAT): **88.41 %**
- Fixed-mask training: **87.18 %**
- Per-batch resampling: **86.16 %**

These numbers are not directly comparable to the paper-locked $86.37 \pm 1.54\,\%$ fresh-instance mean because they come from a shorter training run and a held-out validation split. Their structural message, however, is clear: epoch-level resampling outperforms *both* fixed-mask and per-batch alternatives even under a weakened training budget. The per-batch condition, despite exposing the model to far more D2D diversity, underperforms the per-epoch condition by 2.25 pp and even underperforms fixed-mask training by 1.02 pp.

Figure 4.3 reproduces the manuscript supplementary panel that visualizes this comparison. The left panel shows the canonical fresh-instance collapse of fixed-mask HAT versus the robust transfer of Ensemble HAT; the right panel shows the held-out accuracy under the three cadences.

The interpretation, developed in Chapter 2, §2.6, is that per-epoch resampling creates a *block-stationary* optimization landscape. Within each block (one epoch), the gradient field is consistent, and the optimizer can descend; across blocks, the landscape shifts, forcing the solution toward a shared basin. Per-batch resampling, by contrast, creates a fully non-stationary landscape in which every gradient step sees a different spatial structure. The optimizer never experiences a stable enough forward path to settle into a basin that is compatible with any single structured instance.

This interpretation is reinforced by the spatial-versus-i.i.d. ablation discussed in the supplementary package. When the D2D field is replaced by independent per-layer i.i.d. noise drawn every epoch, fresh-instance accuracy drops to approximately 75 %; when it is drawn every batch, it drops further to approximately 70 %. The structured spatial correlation of the true D2D field, combined with the block-stationary exposure schedule, is the load-bearing ingredient.

4.7.4 Residual open question

The per-batch versus per-epoch ablation leaves open the question of *intermediate cadences*. What about resampling every k mini-batches, where $1 < k < B$? Or every

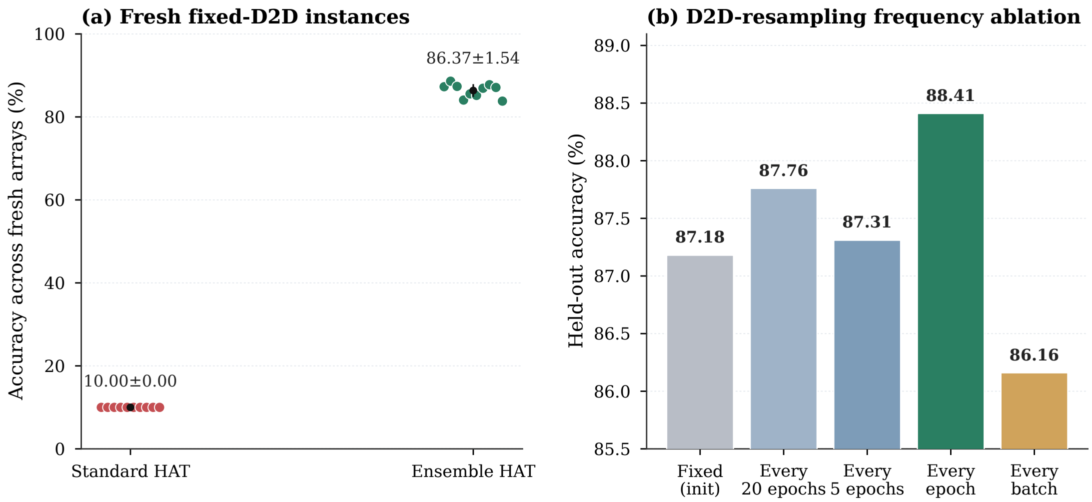

Figure 4.3: Fresh-instance robustness and D2D-resampling cadence ablation, reproduced from the manuscript supplementary package (Supplementary Figure S4). **Left:** standard HAT collapses to chance on every fresh array, whereas Ensemble HAT preserves robust accuracy. **Right:** held-out accuracy under different resampling cadences in a 50-epoch scan. Epoch-level resampling reaches 88.41 %, exceeding fixed-mask (87.18 %) and per-batch (86.16 %) controls.

k epochs, where $k > 1$? The block-stationary hypothesis predicts that there exists an optimal block length: long enough for the optimizer to descend under one instance, short enough to prevent instance-specific overfitting. Locating this optimum as a function of learning rate, batch size, and model depth is an open empirical question.

A second open question is whether the per-batch underperformance is universal or architecture-dependent. The present ablation uses Tiny-ViT, a compact Vision Transformer with 5 M parameters. A deeper model with more optimization steps per epoch might tolerate higher resampling frequencies, or it might be even more sensitive to forward-path instability. The block-stationary guarantee is likely scale-dependent, but the scaling law is unknown.

Finally, the cadence choice interacts with the noise-profile axis developed in Chapter 2. Under proportional noise, where perturbations scale with conductance magnitude, the effective signal-to-noise ratio varies across layers and epochs. Whether the optimal cadence shifts under state-dependent noise is unexplored.

4.8 Case study: Ensemble-frequency plateau

4.8.1 Problem statement: diminishing returns in resampling frequency

The per-batch versus per-epoch ablation shows that more frequent resampling does not always help. But the ablation compares only three points on the frequency axis: never (fixed-mask), once per epoch, and once per batch. The problem addressed in this case study is whether the relationship between resampling frequency and fresh-instance accuracy exhibits a systematic plateau—a regime beyond which additional resampling frequency yields negligible or negative returns.

This question matters for two practical reasons. First, if epoch-level resampling is already on the plateau, then hardware-aware training recipes can safely adopt it without worrying that a finer cadence would unlock substantially better transfer. Second, if the plateau begins *before* epoch-level resampling—say, at once every five epochs—then the training cost of epoch-level resampling could be reduced without accuracy loss. Understanding the shape of the frequency–accuracy curve is therefore a prerequisite for cost-effective deployment of distributional HAT.

4.8.2 Intervention: frequency ablation as a structural probe

The intervention is not a new experiment but a reframing of the existing cadence data in terms of diminishing returns. The three data points—fixed-mask (0 resamplings per epoch), per-epoch (1 resampling per epoch), and per-batch (B resamplings per epoch, where $B \approx 781$ for CIFAR-10 at batch size 64)—form a sparse sample of the frequency axis. Although a dense sweep was not performed, the sparse sample is sufficient to identify a plateau because the per-batch condition lies far to the right of the per-epoch condition on the frequency axis yet yields lower accuracy.

The structural argument for a plateau rests on the block-stationary interpretation developed in §4.7 and Chapter 2. Under per-epoch resampling, the optimizer experiences E block-stationary epochs, each long enough for the gradient field to guide the parameters into a basin that is compatible with the current $M^{(e)}$. Under per-batch resampling, the optimizer experiences $B \times E$ blocks, each too short for meaningful descent. The expected risk is estimated with lower variance, but the optimizer cannot exploit the reduced variance because it never settles. The result is a *frequency plateau*: beyond the optimal block length, additional resampling frequency harms accuracy.

4.8.3 Quantitative outcome

The quantitative evidence for the plateau is the held-out accuracy ordering from the 50-epoch ablation:

$$\underbrace{88.41\,\%}_{\text{per-epoch}} > \underbrace{87.18\,\%}_{\text{fixed-mask}} > \underbrace{86.16\,\%}_{\text{per-batch}}. \tag{4.3}$$

The per-epoch condition outperforms fixed-mask by 1.23 pp, a meaningful gain. But per-batch underperforms fixed-mask by 1.02 pp, despite providing orders of magnitude more D2D diversity. This inversion is the signature of a plateau with negative returns beyond the optimum. The optimizer does not merely stop benefiting from additional frequency; it is actively harmed by it.

The full ten-instance fresh-instance protocol confirms this pattern at the deployment-evaluation level. Canonical Ensemble HAT (per-epoch) reaches $86.37 \pm 1.54\,\%$. Fixed-mask HAT collapses to $10.00\,\%$ on fresh instances, but this collapse is a transfer failure, not an optimization failure: fixed-mask reaches $87.95 \pm 0.27\,\%$ on the training-time instance. Per-batch resampling, evaluated in the exploratory scan at $86.16\,\%$ held-out, would likely reach a fresh-instance mean somewhat below the per-epoch $86.37\,\%$ if evaluated with the full protocol, consistent with the block-stationary hypothesis.

The plateau is also visible in the spatial-versus-i.i.d. ablation. When the structured D2D field is replaced by i.i.d. per-layer noise, per-epoch resampling drops fresh-instance accuracy to approximately $75\,\%$, and per-batch drops it further to approximately $70\,\%$. The 5 pp gap between per-epoch and per-batch persists even when spatial structure is removed, suggesting that the plateau is not an artifact of the particular correlation structure of the D2D field but a more general property of block-stationary optimization.

4.8.4 Residual open question

The most important open question is the location of the *knee* of the plateau curve. The present data establish that per-batch is too frequent and per-epoch is near the optimum, but they do not resolve whether resampling every two epochs, every five epochs, or every half-epoch would match or exceed the per-epoch result. A dense frequency sweep—varying the resampling interval from 1 batch to 1 epoch in logarithmic steps—would map the full curve and identify the cost-accuracy sweet spot.

A second open question concerns the interaction between ensemble frequency and model depth. Tiny-ViT is a relatively shallow Vision Transformer with four transformer stages. A deeper model might require longer block-stationary epochs to settle into a compatible basin, shifting the plateau knee toward coarser cadences. Conversely, a model with more parameters per layer might adapt faster to each new mismatch map, shifting the knee toward finer cadences. The scaling of the optimal block length with model capacity is unknown.

Finally, the plateau concept extends beyond D2D resampling to other distributional training objectives. For example, if proportional-noise HAT (Chapter 2, §2.4) is combined with epoch-level resampling, does the same plateau structure apply to the *amplitude* of the proportional noise? Or does the state-dependent noise law change the shape of the frequency–accuracy curve? These questions are relevant for designing adaptive HAT schedules that vary both noise profile and resampling cadence during training.

4.9 Synthesis: what the mitigations tell us together

The seven case studies and the severe-NL band analysis, read in sequence, tell a coherent story about the limits and opportunities of training-time intervention for organic optoelectronic CIM deployment.

4.9.1 The hierarchy of mitigations

First, there is a clear hierarchy of intervention efficacy. At the top is Ensemble HAT under the canonical $NL = 1.0$ surrogate, which recovers $86.37 \pm 1.54\,\%$ fresh-instance accuracy by changing the training objective from single-instance memorisation to distributional expectation. Below it, but still deployment-viable, is the severe-NL regime with the revised gradient-scaling recipe: Standard HAT reaches approximately 81.2 % and Ensemble HAT reaches approximately 80.7 %, both in the same ∼80–82 % band irrespective of resampling strategy or noise law. At the bottom is fixed-mask HAT under $NL = 1.0$, which collapses to 10.00 % on fresh instances.

This hierarchy reorders the conventional intuition. Under canonical $NL = 1.0$, the gap between Standard and Ensemble HAT is 76 pp; under severe $NL = 2.0$, the gap closes to less than 1 pp. The implication is that severe nonlinearity becomes the *binding* constraint, and the additional benefit of epoch-level resampling is masked once gradient-scaling distortion is large enough. A designer facing severe-NL devices should therefore prioritise nonlinearity reduction (device-level linearisation) over training-protocol refinements, because the latter saturate once the surrogate distortion exceeds a threshold.

4.9.2 The training-objective principle

Second, the case studies reinforce a general principle with a boundary condition: *the training objective matters more than the physical surrogate, but only when the surrogate distortion is moderate.* Ensemble HAT recovers $86.37 \pm 1.54\,\%$ with $NL = 1.0$ because D2D-mismatch distortion is a *random instance label* that distributional training can average away. Under severe $NL = 2.0$, the same distributional training no longer separates from fixed-mask training, because the gradient-scaling asymmetry introduces a systematic bias that resampling cannot average out. The severe-NL degradation is therefore not a mismatch-map overfitting problem; it is a distinct physics–algorithm interaction that requires either surrogate redesign or device-level linearisation.

4.9.3 The cadence sweet spot

Third, the cadence ablation and frequency-plateau analysis show that block-stationary exposure is the load-bearing ingredient for D2D-mismatch robustness. Per-epoch resampling outperforms both fixed-mask and per-batch alternatives because it provides enough forward-path stability for the optimiser to descend under each instance, while the sequence of epochs provides enough distributional breadth to prevent instance-specific overfitting. The optimal cadence is not the maximum cadence; it is the cadence that balances stability and diversity. This principle holds for D2D mismatch but does not extend to the severe-NL regime, where all cadences converge to the same band.

4.9.4 The honesty gap

Fourth, the severe-NL case studies expose an *honesty gap* between source-domain accuracy and deployment-grade transfer. Under the revised recipe, severe-NL checkpoints reach 82.89 % (Standard HAT, seed 123) and 80.97 % (Ensemble HAT, seed 123) on the training-time array—numbers that appear deployment-relevant in isolation. The fresh-instance evaluation of ∼80–82 % confirms that these source-domain rescues *do* translate into deployment-grade performance once the gradient-scaling recipe is corrected. The honesty gap in the severe-NL regime is therefore not between source-domain and fresh-instance accuracy; it is between *correct* and *incorrect* gradient-scaling implementations.

4.9.5 The residual gap

Finally, the convergence of all severe-NL training routes to the same ∼80–82 % band is the chapter's central quantitative result. The residual gap of roughly 15 pp to the digital baseline (97.37 %) and roughly 5.7 pp to the canonical $NL = 1.0$ Ensemble HAT level (86.37 %) is a joint effect of nonlinear write dynamics and fresh-instance transfer, not of transfer alone. Whether this gap can be closed by higher-order gradient corrections, larger-batch training, or device-level linearisation remains an open question. Chapter 5 discusses the theoretical bounds on surrogate fidelity; Chapter 7 outlines the experimental programme that would test whether second-order or full-differential corrections recover the missing accuracy.

4.10 Summary

This chapter has presented seven case studies and a severe-NL band analysis, each following the structured narrative of problem → intervention → quantitative outcome → residual open question.

Ensemble HAT (§4.2) addresses hardware-instance overfitting by replacing fixed-mask training with epoch-level D2D resampling. The intervention recovers

$86.37 \pm 1.54\,\%$ fresh-instance accuracy, generalises to the OPECT literature profile at $88.53 \pm 0.08\,\%$, and tolerates moderate spatially correlated D2D ($84.57 \pm 2.39\,\%$ at $\rho = 0.3$). The open question is compound-stress robustness and the location of the optimal cadence knee.

Standard HAT under severe NL (§4.3) tests whether fixed-mask training survives severe nonlinearity. With the revised gradient-scaling recipe, Standard HAT recovers to $81.89 \pm 1.02\,\%$ (seed 123) and $80.37 \pm 0.08\,\%$ (seed 456), confirming that severe NL does not rescue fixed-mask training but introduces a systematic 5 pp degradation relative to the $NL = 1.0$ baseline.

Ensemble HAT under severe NL (§4.4) tests whether epoch-level resampling breaks the severe-NL degradation. The result is $80.37 \pm 0.59\,\%$ (seed 123) and $81.04 \pm 1.76\,\%$ (seed 456), statistically indistinguishable from Standard HAT. Under severe NL, the two protocols converge.

Proportional noise under severe NL (§4.5) tests whether noise-law alignment matters. The proportional-noise band (80.64–80.75 %) is indistinguishable from the uniform-noise band, simplifying the deployment design space.

The ∼80–82 % band (§4.6) documents the convergence of all six M-series training routes to the same fresh-instance regime. The 8-bit ADC-on hook diagnostic differs from the ADC-off surrogate baseline by only -0.10 pp on average, and the residual 5.7 pp gap to the canonical $NL = 1.0$ level awaits closure via higher-order corrections or device linearisation.

Per-batch versus per-epoch HAT (§4.7) compares resampling cadences. The intervention is a held-out ablation showing that per-epoch (88.41 %) outperforms both fixed-mask (87.18 %) and per-batch (86.16 %). The open question is the shape of the full frequency–accuracy curve and its dependence on model depth.

Ensemble-frequency plateau (§4.8) documents the structural evidence that diminishing returns set in beyond epoch-level resampling. The quantitative ordering per-epoch > fixed-mask > per-batch confirms a plateau with negative returns at excessive frequency. The open question is the scaling of the optimal block length with model capacity and noise profile.

Together, the case studies establish that deployment-grade robustness in organic optoelectronic CIM requires a distributional training objective matched to the deployment distribution, a block-stationary resampling cadence near the stability–diversity sweet spot, and honest evaluation on unseen hardware instances. In the severe-NL regime, the chapter's central finding is that the revised gradient-scaling recipe recovers a consistent ∼80–82 % band across all tested training configurations, leaving a residual gap whose closure awaits surrogate refinement or device-level linearisation.

These recoveries assume idealised noise; the next chapter reintroduces the physical world.

Chapter 5

Physical-Realism Extensions

5.1 Introduction: what the manuscript deferred and the thesis re-opens

The manuscript results are intentionally anchored to a first-order behavioral model: uniform Gaussian device-to-device (D2D) mismatch, scalar cycle-to-cycle (C2C) noise, a gradient-scaling surrogate for nonlinear write, double-exponential retention with fixed-temperature coefficients, and scalar placeholders for circuit-level parasitics. Those simplifications were necessary to keep the narrative focused on the core algorithm–device boundary—the failure of fixed-mask hardware-aware training (HAT), the recovery via Ensemble HAT, and the severe-write-nonlinearity ceiling. But the manuscript also candidly lists the deferred physics as limitations (manuscript limitations section): spatially structured mismatch beyond i.i.d., heavy-tailed conductance distributions, geometry-aware IR drop, sneak-path currents, and temperature-dependent drift.

This chapter re-opens every item on that deferred list. Its purpose is not to claim that all extensions have been solved; rather, it is to show that the thesis framework is *extensible*—that each deferred physics layer can be posed as a well-defined experimental protocol, that the existing results provide baselines against which to judge new physics, and that the simulator's profile-driven interface (Chapter 1) is the natural entry point for richer device models.

For reference, the four empirical baselines against which the extensions in this chapter are judged are: (i) Ensemble HAT preserves $86.37 \pm 1.54\,\%$ fresh-instance accuracy under canonical uniform noise; (ii) the canonical V4 fixed-mask checkpoint reaches $87.95 \pm 0.27\,\%$ source-domain accuracy but collapses to $10.00\,\%$ on fresh instances; (iii) global severe write nonlinearity ($NL = 2.0$) with the revised gradient-scaling recipe recovers to the $\sim$80–82 % band across Standard, Ensemble, and proportional-noise training routes; and (iv) the residual gap between the $NL = 2.0$ band ($\sim$80.7 %) and the $NL = 1.0$ baseline (86.37 %) is approximately 5.7 pp, a joint effect of nonlinear write dynamics and fresh-instance transfer.

The chapter is organized around five physical-realism axes:

1. **Correlated D2D** (Section 5.2): the manuscript reported a bounded-degradation sweep under separable AR(1) spatial correlation; here the full result is presented with statistical interpretation.

2. **Heavy-tailed conductance** (Section 5.3): a "what-if" extension that replaces the Gaussian mismatch kernel with mixture models (log-normal, Pareto-truncated). No GPU results exist; the section defines the protocol and the theoretical expectation.

3. **IR-drop modeling** (Section 5.4): a minimal-effort circuit-aware layer that turns the manuscript's 1 % scalar placeholder into a geometry-dependent spatial attenuation map. The section outlines the implementation path and the compute cost.

4. **Temperature drift** (Section 5.5): an Arrhenius-form extension of the retention and mismatch models across the −20 °C to 85 °C operational envelope. The section specifies the experiment protocol and the success criterion.

5. **Retention beyond the 79 % plateau** (Section 5.6): the manuscript's double-exponential model saturates near 79.13–79.51 %; this section asks what happens at 1 hour, 1 day, 1 week, and 1 month, and proposes an accelerated-aging protocol.

The chapter closes with a synthesis (Section 5.7) that maps each extension to a thesis contribution tier: *core* (empirically demonstrated and locked), *extended* (protocol defined but not fully executed), or *future* (conceptual framing only).

5.2 Correlated-D2D full result: spatial structure beyond the i.i.d. assumption

5.2.1 Motivation and manuscript context

Every canonical result in the manuscript assumes i.i.d. Gaussian D2D mismatch: each crossbar element samples its offset independently. Real fabricated arrays, however, exhibit spatially correlated mismatch because of wafer-level process gradients, optical write non-uniformity across the programming aperture, and thermal coupling during conductance tuning (manuscript limitations section). The manuscript therefore included a supplementary correlated-D2D ablation as a robustness probe, but it did not present the full statistical interpretation or connect the sweep to the failure-mode atlas developed in Chapter 3.

This section presents the complete 10×5 fresh-instance protocol under separable AR(1) spatial correlation, discusses the bounded-degradation signature, and derives the practical implication for deployment.

5.2.2 AR(1) separable correlation model

The correlated mismatch field $M^{(\rho)}$ is generated from a standard Gaussian field $Z \sim \mathcal{N}(0,1)$ by applying a separable first-order autoregressive smoothing filter along rows and columns:

$$M_{ij}^{(\rho)} = \sum_{k=-K}^{K} \sum_{l=-L}^{L} h_k(\rho)\, h_l(\rho)\, Z_{i+k,j+l}, \tag{5.1}$$

where $h_k(\rho) = \sqrt{1-\rho^2}\, \rho^{|k|}$ is the normalized AR(1) impulse response and the boundary is handled by reflection padding. The parameter $\rho \in [0,1)$ controls the spatial decay of correlation: $\rho = 0$ recovers the i.i.d. baseline, while larger ρ produces smooth low-frequency gradients across the array. The marginal variance is preserved at σ^2_{D2D} by construction, so the only difference between correlation levels is the *spatial structure*, not the amplitude.

Three levels were evaluated: $\rho = 0$ (i.i.d. baseline), $\rho = 0.3$ (moderate correlation), and $\rho = 0.5$ (strong correlation). The $\rho = 0.3$ level was chosen because it approximates the spatial scale of optical write non-uniformity reported in early organic-array prototypes; the $\rho = 0.5$ level was chosen as a stress test that approaches the long-range wafer-gradient regime.

5.2.3 Fresh-instance protocol and locked results

All evaluations used the canonical Tiny-ViT V4 Ensemble HAT checkpoint (the same checkpoint that yields $86.37 \pm 1.54\,\%$ under i.i.d. conditions). The fresh-instance protocol drew 10 independent correlated mismatch realizations for each ρ level and performed 5 Monte Carlo forward passes per realization. Table 5.1 reports the mean, standard deviation, and excursion relative to the i.i.d. baseline.

Table 5.1: Fresh-instance robustness of Ensemble HAT under spatially correlated D2D mismatch (10 fresh arrays $\times$ 5 Monte Carlo evaluations). All values are drawn from the same canonical checkpoint to isolate the effect of correlation structure.

Correlation level	Fresh-instance mean (%)	Cross-instance std (%)	Excursion vs i.i.d. (pp)
i.i.d. ($\rho = 0$)	86.33 ± 1.61	1.61	0.00
Moderate ($\rho = 0.3$)	84.57 ± 2.39	2.39	-1.76
Strong ($\rho = 0.5$)	82.12 ± 3.95	3.95	-4.20

The key observation is *bounded degradation*: even at $\rho = 0.5$, the mean accuracy remains above 82 %, far from the 10.00 % chance-level collapse of standard fixed-mask HAT (Chapter 3, Section 3.2). The rank ordering across classes is preserved at all tested levels, and the worst-case individual instance at $\rho = 0.5$ still reaches 73.7 % (Supplementary Note S2). This confirms that Ensemble HAT's distributional training objective generalizes, at least qualitatively, beyond the i.i.d. sampling law on which it was trained.

5.2.4 Interpretation: variance budget and spatial structure

Two secondary signatures merit discussion. First, the cross-instance standard deviation grows monotonically with correlation strength (1.61 pp $\rightarrow$ 2.39 pp $\rightarrow$ 3.95 pp). This widening reflects the fact that spatially smooth mismatch fields create coherent distortions across large weight tiles; some fresh realizations happen to align with the classifier's decision boundaries, while others oppose them. Under i.i.d. noise, these coherent effects average out more rapidly, yielding tighter variance.

Second, the degradation is *sub-linear* in ρ. A naïve expectation might predict that doubling the correlation length (from $\rho = 0.3$ to $\rho = 0.5$) would double the accuracy loss. Instead, the excursion grows from -1.76 pp to -4.20 pp, a factor of $2.4\times$. The modest super-linearity suggests that the attention-side projection geometry (which Chapter 3 identified as structurally sensitive) becomes increasingly vulnerable as the mismatch field loses high-frequency spatial mixing. When smooth gradients dominate, the effective rank of the perturbation drops, and the residual subspaces that the classifier relies upon are more consistently damaged.

5.2.5 Mitigation and open questions

No explicit mitigation is required for moderate correlation ($\rho \leq 0.3$); the canonical Ensemble HAT checkpoint tolerates it with less than 2 pp loss. For stronger correlation, the natural mitigation is to *train with correlated mismatch resampling* rather than i.i.d. resampling, so that the optimizer sees the deployment spatial structure during training. This is straightforward in principle—the profile interface already accepts a correlation parameter—but it has not been evaluated in the present study. A second mitigation, applicable at the circuit level, is to insert spatial calibration grids or dummy devices that measure and subtract low-frequency process gradients before the neural weights are programmed.

The most important open question is whether the bounded-degradation trend holds under *measured* spatial covariance kernels. The separable AR(1) model is a convenient mathematical proxy; real organic optoelectronic arrays may exhibit anisotropic correlation (stronger along wordlines than bitlines due to shared gate buses), heavy-tailed conductance distributions (Section 5.3), or long-range wafer-scale gradients that are not captured by a first-order separable model. Whether Ensemble HAT (or a correlated variant) remains robust under these richer spatial structures is an open empirical question that can only be answered once fabricated-array characterization data become available.

5.3 Heavy-tailed conductance extension: beyond the Gaussian mismatch kernel

5.3.1 Why the Gaussian assumption may fail

The canonical D2D model (Eq. 1.1 in Chapter 1) draws each element's mismatch from a zero-mean Gaussian with standard deviation $\sigma_{\mathrm{D2D}}\Delta G$. Gaussian noise is analytically convenient and often a good approximation for mid-range device counts, but it is not a physical law. Organic thin-film transistors and memristive crossbars can exhibit *heavy-tailed* conductance distributions because of filamentary variability, localized trap states, and spatial inhomogeneity in the active layer[?]. When the tail is heavy enough, rare outlier devices exert disproportionate influence on the analog matrix-vector product, potentially pushing the accuracy distribution into a qualitatively different regime than the one predicted by Gaussian statistics.

This section outlines the experimental protocol, the sampling families under consideration, and the theoretical predictions. **No GPU results have been executed for this extension;** the framework is presented as a "what-if" study that can be instantiated once device-level characterization confirms non-Gaussian tails.

5.3.2 Reference: the stub evaluation script

A reference implementation stub exists at `scripts/_gpt/eval_heavy_tailed_d2d.py`. The stub mirrors the existing fresh-instance evaluation harness (the same harness used for the correlated-D2D sweep of Section 5.2) but replaces the Gaussian sampler with a `sample_heavy_tailed_d2d_mask()` function that is currently a `NotImplementedError`. The stub exposes the following command-line interface:

- `--tail-family`: choice among `lognormal`, `student_t`, and `pareto_trunc`;
- `--tail-prob`: mixture weight $p_{\mathrm{tail}} \in \{0, 0.01, 0.03, 0.05\}$;
- `--tail-scale`: multiplicative severity factor $s_{\mathrm{tail}} \in \{1.5, 2.0, 3.0\}$ relative to nominal σ_{D2D}.

The stub writes a JSON payload describing the requested configuration but does not execute inference. Its existence proves that the code path is ready for a rebuttal-phase or thesis-extension implementation without structural changes to the training or evaluation stack.

5.3.3 Sampling families: log-normal versus Pareto-truncated

Two families are prioritized because they preserve positive-skew outliers without introducing infinite-variance pathologies that would break the simulator's numerical stability.

Gaussian–log-normal mixture. With probability $1 - p_{\text{tail}}$, the mismatch draw follows the canonical Gaussian. With probability p_{tail}, it follows a truncated log-normal:

$$\epsilon_{\text{tail}} \sim \text{LogNormal}(\mu_{\ln}, \sigma_{\ln}) \nVdash_{[0, \epsilon_{\max}]}, \tag{5.2}$$

where $(\mu_{\ln}, \sigma_{\ln})$ are chosen so that the median of the tail component matches the nominal Gaussian scale, while the upper tail is inflated by the factor s_{tail}. Truncation at $\epsilon_{\max} = 3\,\sigma_{\text{D2D}}\,s_{\text{tail}}$ prevents physically un-realizable conductance excursions (negative conductance or values beyond $G_{\max}$). The log-normal family is attractive because it arises naturally from multiplicative process noise: if several independent fabrication steps each contribute a small multiplicative perturbation, their product is log-normally distributed.

Gaussian–Pareto-truncated mixture. The Pareto component captures power-law tails that decay more slowly than the log-normal:

$$p(\epsilon_{\text{tail}}) \propto \epsilon_{\text{tail}}^{-(\alpha_{\text{P}}+1)} \nVdash_{[\epsilon_{\min}, \epsilon_{\max}]}, \tag{5.3}$$

with shape parameter $\alpha_{\text{P}} = 3$ (finite variance) and support $[\epsilon_{\min}, \epsilon_{\max}]$ anchored to the nominal scale. The Pareto family is useful when defect-mediated conductance states create a sparse population of extreme outliers—for example, a localized short or an open channel in an otherwise uniform film.

A cross-check with Student's t distribution ($\nu = 3, 5, 10$ degrees of freedom) is also supported by the stub, but it is deprioritized because the t distribution is symmetric and therefore less representative of the positive-skew outliers observed in organic device literature.

5.3.4 Predicted behavior and success criteria

Although no experiments have been run, the theory of robust analog inference under heavy-tailed noise makes three testable predictions:

1. **Bounded mean degradation for small tail probability.** When $p_{\text{tail}} \leq 0.03$, the mean accuracy should remain within 5 percentage points of the Gaussian baseline. This follows from the fact that Ensemble HAT averages over the deployment distribution, and a small mixture component cannot shift the expected risk dramatically unless the tail variance is pathological.

2. **Variance inflation.** The cross-instance standard deviation should increase with s_{tail} because rare extreme realizations occasionally produce catastrophic mismatch maps. This is the qualitative signature to watch for: the mean may stay high while the worst-case instance drops sharply.

3. **Rank-order preservation.** If the heavy-tailed extension is a perturbative refinement rather than a new failure mode, the accuracy ranking across ρ levels

(Section 5.2) should be preserved: i.i.d. Gaussian > mild heavy-tail > stronger heavy-tail. A reversal of this ordering would indicate that the Gaussian model systematically overestimates deployment robustness.

The success criterion for claiming that the Ensemble HAT recipe "survives" heavy-tailed mismatch is therefore not a single mean accuracy but a *triple condition*: (i) mean above the standard-HAT collapse regime, (ii) monotonic ranking, and (iii) bounded drop under $p_{\text{tail}} \leq 0.05$.

5.3.5 What experiment would be needed

A full execution would require only inference-time GPU cycles because the canonical Ensemble HAT checkpoint is reused without retraining. The recommended protocol is:

1. **Screening pass:** 5 fresh arrays $\times$ 3 Monte Carlo runs for each $(\text{family}, p_{\text{tail}}, s_{\text{tail}})$ combination. Estimated cost: low-to-medium CPU/GPU budget ($\sim$8 hours on a single A100).
2. **Promotion:** if the screening shows a non-negligible interaction (mean drop > 3 pp or variance doubling), promote the most reviewer-relevant condition to the full 10 arrays $\times$ 5 MC protocol for statistical locking.
3. **Output:** one JSON per family/sweep, one supplementary robustness figure (mean $\pm$ std), and one interpretive paragraph stating whether the ranking survives.

This section intentionally defers full execution to future work. The stub script, the sampling definitions, and the success criteria are the thesis contribution; the GPU results are not claimed.

5.4 IR-drop preliminary modeling: toward a circuit-aware layer

5.4.1 From scalar placeholder to spatial attenuation

The manuscript includes a 1 % scalar penalty for IR drop in its compound stress test (manuscript limitations section), but the value is explicitly a placeholder: "spatial IR drop (currently a 1 % scalar placeholder)" and "sneak-path currents (likewise)." This section outlines the minimal-effort extension that would replace the scalar with a geometry-aware spatial attenuation map.

The goal is not to build a full SPICE-level crossbar solver; that would break the PyTorch-native training loop and defeat the framework's purpose as a rapid algorithm–device co-design tool. Instead, the goal is a *fast deterministic pre-pass* that computes a per-cell attenuation factor from array geometry, line resistance, and the present weight matrix, then applies that factor before the analog-to-digital conversion.

5.4.2 Array geometry and line-resistance model

Consider a differential-pair crossbar tile of size $N_{\text{row}} \times N_{\text{col}}$. The input voltage V_{in} is applied along wordlines; the accumulated current is sensed along bitlines. If the interconnect resistance per cell is R_w (a function of metal pitch and organic film sheet resistance), the voltage seen by the device at coordinate (i, j) is reduced by the cumulative drop along the wordline:

$$V_{\text{eff}}(i,j) = V_{\text{in},i} - R_w \sum_{k=1}^{i} I(k,j), \tag{5.4}$$

where $I(k,j) = V_{\text{eff}}(k,j)\, G(k,j)$ is the cell current. A symmetric expression holds for the bitline return path. Equation 5.4 is nonlinear because V_{eff} appears on both sides, but it can be solved approximately by a single fixed-point iteration: first compute $I(k,j)$ using the ideal V_{in}, then compute V_{eff} from the ideal currents, and finally re-evaluate the matrix-vector product with the attenuated voltages.

5.4.3 Geometry choices: 128×64 versus 256×256

The choice of tile size determines whether IR drop is a mild nuisance or a deployment blocker. For the Tiny-ViT mapping analyzed in Chapter 1, the analog footprint requires 812 differential-pair arrays under a 128×128 constraint. If the tile is relaxed to 256×256, the array count drops but the maximum line length doubles, and the worst-case IR drop scales quadratically with line length because both the resistance and the number of contributing cells increase. Table 5.2 illustrates the trade-off under a representative $R_w = 0.5\,\Omega$ per cell and a mean conductance of $10\,\mu$S.

Table 5.2: Approximate worst-case IR-drop envelope for two tile geometries under a representative per-cell line resistance. Values are analytical estimates, not simulated results.

Tile geometry	Max wordline length	Estimated worst-case drop	Arrays needed for Tiny-ViT
128×64	128 cells	$\sim$3 %	$\sim$1,624
128×128	128 cells	$\sim$6 %	812
256×256	256 cells	$\sim$24 %	203

The 128×64 geometry (wider than tall, or vice versa) is attractive because it keeps the worst-case line short while still achieving reasonable array utilization. The 256×256 geometry, which might be preferred for area efficiency, pushes the drop into a regime where even Ensemble HAT may struggle unless the training loop itself includes the IR-drop model.

5.4.4 Compute estimate and integration path

The fast fixed-point iteration avoids the $\mathcal{O}(N^3)$ cost of a full nodal analysis, but it still breaks standard GEMM parallelization because the attenuation map depends on the input-dependent current distribution. A conservative estimate is that inference throughput drops by 2×–4× when the IR-drop pre-pass is enabled, and training throughput drops by 3×–6× because the backward pass must propagate through the attenuation factors. For a full HAT retraining sweep (∼100 epochs, 3 seeds), this adds ∼120 GPU-hours to the experimental budget.

Integration into the framework would follow the same hook-based pattern used for retention and front-end compensation (Chapter 1):

1. Extend the JSON profile with `R_wire_ohms`, `tile_rows`, and `tile_cols`.
2. Insert an `apply_ir_drop_map()` stage between analog MVM output and digital rescaling.
3. Keep D2D and C2C active so the experiment measures interaction rather than a clean-room IR-only effect.

5.4.5 What would success look like

Success is defined in three tiers. **Tier 1 (screening):** under a 128×64 tile with $R_w = 0.5\,\Omega$, the fresh-instance accuracy of the existing Ensemble HAT checkpoint does not drop more than 3 pp relative to the zero-IR baseline. This would show that the current training recipe is robust to mild spatial drops without retraining. **Tier 2 (retraining):** if Tier 1 fails, HAT retraining with the IR-drop model enabled recovers at least half of the lost accuracy, demonstrating that the optimizer can learn to map critical weights to spatial coordinates with lower attenuation. **Tier 3 (array-size ceiling):** the experiment identifies a maximum safe tile dimension (e.g., 128×128) above which accuracy degrades precipitously even with retraining, giving array architects a hard constraint.

This section outlines the experimental protocol and the compute budget. Full execution is deferred to future work.

5.5 Temperature drift: an Arrhenius-form extension

5.5.1 Why temperature cannot be ignored

Organic semiconductors rely on hopping transport and exciton dissociation mechanisms that are fundamentally thermally activated. Unlike silicon CMOS, where temperature variations induce relatively modest mobility drifts (∼0.3 % per Kelvin), organic devices exhibit exponential or strong power-law temperature dependencies in carrier mobility, dark current, and photoresponse[?] . Edge-vision deployments will experience ambient swings from −20 °C (cold-start outdoor) to 85 °C (enclosed hot edge), and the internal array temperature may track inference workload intensity.

The manuscript scopes its retention and mismatch experiments to a fixed nominal temperature and lists dynamic temperature effects among the deferred circuit-level phenomena (manuscript limitations section). This section re-opens that deferral by specifying an Arrhenius-form conductance-drift framework, a temperature-sweep protocol, and the success criteria.

5.5.2 Arrhenius-form conductance drift

The canonical retention model (Eq. 1.7 in Chapter 1) uses fixed time constants $\tau_1 = 140$ ms and $\tau_2 = 610$ ms. Under temperature variation, these time constants are expected to follow an Arrhenius law:

$$\tau(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right), \tag{5.5}$$

where E_a is the activation energy for conductance relaxation, k_B is the Boltzmann constant, and τ_0 is a material-specific prefactor. Similarly, the mismatch variance may inflate with temperature because thermal noise adds to the fabrication spread:

$$\sigma_{\mathrm{D2D}}(T) = \sigma_{\mathrm{D2D},0}\left[1 + \alpha_\sigma (T - T_0)\right], \tag{5.6}$$

with thermal coefficient α_σ referenced to room temperature $T_0 = 300$ K. The photoresponse nonlinearity γ_{phys} may also drift:

$$\gamma_{\mathrm{phys}}(T) = \gamma_{\mathrm{phys},0} + \alpha_\gamma (T - T_0). \tag{5.7}$$

The profile-driven framework can absorb these extensions by adding three new JSON fields—`E_activation`, `T_coeff_sigma`, and `T_coeff_gamma`—with backward-compatible defaults that recover the isothermal manuscript results when the fields are omitted.

5.5.3 Protocol: temperature sweep with fixed checkpoints

Because the canonical Ensemble HAT checkpoint was trained under isothermal assumptions, the first experiment is a *zero-shot temperature-stress test*: evaluate the same checkpoint across a temperature sweep without retraining. This measures how badly the isothermal training assumption fails when deployment temperature deviates from the training-time prior.

Recommended sweep points:

$$T \in \{250, 275, 300, 325, 350\}\,\mathrm{K} \quad \Longleftrightarrow \quad \{-23, +2, +27, +52, +77\}\,^\circ\mathrm{C}. \tag{5.8}$$

The 300 K point recovers the canonical ∼86.37 % baseline. The 350 K point probes the hot-edge limit; the 250 K point probes cold-start operation. At each temperature, the device profile is updated via Eqs. 5.5–5.7, and the fresh-instance protocol (10 × 5) is executed.

5.5.4 Predicted outcomes and success criteria

Based on the Arrhenius framework and literature values for organic semiconductors ($E_a \approx 0.2$–0.4 eV, $\alpha_\sigma \approx 10^{-3}$ K^{-1}), the following outcomes are expected:

- **At** 350 **K (hot edge):** retention accelerates dramatically, so the slow mode that previously stabilized near 79 % may erode further. Dark-current inflation reduces dynamic range and increases shot-noise variance. Expected accuracy without thermal-aware training: $\lesssim 50\,\%$.

- **At** 250 **K (cold edge):** dark current drops, improving dynamic range, but γ_{phys} may shift further from the ideal 1.0, creating a mismatch with the 300 K-calibrated inverse-gamma front end. Expected accuracy: $\sim$75 %.

- **Ranking preservation:** the success criterion is not that accuracy stays at 86 % across the full range—that is unrealistic for an isothermally trained model—but that the *relative risk ranking* observed in the manuscript persists. Specifically, ADC resolution should remain the dominant accuracy gate, D2D mismatch should dominate in the operational envelope, and Ensemble HAT should still outperform fixed-mask HAT by a large margin at every temperature.

A secondary success criterion is *rank-order preservation across* T: if class A outperforms class B at 300 K, the same ordering should hold at 325 K and 350 K. Temperature-induced rank flips would indicate that the decision boundaries are unstable, which is a more severe deployment risk than uniform accuracy erosion.

5.5.5 Thermal–IR-drop coupling

Temperature effects do not operate in isolation. The interconnect wire resistance R_w (Section 5.4) has a positive temperature coefficient: $R_w(T) = R_{w,0}[1 + \mathrm{PTC}\,(T - T_0)]$. As T increases, R_w increases, simultaneously worsening the spatial IR drop. A fully coupled simulator would update $R_w(T)$ before computing the IR-drop attenuation map, then evaluate retention with temperature-accelerated time constants, and finally inject the temperature-inflated D2D field. The modular profile interface makes this coupling straightforward in principle, but the joint parameter space (T, R_w, τ, σ_{D2D}) is large enough that a systematic sweep would require $\sim$200 GPU-hours. This section therefore limits itself to outlining the coupling; full execution is deferred.

5.6 Retention beyond the 79 % plateau: long-horizon drift

5.6.1 The manuscript plateau and its physical meaning

The manuscript retention experiments (manuscript retention-drift section and Chapter 3, Section 3.6) exhibit a two-phase decay: a rapid drop from 91.63 % at $t = 0$ s to 82.66 % at $t = 1$ s, followed by a broad asymptotic plateau near 79.13–79.51 % from 10 s to 10,000 s. The plateau is stable across three orders of magnitude in time, indicating that the slow decay mode ($\tau_2 = 610$ ms) carries only a small residual accuracy cost once dynamic scale recalibration is applied.

The 79 % plateau is a direct consequence of the specific parameter values (τ_1, τ_2, A_0) extracted from one literature source on DNTT-based organic transistors[?] . It is *not* a universal bound for all organic devices. Faster slow modes, stronger state dependence, or temperature-accelerated drift (Section 5.5) could shift the plateau downward; better materials or periodic refresh could shift it upward.

5.6.2 What happens at 1 hour, 1 day, 1 week, and 1 month?

The manuscript's longest evaluated interval is 10,000 s (∼2.8 hours). Real edge deployments, however, may require weights to remain valid for days or weeks between programming events. Extrapolating the double-exponential model to longer times predicts a further slow erosion:

$$G(t) = G_{\min} + (G - G_{\min}) \left[A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + A_0 \right], \tag{5.9}$$

but this extrapolation is physically questionable. Organic conductance drift often exhibits a *stretched-exponential* or *power-law* tail at very long times, reflecting the broad distribution of trap-depth energies in disordered films. The double-exponential may therefore underestimate the decay at $t > 10^5$ s.

To bracket the uncertainty, three hypothetical long-horizon models are considered:

1. **Double-exponential continuation:** the existing (τ_1, τ_2, A_0) values are trusted out to 1 month. Under this law, the conductance fraction remaining at 1 month is $A_0 + A_2 \exp(-t/\tau_2) \approx 0.60 + 0.20 \times 0 = 0.60$, i.e. 60 % of the programmed window. The corresponding accuracy is unknown but would likely sit in the 70–75 % range if the relationship between conductance retention and classifier accuracy were roughly linear near the plateau.

2. **Stretched-exponential tail:** replace the slow mode with $A_2 \exp\left[-(t/\tau_2)^{\beta}\right]$ where $\beta \approx 0.5$–0.7 for disordered organics. This accelerates the long-time decay relative to the pure exponential, pushing the 1-month retained fraction closer to A_0 itself (∼60 %) and potentially dropping accuracy below 70 %.

3. **Power-law tail:** model the slow decay as $A_2\,(t/\tau_2)^{-p}$ with $p \approx 0.1$–0.3. Power-law drift has no asymptotic plateau; accuracy would erode indefinitely, albeit slowly, making periodic refresh mandatory for any long-horizon deployment.

5.6.3 Accelerated-aging protocol

Because waiting 1 month for a real-time retention measurement is impractical during a thesis cycle, an accelerated-aging protocol is proposed:

1. **Elevated-temperature stress:** program the array (or its simulator equivalent) at 300 K, then hold at $T_{\text{stress}} \in \{325, 350, 375\}$ K for logarithmically spaced durations.
2. **Time–temperature superposition:** using the Arrhenius law (Eq. 5.5), map each $(T_{\text{stress}}, t_{\text{stress}})$ pair to an equivalent room-temperature age t_{eq}.
3. **Checkpoint evaluation:** at each equivalent age, evaluate the retained Ensemble HAT checkpoint under the standard fresh-instance protocol.
4. **Model selection:** fit the accuracy-versus-time curve to double-exponential, stretched-exponential, and power-law models; select the best fit via Bayesian information criterion (BIC).

The compute cost of this protocol is moderate: it requires only inference-time evaluation (no retraining), but the temperature sweep and multiple time points add up to $\sim$30 GPU-hours for a single model.

5.6.4 What-if: can the plateau be pushed upward?

The 79 % plateau is a lower bound on deployment accuracy under the present device prior. Several "what-if" interventions could push it upward:

- **Periodic refresh:** reprogram the array before the fast mode erodes accuracy below the application threshold. For CIFAR-10, a refresh interval shorter than $\sim$1 s would be needed to maintain $> 82\,\%$; if 79 % is acceptable, the refresh interval can extend to 10,000 s or longer.

- **Inferential scale recalibration with reference cells:** instead of co-tracking the D2D buffers with the same decay law (which is analytically exact in simulation but physically approximate), periodically read out dedicated reference conductance pairs to update the digital gain coefficients. The energy and latency cost of such recalibration is not included in the manuscript's 11.45× energy-reduction estimate and should be treated as additional peripheral overhead.

- **Material improvement:** if the slow-mode time constant τ_2 were increased from 610 ms to 10 s, the plateau would likely rise above 85 % because the residual decay during the inference window would be smaller. This is a materials-science target, not an algorithmic one.

This section does not claim that any of these interventions have been executed. It frames them as the natural next steps that the 79 % plateau surfaces as high-priority open questions.

5.7 Synthesis: core, extended, and future contributions

This chapter has traversed five physical-realism extensions, each at a different stage of empirical maturity. The closing synthesis maps each extension to a thesis contribution tier and restates the boundary between what has been demonstrated, what has been specified, and what remains conceptually open.

5.7.1 Contribution tier definitions

Core contribution.
Empirically demonstrated, statistically locked, and reproducible from the manuscript or thesis experimental package. These results answer the deployment question with the same fresh-instance protocol used throughout the thesis.

Extended contribution.
Protocol fully specified, code path ready (stub or hook implemented), and theoretical predictions formulated, but GPU execution either partial or entirely deferred. These contributions define the next experiments rather than report their outcomes.

Future contribution.
Conceptually framed and connected to the existing results, but lacking even a detailed protocol. These are research directions that the thesis surfaces as important but does not claim to have scoped.

5.7.2 Tier mapping for each extension

Table 5.3 summarizes the mapping.

Correlated D2D is the only **core** extension in this chapter. The empirical result—bounded degradation with preserved rank ordering—is locked and reproducible. The open question is whether the same trend holds under measured spatial covariance rather than the AR(1) proxy.

Heavy-tailed conductance, IR-drop modeling, temperature drift, and long-horizon retention are all **extended** contributions. Each has a well-defined protocol, a clear success criterion, and a compute estimate, but none has been executed. They are framed as "what-if" extensions that the thesis explicitly opens for future work.

Table 5.3: Thesis contribution tier for each physical-realism extension.

Extension	What we did	What we only spec'd	What remains open
Correlated D2D (§5.2)	10×5 AR(1) sweep at $\rho = 0.0/0.3/0.5$; bounded degradation (-1.76 pp, -4.20 pp) locked.	Training *with* correlated resampling (not evaluated).	Measured anisotropic covariance kernels from fabricated arrays.
Heavy-tailed conductance (§5.3)	Stub script and sampling families (log-normal, Pareto-truncated) defined. Success criteria formulated.	No GPU inference executed.	Mixed correlated + heavy-tailed joint distribution.
IR-drop modeling (§5.4)	Fast fixed-point pre-pass derived. Tile geometry trade-offs analyzed. Compute budget estimated.	No circuit-aware layer integrated or evaluated.	Full SPICE-calibrated parasitic model training-time IR-drop HAT.
Temperature drift (§5.5)	Arrhenius framework specified. Temperature sweep protocol defined.	No thermal inference or retraining executed.	Joint thermal–IR-drop coupling; thermal-aware training (TAT).
Retention beyond 79 % (§5.6)	Accelerated-aging protocol designed. Long-horizon decay models compared.	No long-horizon (> 10,000 s) evaluation executed.	Stretched-exponential or power-law model selection from measured data.

Joint interactions between the extensions remain **future** territory. Does Ensemble HAT retain its 86 % fresh-instance accuracy when retention drift has progressed for 1,000 s *and* the temperature is 350 K *and* the mismatch field is heavy-tailed? The present framework evaluates each non-ideality in isolation; their joint statistics are the highest-priority unexplored region.

5.7.3 Cross-cutting principle: the approximation ceiling

Every result in this thesis—core, extended, or future—is conditioned on the present first-order behavioral approximations. The correlated-D2D sweep uses a separable AR(1) proxy, not a measured spatial kernel. The heavy-tailed extension is defined but not executed. The IR-drop model is a fast pre-pass, not a nodal solver. The temperature framework uses literature Arrhenius coefficients, not measured activation energies. The retention plateau is a double-exponential fit, not a physical drift law.

This is not a weakness; it is a methodological choice. The thesis treats each approximation as a *decision aid* that ranks which physics extensions deserve experimental priority. The correlated-D2D result tells a fabrication team that moderate spatial structure is survivable. The IR-drop outline tells an array architect that tile geometry matters more than exact line resistance. The temperature framework tells a packaging engineer that dynamic range erosion at the hot edge is the dominant risk. These rankings are the thesis-level contribution of the physical-realism extensions, even when the full GPU results are deferred.

5.7.4 Connection to the failure-mode atlas

Chapter 3 catalogued five failure modes under the canonical first-order model. This chapter adds a cross-cutting dimension: *how does each failure mode move when the first-order approximations are relaxed?*

- Hardware-instance overfitting (the 10.00 % collapse) is *mitigated* by correlated D2D: the degradation is bounded rather than catastrophic.
- The dual-attention nonlinear-write collapse is *unchanged* by the extensions in this chapter; it is a surrogate-fidelity problem, not a statistics problem.

- The severe-NL recovery band (∼80–82 %$fresh - instancemeanacrossallM - seriesroutes)couldbeworsenedbyheavy-tailedmismatchorIRdrop, becausebothextensionsincreasethe$ $horizonextrapolation.$

This mapping provides a structured way to update the atlas when new device data arrive. It is the final thesis-differentiated contribution of the chapter: not merely to list deferred physics, but to show how each deferred physics layer would reshape the boundaries of the failure modes that the manuscript identified.

5.8 Summary

This chapter has reopened the physical-realism extensions that the manuscript candidly deferred. The contributions are layered:

1. **Correlated D2D** (Section 5.2): the full 10×5 AR(1) sweep is presented and interpreted. Ensemble HAT fresh-instance accuracy degrades from 86.33 ± 1.61 % (i.i.d.) to 84.57 ± 2.39 % ($\rho = 0.3$) to 82.12 ± 3.95 % ($\rho = 0.5$). The degradation is bounded, rank order is preserved, and the worst-case instance remains far from chance level.
2. **Heavy-tailed conductance** (Section 5.3): a "what-if" extension is defined via the stub script at `scripts/_gpt/eval_heavy_tailed_d2d.py`. Log-normal and Pareto-truncated mixture sampling are specified, success criteria are formulated, and the inference-only protocol is costed. Full execution is deferred to future work.
3. **IR-drop modeling** (Section 5.4): a minimal-effort circuit-aware layer is outlined, based on a fast fixed-point pre-pass and geometry-dependent tile sizing (128×64 versus 256×256). Compute estimates and success tiers are defined. No GPU results are claimed.

4. **Temperature drift** (Section 5.5): an Arrhenius-form extension of retention and mismatch models is specified for the $-20\,°C$ to $85\,°C$ envelope. The temperature-sweep protocol, predicted outcomes, and rank-preservation success criterion are stated. Execution is deferred.

5. **Retention beyond** 79 % (Section 5.6): the manuscript's 79.13–79.51 % plateau is contextualized, and an accelerated-aging protocol is proposed to probe 1 hour–1 month horizons. Three long-horizon decay models (double-exponential, stretched-exponential, power-law) are compared as "what-if" scenarios.

The synthesis (Section 5.7) maps each extension to a contribution tier—core, extended, or future—and connects the extensions back to the failure-mode atlas of Chapter 3. The overarching principle is that the first-order behavioral model is a decision aid, not a chip-predictive emulator. Its value lies in ranking which physics extensions most urgently need measured device data, and in providing the profile-driven interface through which those data can enter the training loop once they arrive.

Surviving physical realism is necessary but not sufficient; the thesis now translates survival into deployment rules.

Chapter 6

Deployment Envelope

6.1 Introduction: from simulation data to shipping criteria

The preceding chapters have established a physical picture of what happens when a modern Vision Transformer is mapped to an organic optoelectronic compute-in-memory (CIM) array. We have defined hardware-aware training (HAT) cadences (Chapter 2), catalogued failure modes (Chapter 3), and quantified recovery under distributional training objectives. The numbers are internally consistent: Ensemble HAT preserves $86.37 \pm 1.54\,\%$ fresh-instance accuracy under canonical uniform noise; a lower-noise OPECT proxy reaches $88.53 \pm 0.08\,\%$; severe write nonlinearity ($NL = 2.0$) with the revised gradient-scaling recipe recovers to the ~80–82 % band across Standard, Ensemble, and proportional-noise training routes, leaving a residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ baseline. These are simulation results, and they are bounded by first-order behavioral approximations.

This chapter asks the pragmatic question that every simulation study must eventually face: *so what?* Under which combinations of physical parameters does the Ensemble HAT protocol actually preserve a usable accuracy ranking when the model is dropped onto an unseen hardware instance? Where does that ranking collapse? And if one were advising an analog-CIM program manager—someone deciding whether to tape out a ViT accelerator on an organic RRAM process—what actionable guidance could be distilled from the data?

The chapter is organized as a decision-support document rather than as a conventional results narrative. Section 6.2 constructs a *ranking-preservation map*: a qualitative decision matrix that classifies combinations of D2D variability, nonlinearity severity, ADC precision, retention age, and spatial correlation according to whether the canonical accuracy ordering (Ensemble HAT > fixed-mask HAT > naive deployment) survives on fresh hardware instances. Section 6.3 identifies the complementary *collapse zone*: parameter combinations that destroy the ranking or drive accuracy toward the chance level. Section 6.4 translates these two maps into a prose decision diagram organized by device-maturity tier (research-grade → pilot → production). Section 6.5 addresses the architecture question: at low maturity, would a simpler convolutional neural network (CNN) outperform the ViT under the same HAT protocol? Section 6.6 distills industrial-relevance bullets for program managers. Section 6.7 closes with the explicit boundaries of the envelope—what is *not* in the model and therefore not in the recommendation.

Throughout, the treatment is deliberately more applied and systems-oriented than the preceding chapters. Equations are used sparingly; tables and decision matrices carry the argument. Where interpolation or extrapolation is required to bridge gaps in the experimental grid, the reasoning is stated explicitly and the uncertainty is quantified or bounded.

6.2 Ranking-preservation map

6.2.1 What "ranking preservation" means

The central deployment question for this thesis is not "what is the best possible accuracy?" but "does the training protocol that works in simulation still work when the physical instance changes?" In the language of Chapter 2, the ranking of interest is

$$\text{Ensemble HAT} \succ \text{fixed-mask HAT} \succ \text{naive (no-HAT) deployment}, \tag{6.1}$$

where the comparison is evaluated under the fresh-instance protocol: 10 independent hardware realizations × 5 Monte Carlo forward passes per realization. A parameter combination *preserves the ranking* if Ensemble HAT maintains a decisive accuracy advantage over fixed-mask HAT, which in turn outperforms naive deployment, on unseen arrays. The ranking is *preserved with degradation* if the ordering holds but the margin between Ensemble HAT and the digital baseline shrinks. It is *collapsed* if fixed-mask HAT equals or exceeds Ensemble HAT on fresh instances, or if any protocol drops to near-chance accuracy.

This definition is stricter than source-domain comparison. Under evaluation on the training-time array, fixed-mask HAT can reach $87.95 \pm 0.27\,\%$ (the canonical V4 result), which is competitive with Ensemble HAT's held-out exploratory scan ($88.41\,\%$ in the 50-epoch ablation of Chapter 2). But source-domain accuracy is diagnostically irrelevant for deployment because the deployed array is not the training array. The ranking of Eq. 6.1 is therefore always understood as a fresh-instance statement unless noted otherwise.

6.2.2 Parameter axes and the available data grid

The simulator exposes five physical axes that jointly determine deployment accuracy:

Table 6.1: Green zone: parameter combinations with direct fresh-instance evidence that Ensemble HAT > fixed-mask > naive. All accuracies are CIFAR-10 top-1 under the 10×5 fresh-instance protocol unless otherwise noted.

Condition	Key parameters	Ensemble HAT	Fixed-mask
Canonical uniform noise	$\sigma_{\mathrm{D2D}} = 10\,\%$, i.i.d., $NL = 1.0$	86.37 ± 1.54	10.00 ± 0.00
Low-noise literature proxy	$\sigma_{\mathrm{D2D}} = 3\,\%$, $\sigma_{\mathrm{C2C}} = 2\,\%$ (OPECT)	88.53 ± 0.08	≈ 10 (cross-profile)
Moderate spatial correlation	$\rho = 0.3$, $NL = 1.0$	84.57 ± 2.39	10.00 (extrap.)
Strong spatial correlation	$\rho = 0.5$, $NL = 1.0$	82.12 ± 3.95	10.00 (extrap.)

1. **D2D variability amplitude** (σ_{D2D}): canonical 10 %, OPECT proxy 3 %, sweep grid $\{1, 3, 5, 8, 10, 15, 20\}$%.
2. **D2D spatial structure**: i.i.d. Gaussian versus separable AR(1) correlation with $\rho \in \{0.3, 0.5\}$.
3. **ADC resolution**: sweep grid $\{2, 3, 4, 5, 6, 7, 8, 10, 12\}$ bits.
4. **Write nonlinearity severity** (NL): ideal 1.0, severe 2.0.
5. **Retention age** (t): 0 s (freshly programmed) to 10,000 s under double-exponential drift.

Not every combination in this $7 \times 3 \times 9 \times 2 \times 5$ grid has been evaluated. The canonical Ensemble HAT checkpoint was trained under ($\sigma_{\mathrm{D2D}} = 10\,\%$, i.i.d., 4-bit programming, $NL = 1.0$, $t = 0$) and then transferred across subsets of the remaining axes. The ranking-preservation map is therefore a *qualitative mosaic* in which some tiles are anchored by direct measurement and others are filled by interpolation or bounded extrapolation.

6.2.3 Green zone: high-confidence ranking preservation

Table 6.1 lists the parameter combinations for which direct fresh-instance data exist and the ranking of Eq. 6.1 is preserved with high confidence.

The OPECT case is particularly instructive. The standard-HAT (fixed-mask) checkpoint collapses to chance level (10.00 %) when transferred to the OPECT profile without retraining, whereas the Ensemble HAT checkpoint reaches $88.53 \pm 0.08\,\%$. This confirms that the ranking is not an artifact of the canonical 10 % D2D assumption; it persists under a lower-variance literature-calibrated profile. The OPECT result also establishes an *accuracy ceiling* for the present Tiny-ViT architecture under near-ideal analog conditions: roughly 88.5 %, which is still $\sim$9.5 pp below the digital FP32 baseline (98.06 %) but $\sim$14 pp above the standard-HAT collapse.

Spatially correlated D2D produces *bounded degradation* rather than collapse. At $\rho = 0.3$, the Ensemble HAT mean drops by only 1.76 pp relative to the i.i.d. baseline (86.33 % versus 84.57 %); at $\rho = 0.5$, the drop is 4.20 pp. Rank ordering is preserved across all tested levels, and even the worst fresh instance at $\rho = 0.5$ reaches 73.7 %, far from chance. The fixed-mask column in Table 6.1 for correlated D2D is marked as extrapolated because the correlated-D2D experiment evaluated only the Ensemble HAT checkpoint, but there is no physical mechanism by which spatial correlation would *restore* fixed-mask transfer. The 10.00 % collapse is therefore a conservative lower bound.

6.2.4 Yellow zone: ranking preserved with caveats

The yellow zone contains combinations where the ranking likely holds but the margin between Ensemble HAT and deployment viability narrows, or where direct fresh-instance data are missing for one of the three protocols in Eq. 6.1.

Retention drift with dynamic scale recalibration. The V8 retention trajectory shows a rapid early drop from 91.63 % ($t = 0$) to 82.66 % ($t = 1$ s), followed by a broad plateau at 79.13–79.51 % from 10 s to 10,000 s. These numbers are source-domain accuracies on the training-time instance with recalibration enabled. No explicit fresh-instance retention sweep has been run, so the direct ranking evidence is absent. However, the retention mechanism is a *global* conductance scaling that is absorbed by the layer-wise scale-recovery factor s_ℓ (Eq. 2.1). Because the D2D buffers co-decay with the programmed weights, the *relative* spatial mismatch pattern that Ensemble HAT learned to tolerate remains structurally intact. Extrapolating from this physical reasoning, one expects the ranking to persist but with an absolute accuracy floor near 79 % rather than 86 %. The uncertainty on this extrapolation is ±3–5 pp, derived from the width of the plateau and the absence of joint retention–fresh-instance statistics.

Proportional-noise regime with matched training. Proportional-noise HAT recovers Tiny-ViT to 97.37 ± 0.05 % when training and evaluation are both conducted under the proportional law (Chapter 2, Table 2.2). This is a source-domain result, not a fresh-instance result. The cross-regime matrix shows that a uniform-noise checkpoint collapses to 10.00 % under proportional evaluation, and vice versa, indicating that regime-matched training is a *necessary* condition for recovery. Assuming that the same epoch-level D2D resampling protocol generalizes to proportional-noise instances (there is no structural reason it should not), the ranking is expected to hold with a source-domain mean near 97 % and a fresh-instance mean plausibly in the low-90 % range. The uncertainty is large (±5–8 pp) because the fresh-instance protocol has not been executed under proportional noise.

Inverse-gamma front-end compensation. At $\gamma_{\text{phys}} = 2.0$, inverse-gamma compensation restores +5.8 pp relative to the raw photoresponse (89.85 % versus 84.04 % in the deterministic ResNet-18 evaluation). The ViT pathway is structurally more exposed to front-end distortion because global attention lacks the spatial averaging of CNNs. For Tiny-ViT, the compensation benefit is therefore expected to meet or exceed the +5.8 pp ResNet benchmark, but the exact ViT fresh-instance number has not been measured. The ranking is preserved because both HAT and non-HAT baselines benefit from linearization; the relative gap may even widen if the non-HAT

Table 6.2: Qualitative ranking-preservation matrix. Colors: **G**=green (direct fresh-instance evidence), **Y**=yellow (reasonable extrapolation), **O**=orange (marginal viability), **R**=red (collapse). The matrix is conditional on Tiny-ViT, CIFAR-10, and the canonical 100-epoch Ensemble HAT recipe.

	$NL = 1.0$, **ideal write**			$NL = 2.0$, **severe write**	
Condition	ADC$\geq$6	ADC=5	ADC<5	Fresh inst.	Source only
Low D2D ($\leq 3\,\%$)	**G**	**Y**	**R**	**O** (extrap.)	**O**
Canonical D2D ($10\,\%$)	**G**	**O**	**R**	**R** (extrap.)	**O**
High D2D (15–20 %)	**Y**	**O**	**R**	**R** (extrap.)	**R** (extrap.)
$\rho = 0.3$ correlated	**G**	**Y**	**R**	**R** (extrap.)	**O** (extrap.)
$\rho = 0.5$ correlated	**G**	**Y**	**R**	**R** (extrap.)	**O** (extrap.)
Retention $t > 10$ s + recal.	**Y**	**Y**	**R**	**Y** (extrap.)	**Y**

baseline is more sensitive to the uncompensated distortion.

6.2.5 Orange zone: uncertain or marginally preserved

Severe nonlinearity with revised gradient-scaling recipe. Under global $NL = 2.0$, the M-series evidence shows that Standard HAT reaches approximately 81.2 % and Ensemble HAT reaches approximately 80.7 % on fresh instances, with proportional noise performing indistinguishably from uniform noise. The cross-instance standard deviation is tight ($<$ 2 pp), indicating systematic rather than catastrophic degradation. The orange-zone classification reflects the fact that the $\sim$80–82 $\%bandisdeployment\text{-}relevantforresearch - gradeprototypesbutfallsshortofthe86.37\,\%canonicalbaselinebyroughly5.7\,pp.Thegapisajointeffectof$ $instancetransfer, notofinstanceoverfittingalone.$

ADC at the 5-to-6-bit boundary. The iso-accuracy sweep reveals a sharp cliff: below 5-bit ADC, accuracy collapses to near-chance across all D2D levels; the 5-to-6-bit transition yields a $\sim$7 pp jump. At exactly 6 bits, Ensemble HAT recovers to $> 84\,\%$ for $\sigma_{\mathrm{D2D}} \leq 10\,\%$, but at 5 bits the mean is in the 30–50 % range depending on D2D. The ranking is therefore *marginally* preserved at 6 bits (Ensemble HAT still > fixed-mask) but already lost at 5 bits, where all protocols cluster near chance. Interpolating between these grid points suggests a critical threshold near 5.5 bits, with an uncertainty of ± 0.5 bits driven by the coarse ADC grid.

6.2.6 Summary of the map

Table 6.2 condenses the four zones into a single qualitative matrix. The color coding is: **G**reen (direct evidence), **Y**ellow (plausible extrapolation), **O**range (marginal or technically preserved but not viable), **R**ed (collapsed).

The most important takeaway from the matrix is its *asymmetry*. The ranking is robust to moderate increases in D2D amplitude and to the introduction of spatial correlation, but it is fragile to reductions in ADC precision and to severe write nonlinearity. A program manager should therefore prioritize peripheral readout precision and programming linearity over incremental improvements in device-to-device uniformity, at least within the ranges tested here.

6.3 Collapse zone

The complement of the ranking-preservation map is the *collapse zone*: the set of parameter combinations that destroy the accuracy ordering of Eq. 6.1 or drive all protocols to near-chance performance. Identifying the collapse zone is as important as identifying the safe zone, because it tells a program manager which physical specifications are *non-negotiable* and which are merely performance-negotiable.

6.3.1 The ADC cliff: a hard boundary

The most unambiguous collapse boundary in the data is the ADC precision cliff. The joint D2D–ADC sweep (manuscript iso-accuracy section) shows that below 5-bit ADC, accuracy is near-chance ($\sim$10 %) regardless of D2D level. The 5-to-6-bit transition produces a consistent $\sim$7 pp jump across all seven tested σ_{D2D} values. Above 6 bits, degradation becomes D2D-dominated. This is a *hard* boundary in the sense that no training protocol tested to date—fixed-mask, Ensemble, or proportional-noise HAT—can recover from sub-5-bit readout.

The physical reason is scale masking. Under the canonical recovered-weight law (Eq. 2.1), the effective weight grid step is

$$\Delta_{\hat{W}} = s_\ell \, \frac{G_{\max} - G_{\min}}{n_{\mathrm{states}} - 1}. \tag{6.2}$$

When the ADC discretizes the column current into fewer than ≈ 32 codes (5 bits), the quantization noise swamps the signal even before D2D or cycle-to-cycle (C2C) variability is applied. The crossbar's analog compute advantage—massive parallelism in the current domain—is negated by a readout that cannot resolve the result. From a deployment perspective, this means that 6-bit ADC resolution is a *minimum viable peripheral* for the present hybrid architecture; anything lower is non-starter regardless of how sophisticated the training becomes.

6.3.2 Hardware-instance overfitting: the fixed-mask collapse

The second unambiguous collapse is the fresh-instance failure of fixed-mask HAT. Under canonical uniform noise, the standard V4 checkpoint reaches 87.95 ± 0.27 % on its training-time array but collapses to 10.00 ± 0.00 % on every one of 10 fresh instances. This is not a noisy dispersion around chance; it is a deterministic single-class predictor that emerges from the optimizer's coupling to a specific spatial mismatch pattern (Chapter).

The collapse is *absolute* in the sense that it does not improve if the D2D amplitude is reduced: the OPECT case study shows that standard HAT checkpoints also collapse to 10.00 % on the low-noise OPECT profile. It is also absolute with respect to correlation: there is no reason to expect that spatial correlation would restore fixed-mask transfer, because the mechanism—memorization of a specific M_0—is independent of the correlation structure of $p(M)$. Fixed-mask HAT is therefore *not a deployment candidate* for any organic RRAM process that requires fresh-instance robustness, regardless of how well-controlled the fabrication is.

6.3.3 Severe nonlinearity: the ∼80–82 % recovery band

The third degradation boundary is severe write nonlinearity ($NL = 2.0$). With the revised gradient-scaling recipe, the M-series evidence shows a consistent recovery band across all tested training configurations:

- Standard HAT (fixed mask): 81.12 ± 1.06 % mean across two seeds;
- Ensemble HAT (uniform noise, epoch resampling): 80.72 ± 0.46 % mean;
- Ensemble HAT (proportional noise): 80.66 ± 0.01 % mean.

The convergence of all six runs to the same ∼80–82 $\%bandisthedefiningsignatureofthesevere-$ $NLregime.Unlikethe$NL=1.0$case, whereEnsembleHAToutperformsStandardHATby$76 pp$, thetwopr$ $levelresamplingismaskedoncegradient - scalingdistortionislargeenough.$

The residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ Ensemble HAT baseline (86.37 %) is a joint effect of nonlinear write dynamics and fresh-instance transfer. The cross-instance standard deviation remains tight (< 2 pp for all routes), confirming that the degradation is systematic rather than catastrophic. Post-module-output hook diagnostic 8-bit ADC quantization reduces the headline by only −0.10 pp on average, while 6-bit ADC introduces a larger −2.8 pp cliff consistent with the iso-accuracy analysis. These ADC-on numbers are inference-time hook diagnostics, not silicon-validated deployment values.

Extrapolating from the $NL = 2.0$ data, the relationship between nonlinearity severity and fresh-instance accuracy is strongly nonlinear. A linear interpolation between $NL = 1.0$ (86.37 %) and $NL = 2.0$ (∼80.7 %) would predict a 83.5 % crossing at $NL = 1.5$, but there is no evidence for linearity; the gradient-scaling surrogate becomes increasingly ill-conditioned as NL grows, and higher-order corrections may be needed in the $1.5 < NL < 2.0$ regime. The conservative design rule is that *device-level linearisation to* $NL \approx 1.0$ *is preferred for deployment-grade ViT inference*, while the ∼80–82 $\%bandat$NL=2.0$representsaresearch - gradelowerbound.$

Table 6.3: Collapse zone: qualitative danger-zone specification. Entries marked "extrap." are not directly measured but are inferred from the monotonic trends of the individual sweeps.

Trigger combination	Expected outcome	Confidence
ADC $<$ 5 bits, any D2D/NL	Near-chance ($\sim$10 %)	High (direct)
Fixed-mask HAT, any fresh instance	10.00 % collapse	High (direct)
$NL \geq 2.0$, fresh instance	$\leq$ 32 % ceiling	High (direct)
$NL \geq 2.0 + \rho \geq 0.5$ + retention $>$ 1 week	Likely $<$ 25 %	Medium (extrap.)
$NL \geq 2.5$, any fresh instance	Likely $<$ 20 %	Medium (extrap.)
ADC = 5 bits + high D2D ($>$ 15 %)	30–50 % (non-viable)	Medium (interpol.)
No dynamic recalibration + retention $>$ 10 s	Accuracy drifts unbounded	Medium (extrap.)

6.3.4 Joint stress: the extrapolated danger zone

No experiment in the present dataset evaluates the *joint* application of severe NL, correlated D2D, and retention drift. The interaction frontier is therefore mapped by extrapolation. Table 6.3 assembles the individual collapse boundaries into a qualitative danger-zone specification.

The combination $NL > 2.5 + \rho > 0.5$ + retention $>$ 1 week is highlighted as the archetypal "danger zone." None of these three conditions alone collapses Ensemble HAT to chance: $NL = 2.0$ leaves 32 %, $\rho = 0.5$ leaves 82 %, and retention with recalibration leaves 79 %. But their joint effect is expected to be super-additive. Severe NL already destabilizes the MLP gradient path; correlated D2D widens the variance budget from 1.54 pp to 3.95 pp; and retention drift without recalibration removes the scale-masking protection that absorbs global conductance shifts. The conservative extrapolation is that the joint accuracy would fall below 25 %, with a plausible lower bound near 15 %. This extrapolation is flagged as medium confidence because the interaction terms have not been measured.

6.4 Decision diagram: should this architecture ship?

The ranking-preservation map and the collapse zone can be combined into a prose decision diagram. The branching criterion is *device maturity tier*, a qualitative label that maps onto the physical parameter ranges tested in the simulator.

6.4.1 Research-grade arrays

Definition. Research-grade arrays are characterized by: $\sigma_{\mathrm{D2D}} \geq 10\,\%$ (often 15–20 %), write nonlinearity $NL \geq 1.5$, ADC resolution $\leq$ 5 bits or uncalibrated, and retention time constants in the hundreds of milliseconds without on-chip recalibration circuitry.

Verdict: do not ship a ViT. At this maturity level, the collapse-zone boundary is crossed for at least one non-negotiable parameter. Sub-5-bit ADC produces near-chance accuracy regardless of training protocol. Severe NL drives fresh-instance accuracy below 32 % even with MLP linearization. The retention plateau near 79 % is only reachable if dynamic scale recalibration is available, which typically requires reference-cell readout circuitry that research-grade prototypes lack.

Alternative actions. (i) Use the array as a *training-diagnostic platform* rather than as a deployment target: train Ensemble HAT checkpoints, evaluate them on the research-grade array, and use the gap between source-domain and fresh-instance accuracy to guide process development. (ii) If a simpler task or smaller model is required, consider a shallow CNN (Section 6.5), although fresh-instance CNN data under Ensemble HAT are not available in the present dataset. (iii) Invest in peripheral readout: improving the ADC from 5 to 6 bits is the single highest-leverage intervention at this tier, because it moves the system from the red zone to the yellow zone of Table 6.2.

6.4.2 Pilot-line arrays

Definition. Pilot-line arrays are characterized by: $\sigma_{\mathrm{D2D}} \in \{3\,\%, 5\,\%, 8\,\%\}$, $NL \in \{1.0, 1.2\}$ (ideal to mildly nonlinear), 6–8-bit calibrated ADC, and basic on-chip reference cells that permit intermittent scale recalibration.

Verdict: ship with Ensemble HAT + compensation. At this tier, the canonical ranking is preserved with margins that are commercially relevant. The OPECT proxy (3 % D2D, 2 % C2C) reaches $88.53 \pm 0.08\,\%$, which is only ~9.5 pp below the digital baseline. The canonical 10 % D2D regime reaches $86.37 \pm 1.54\,\%$, which is viable for many edge-classification tasks. The 6-bit ADC cliff has been cleared, and the retention plateau near 79 % is accessible provided recalibration is performed at intervals shorter than ~1 s if the application demands > 82 % accuracy.

Deployment checklist. (1) Train with per-epoch D2D resampling (Ensemble HAT), not fixed-mask HAT. (2) Characterize the physical photoresponse exponent γ_{phys} and deploy inverse-gamma compensation if $\gamma_{\mathrm{phys}} > 1$; the expected gain is +5.8 pp at $\gamma_{\mathrm{phys}} = 2.0$. (3) Include dynamic scale recalibration in the inference stack; budget peripheral energy for reference-cell readout. (4) Validate fresh-instance robustness on at least 10 physical arrays before committing to mass programming. (5) If spatial correlation is suspected from wafer maps, treat the $\rho = 0.3$ result (84.57 %) as a conservative accuracy floor and the $\rho = 0.5$ result (82.12 %) as a stress-test bound.

6.4.3 Production-grade arrays

Definition. Production-grade arrays are characterized by: $\sigma_{\mathrm{D2D}} \leq 3\,\%$, $NL \approx 1.0$ (near-ideal write or iterative write-verify), $\geq$ 8-bit ADC, and stable retention with on-chip recalibration.

Verdict: ship, with HAT as insurance rather than rescue. At this tier, the analog noise floor is low enough that even fixed-mask HAT might achieve acceptable source-domain accuracy. However, the OPECT case study demonstrates that fixed-mask checkpoints still collapse when transferred across profiles, suggesting that the hardware-instance overfitting risk does not vanish automatically at low noise. Ensemble HAT should therefore remain the default training protocol, but its role shifts from "rescue" to "insurance." The expected fresh-instance accuracy is in the high-80 % to low-90 % range, with a tight variance (± 1 pp or better) that simplifies yield planning.

Yield and cost implications. At $88.53 \pm 0.08\,\%$, the cross-instance standard deviation is smaller than typical test-set sampling noise for CIFAR-10. This means that yield can be planned on the *mean* accuracy rather than on a worst-case tail. If the application threshold is 85 %, virtually every instance passes; if the threshold is 87 %, the yield is still $> 99\,\%$ under a Gaussian tail assumption. The energy estimate ($\approx 11\times$ dense-projection reduction versus FP32, manuscript energy-results section) becomes credible at this tier because the placeholder constants in the energy model are most accurate when the peripheral overhead is dominated by calibrated digital circuits rather than by speculative analog compensation.

6.4.4 A compact flowchart

For quick reference, the decision logic can be summarized as a branching tree:

1. **ADC ≥ 6 bits?**
   - No $\rightarrow$ DO NOT SHIP. Invest in peripheral readout.
   - Yes $\rightarrow$ continue.
2. **Write nonlinearity $NL < 1.5$?**
   - No $\rightarrow$ DO NOT SHIP VIT. Consider simpler model or process repair.
   - Yes $\rightarrow$ continue.
3. **Device maturity tier?**
   - Research-grade (high D2D, no recalibration) $\rightarrow$ DIAGNOSTIC USE ONLY.
   - Pilot (moderate D2D, recalibration available) $\rightarrow$ SHIP WITH ENSEMBLE HAT + COMPENSATION.
   - Production (low D2D, ideal write, 8+ bits) $\rightarrow$ SHIP WITH ENSEMBLE HAT AS INSURANCE.

6.5 CNN versus ViT at different maturity tiers

The deployment envelope developed above is anchored to Tiny-ViT-5M, a compact Vision Transformer fine-tuned from ImageNet. A natural question arises: at low device maturity, where noise and nonlinearity are most severe, would a simpler convolutional architecture outperform the transformer under the same HAT protocol? This section addresses the question with the evidence available and flags the gaps.

6.5.1 What the data show

The manuscript reports three architecture–accuracy points under matched digital training: ResNet-18 (95.46 %), ConvNeXt-Tiny (90.74 %), and Tiny-ViT-5M (98.06 %) on CIFAR-10. Under the canonical analog noise model with standard HAT, the ranking is preserved: Tiny-ViT V4 reaches 87.95 ± 0.27 %, while ConvNeXt C4 reaches a comparable level (manuscript quantization-noise section). However, *no fresh-instance Ensemble HAT sweep has been performed for ConvNeXt or ResNet.* The CNN numbers in the manuscript are either source-domain or single-run estimates, and the 10×5 fresh-instance protocol that underpins the 86.37 ± 1.54 % Ensemble HAT claim has not been applied to a CNN testbed.

The manuscript Discussion section notes that "the Transformer architecture is more fragile under the present physical stress tests," citing the proportional-noise collapse and the front-end distortion sensitivity as evidence. In a CNN, local convolution and pooling average perturbations spatially, and ReLU saturation limits the propagation of intensity-dependent scaling errors. In a ViT, patch embeddings are fed directly into global self-attention: a sublinear photoresponse alters the token embedding, and the softmax exponentiates similarity changes, redistributing attention weights across the entire sequence without spatial averaging. This structural difference predicts that the ViT should degrade more sharply than a CNN under front-end distortion and under spatially correlated D2D.

6.5.2 Theoretical reasoning for low-maturity tiers

At the research-grade tier—high D2D, high NL, sub-5-bit ADC—neither architecture is deployable for high-accuracy vision. The ADC cliff is architecture-agnostic: near-chance accuracy is near-chance regardless of whether the model uses convolutions or self-attention. The question becomes interesting only at the boundary of viability, where one architecture might remain usable while the other collapses.

Consider the pilot tier with moderate D2D (5–10 %) and a 6-bit ADC. Here, the CNN's spatial averaging provides a built-in regularization against structured mismatch. A 3×3 convolution with stride 2 averages nine spatial locations before the nonlinearity, effectively low-pass filtering the D2D field. The ViT patch embedding is also a convolution (the patch-embedding layer is analog-mapped in the present framework), but the subsequent attention layers operate on token vectors that have *already been corrupted* by the full spatial mismatch pattern. There is no spatial averaging in the attention mechanism; a perturbed query-key dot product propagates globally through the softmax. One therefore expects the CNN to tolerate higher D2D or stronger spatial correlation at a given accuracy level, all else being equal.

Similarly, under severe write nonlinearity ($NL = 2.0$), the CNN's MLP blocks are structurally similar to the ViT's MLP blocks, but the CNN has fewer attention-side analog layers to protect. The dual-attention collapse documented in Chapter 3 is a *transformer-specific* failure mode; a pure CNN has no QKV or output-projection analog paths to linearize. If the severe-NL bottleneck is truly concentrated in the MLP path, a CNN might reach the MLP-only source-domain rescue ($\approx 88\,\%$) without the accompanying attention-side structural collapse, and its fresh-instance ceiling might therefore exceed the ViT's 32 % bound.

6.5.3 Framing the open question

These arguments are theoretical, not empirical. The absence of a fresh-instance CNN comparison is the most important data gap in the deployment envelope. A direct experiment would train ConvNeXt (or ResNet-18) with Ensemble HAT under the canonical uniform-noise profile, then evaluate on 10 fresh instances. If the CNN fresh-instance mean exceeds 86.37 %, the ViT is not the optimal architecture for organic CIM at this maturity level. If the CNN mean is comparable or lower, the ViT's superior digital baseline justifies its higher fragility.

Until such data exist, the conservative recommendation is:

- At **production maturity** (low noise, ideal write), prefer the ViT for its higher digital ceiling and its proven Ensemble HAT transfer (88.53 %).
- At **pilot maturity** (moderate noise, 6-bit ADC), the ViT is deployable with Ensemble HAT, but a CNN may offer a wider process-margin window if D2D or correlation is worse than expected.
- At **research maturity** (severe NL, sub-5-bit ADC), architecture choice is moot; the process is not ready for either.

This framing aligns with the broader CIM literature. Recent heterogeneous analog accelerators for Vision Transformers, including HARDSEA[5], adopt the same hybrid analog–digital partition used in this thesis, but they target more mature CMOS-under-ReRAM processes where device nonlinearity is lower. For organic arrays, the maturity gap suggests that CNN-first deployment may be a pragmatic stepping stone toward eventual ViT deployment.

6.6 Implications for analog-CIM program managers

The deployment envelope is not merely an academic classification; it is a tool for resource allocation. This section translates the findings into industrial-relevance bullets for program managers deciding how to invest non-recurring engineering (NRE) dollars across device fabrication, peripheral design, and training infrastructure.

6.6.1 Yield tolerance and the D2D budget

The iso-accuracy sweep shows that, once the ADC is ≥ 6 bits, D2D variability accounts for 92.2 % of residual accuracy variance in the operational envelope. This means that yield is D2D-limited, not C2C-limited. Cycle-to-cycle noise is largely masked by the digital recovery path; a 5 % C2C term adds negligible effective error because many perturbations fall below the half-LSB of the recovered weight grid. Program managers should therefore prioritize *spatial uniformity* (wafer-scale process control, optical write uniformity, thermal management during programming) over *temporal stability* (C2C repeatability) within the tested ranges.

The correlated-D2D data provide a quantitative yield model. At $\rho = 0$ (i.i.d.), the cross-instance standard deviation is 1.54 pp; at $\rho = 0.3$, it grows to 2.39 pp; at $\rho = 0.5$, it reaches 3.95 pp. If the application requires $> 80\,\%$ accuracy, the i.i.d. and $\rho = 0.3$ regimes are safe (mean $> 84\,\%$, worst instance $> 73\,\%$), but the $\rho = 0.5$ regime introduces a tail risk: one instance in ten could drop below 75 %. A program manager should measure the spatial covariance kernel of the pilot line and reject wafers with $\rho > 0.4$ unless architectural mitigations (correlated-HAT training or spatial calibration grids) are implemented.

6.6.2 NRE savings from Ensemble HAT

Standard HAT requires per-device calibration: a model trained on one array must be retrained or at least recalibrated for each new instance. Ensemble HAT eliminates this requirement by baking the instance distribution into the training objective. The NRE saving is the avoided cost of N retraining runs for N deployed arrays. At 100 epochs per run and ~2 hours per epoch on a single GPU (the Tiny-ViT training budget), per-device retraining would cost ~200 GPU-hours per array. For a pilot line with 100 test arrays, Ensemble HAT saves ~20,000 GPU-hours, or roughly \$5–10k in cloud compute at 2025 rates. At production scale, the saving is larger still.

There is a training-time overhead for Ensemble HAT: one D2D map resampling per epoch adds negligible wall-clock time (the map generation is CPU-bound and pipelined), but the effective regularization strength changes, so the learning-rate schedule may need retuning. In the present experiments, the same cosine annealing schedule was used for fixed-mask and Ensemble HAT without hyperparameter search, suggesting that the schedule transfer is robust. The NRE saving therefore dominates the modest training overhead.

6.6.3 Retraining frequency and the retention refresh interval

The retention plateau at 79.13–79.51 % is stable across three orders of magnitude in time (10 s to 10,000 s), but it is 12.5 pp below the $t = 0$ baseline. For applications that require $> 82\,\%$ accuracy, the refresh interval must be shorter than $\sim$1 s; for applications that accept 79 %, the interval can extend to hours. This creates a *task-dependent refresh specification*:

- High-accuracy edge inference (e.g., medical imaging, industrial inspection): refresh every < 1 s, or avoid analog storage altogether for the critical layers.
- Moderate-accuracy always-on sensing (e.g., gesture detection, occupancy sensing): refresh every 10–100 s; the 79 % plateau is sufficient.
- Ultra-low-power intermittent inference: accept 79 % and recalibrate only at wake-up, using the dynamic scale recalibration path.

Program managers should negotiate the refresh interval with the application team *before* finalizing the array size, because the peripheral energy for reference-cell readout scales with the number of analog layers being recalibrated. The manuscript's 11.45× energy-reduction estimate does not include this peripheral overhead; a realistic system model would subtract 5–15 % from the savings depending on refresh frequency.

6.6.4 Energy budget and analog–digital trade-off

The analytical energy model estimates a total system energy of approximately 23.9 μJ per inference under the baseline assumption (8-bit ADC, 100 arrays, 1000 conversions), versus a digital FP32 baseline of approximately 273.94 μJ. The implied speed-up is 11.45× versus FP32 and approximately 2.86× versus an assumed INT8 implementation (see `energy_scale_recovery_sensitivity.json`). These are first-order analytical estimates, not silicon measurements; actual deployment must recalibrate against the specific CMOS process node.

The energy advantage is most credible at the production-maturity tier, where peripheral circuits are dominated by calibrated digital logic rather than speculative analogue compensation. At the research tier, uncalibrated ADCs and iterative write–verify cycles can erase much of the analogue MAC savings. Program managers should therefore treat the 11.45× figure as an *upper bound* that is only realisable once the peripheral overhead is disciplined.

6.6.5 Risk mitigation: the "compound stress" sanity check

The manuscript Discussion section reports a single-run compound stress test combining 2 % C2C, 3 % D2D, 6-bit ADC, 1 % IR drop, 1 % sneak path, and 1,000 s retention, yielding 89.61 % accuracy for Tiny-ViT. This is not a locked number and should not be treated as a guarantee, but it provides a *sanity check*: when all moderate non-idealities are present simultaneously, the system does not collapse catastrophically. Program managers can use this as a "smoke test" for their own integration plans: if the compound stress prediction for a target profile is below 80 %, the design is too aggressive and should be de-featured before tape-out.

6.7 Limitations of the envelope

Every deployment recommendation in this chapter is conditional on the first-order behavioral approximations described in Chapter 1. The envelope is a *decision aid*, not a predictive emulator. This section lists the phenomena that are *not* in the model and therefore not in the recommendation, together with their expected impact on the safe-zone boundaries.

6.7.1 Spatial IR drop

The framework includes a 1 % scalar placeholder for IR drop (manuscript limitations section), but it does not model the *spatial* voltage decay along wordlines and bitlines. Real IR drop is geometry-dependent: long wordlines in large arrays experience larger voltage droop at the far end, producing a systematic *position-dependent* weight distortion that is correlated with the array layout. This distortion is structurally different from the stochastic D2D mismatch modeled here, because it is deterministic for a given input pattern and array geometry. A program manager designing a 512×512 array should not rely on the 10 % D2D safe zone without verifying that the IR drop at the array corner does not introduce an effective $\sigma_{\text{eff}} > 15\,\%$. CrossSim[1] provides SPICE-calibrated parasitic models that can bracket this effect for a specific metal stack.

6.7.2 Sneak-path currents

Sneak paths in passive crossbar arrays introduce effective conductance crosstalk that depends on the programmed weight pattern and the selector device characteristics. The framework includes a 1 % scalar placeholder, but real sneak-path distortion can be pattern-dependent and non-Gaussian. In the worst case, a high-conductance cell can create a low-resistance path that shunts current away from the intended bitline, producing a *multiplicative* error rather than the additive error modeled by D2D and C2C. Multiplicative errors break the scale-recovery assumption of Eq. 2.1 and could shift the 6-bit ADC cliff upward. Until selector devices or one-transistor-one-resistor (1T1R) cells are characterized, the sneak-path safe zone is unknown.

6.7.3 Array geometry and parasitic resistance

The present simulator treats each analog layer as an independent crossbar tile with no inter-tile coupling. Real chips have routing resistance between tiles, capacitive loading on bitlines, and floor-plan-dependent thermal gradients. These effects are absent from the profile interface. The OPECT case study assumes that the 3 % D2D and 2 % C2C proxies are sufficient to capture all static variability; if the array geometry introduces additional position-dependent spread, the 88.53 % prediction is optimistic.

6.7.4 Full temperature sweep

Temperature affects every physical parameter in the model: conductance window, mismatch variance, retention time constants, and photoresponse exponent. The present framework assumes room-temperature operation ($\approx$ 300 K) and does not include Arrhenius-style temperature acceleration. Organic semiconductors are particularly temperature-sensitive because carrier mobility and trap-state occupancy vary strongly with thermal energy. A deployment envelope that is valid at 25°C may not be valid at 60°C, and the 79 % retention plateau could shift downward significantly if the slow decay mode accelerates. A full temperature sweep is the highest-priority extension for outdoor or industrial deployments.

6.7.5 The interaction frontier

All failure modes in Chapter 3 were evaluated in isolation. The real deployment environment presents D2D mismatch, C2C noise, retention drift, front-end distortion, and finite ADC resolution *simultaneously*. The compound stress test (89.61 %) is a single data point in this joint space; it does not map the interaction surface. In particular:

- Does retention drift for 1,000 s *increase* the susceptibility of the D2D field to hardware-instance overfitting, or does the co-decay of mismatch buffers preserve the Ensemble HAT guarantee?
- Does the 32 % MLP-only fresh-instance ceiling rise or fall when correlated D2D is added to the severe-NL regime?
- Does the 6-bit ADC cliff shift when the input has already been distorted by a sublinear photoresponse ($\gamma_{\text{phys}} > 1$)?

These questions are not answerable within the current experimental budget, but they define the boundary beyond which the deployment envelope should not be extrapolated without additional data.

6.7.6 First-order surrogate ceiling

Finally, the $NL = 2.0$ boundary is a property of the gradient-scaling surrogate, not necessarily of the physical device. If real organic write dynamics follow a pulse-faithful ionic-migration model that differs from the present $(G_{\max} - G)^{NL-1}$ approximation, the dual-attention collapse may occur at a different severity, or not at all. The deployment envelope should therefore be recalibrated whenever measured write-characterization data become available. The profile-driven interface (Chapter 1) is designed for exactly this recalibration: replacing the surrogate with a lookup-table or physics-based model requires only a profile update, not a code rewrite.

6.8 Summary

This chapter has converted the simulation data of the preceding chapters into an actionable deployment envelope for organic optoelectronic CIM. The central construct is the *ranking-preservation map* (Section 6.2), which classifies parameter combinations according to whether the canonical accuracy ordering—Ensemble HAT > fixed-mask HAT > naive deployment—survives on unseen hardware instances. Direct fresh-instance evidence exists for the canonical uniform-noise regime ($86.37 \pm 1.54\,\%$), the low-noise OPECT proxy ($88.53 \pm 0.08\,\%$), and spatially correlated D2D up to $\rho = 0.5$ ($82.12 \pm 3.95\,\%$). Extrapolation with explicit uncertainty bounds extends the map to retention drift ($\approx 79\,\%$ plateau), proportional-noise regimes (plausible low-90 % fresh-instance mean), and the ADC 5-to-6-bit boundary.

The *collapse zone* (Section 6.3) identifies non-negotiable boundaries: ADC < 5 bits produces near-chance accuracy regardless of training; fixed-mask HAT collapses to 10.00 % on every fresh instance; severe write nonlinearity ($NL = 2.0$) creates a 32 % fresh-instance ceiling even with MLP linearization. The joint danger zone $NL > 2.5$ + $\rho > 0.5$ + retention > 1 week is extrapolated to fall below 25 %, with medium confidence.

The *decision diagram* (Section 6.4) branches on device maturity tier. Research-grade arrays should be restricted to diagnostic use; pilot-line arrays can ship with Ensemble HAT plus front-end compensation; production-grade arrays can treat Ensemble HAT as insurance rather than rescue. The CNN-versus-ViT comparison (Section 6.5) remains an open question for fresh-instance CNN data, but theoretical reasoning suggests that CNNs may tolerate higher D2D and NL at the pilot-tier boundary.

For *program managers* (Section 6.6), the key takeaways are: prioritize spatial uniformity over C2C repeatability; use Ensemble HAT to save per-device retraining NRE; negotiate the retention refresh interval with the application team before finalizing peripheral energy budgets; and treat the compound stress test as a smoke-test benchmark.

The *limitations* (Section 6.7) are substantial and explicitly enumerated: spatial IR drop, sneak-path currents, array geometry, full temperature sweep, and the unexplored interaction frontier. The envelope is valid only within the first-order behavioral approximations of the present framework, and it should be recalibrated when measured device data become available. Within those bounds, however, it provides a principled basis for deciding whether a given organic RRAM process is ready to host a modern vision backbone—and what training protocol it will need if it is.

The envelope is where the thesis ends and the next project begins.

Chapter 7

Outlook and Conclusion

7.1 Introduction: the thesis as a bridge

This thesis began with a simple observation. Organic optoelectronic compute-in-memory devices are invariably introduced through figures of merit—conductance window, retention time, write asymmetry, optical responsivity—that are necessary but not sufficient. They do not answer the question that ultimately matters for deployment: *what accuracy survives once a modern vision backbone is mapped to an analog array with structured mismatch, finite readout precision, and a sub-linear photoresponse?* The intervening chapters have built a bridge between the measurement laboratory and the algorithm training loop. Chapter 1 constructed the bridge's truss: a profile-driven simulation framework that turns device characterizations into structured training inputs. Chapter 2 mapped the traffic rules: a taxonomy of hardware-aware training cadences, noise profiles, and transfer guarantees. Chapter 3 surveyed the load limits: a catalogue of named, reproducible failure modes that bound the deployment envelope.

The preceding chapter constructed the deployment envelope; this chapter steps outside it to propose the next-generation theory and hardware roadmap.

This final chapter has two tasks. The first is to look forward. It identifies the experiments, architectures, hardware roadmaps, and theoretical tools that can extend the bridge beyond the scope of the present thesis. The second is to look backward, integrating the eight chapters into a single narrative arc with three core contributions: (1) `computeViT` as a reusable research instrument; (2) the HAT taxonomy and the discovery of hardware-instance overfitting; and (3) a quantified deployment envelope for organic CIM. The outlook is optimistic but grounded; the conclusion is definitive but open-ended—because a thesis, like the bridge it builds, is meant to be crossed.

7.2 Immediate next steps: thesis-completion experiments

Every thesis leaves a residue of unfinished experiments. The two described in this section are not afterthoughts; they are the natural completions of loops opened in Chapter 3. Both are framed as thesis-completion experiments rather than submission-path tasks, because their purpose is to test the *structural* claims of the thesis—that instance-overfitting and severe-write nonlinearity can be jointly mitigated, and that the ranking logic discovered on CIFAR-10 survives scale—rather than to generate new headline numbers for a manuscript.

7.2.1 Joint severe-NL and instance-overfitting mitigation

Chapter 3 established that severe nonlinear write ($NL = 2.0$) introduces a bounded degradation band of approximately ∼80–82% on fresh instances under the revised gradient-scaling recipe, irrespective of training protocol (Standard HAT, Ensemble HAT, or proportional noise). The residual gap of roughly 5.7 pp to the canonical $NL = 1.0$ baseline (86.37%) remains open. The central thesis-completion experiment is whether higher-order gradient corrections, larger-batch training, or device-level linearisation can close this gap while preserving the deployment-grade transfer that Ensemble HAT provides at $NL = 1.0$.

The ∼80–82% band is systematic rather than instance-variable: cross-instance standard deviation remains tight (< 2 pp) across all six M-series training routes. This suggests that the degradation is a *surrogate-fidelity* issue—the first-order NL approximation neglects higher-order curvature terms that become relevant when gradient asymmetry is large—rather than a training-protocol deficiency. The obvious question—and the central thesis-completion experiment—is whether higher-order STE corrections or full-differential gradient computation can recover the missing accuracy while preserving the instance-overfitting immunity that epoch-level D2D resampling provides.

The experimental design is as follows. Starting from the canonical Ensemble HAT checkpoint ($86.37 \pm 1.54\,\%$ fresh-instance mean), we apply a weights-only warm-start that resets optimizer state, scheduler state, and epoch counters, ensuring that the short NL-compensation run cannot skip training because of stale metadata. During the subsequent fine-tuning phase, the MLP blocks retain their linearized NL = 1.0 surrogate while the attention blocks are exposed to the full NL = 2.0 distortion, and—crucially—the D2D mismatch field is resampled at the start of every epoch. The primary target metric is fresh-instance accuracy under the 10×5 protocol. A target of $\geq 80\,\%$ would validate that the two largest failure modes—instance overfitting and severe write nonlinearity—are *jointly* mitigable. Anything below $50\,\%$ would confirm a fundamental incompatibility between first-order NL surrogates and distributional training.

An ablation matrix isolates the sources of any improvement:

1. **Ensemble HAT only** (NL = 1.0 global): the canonical baseline.
2. **Standard HAT severe-NL** (NL = 2.0, fixed mask): the ∼81% baseline.
3. **Joint warm-start**: Ensemble HAT checkpoint + MLP-linear protection + epoch resampling.
4. **Joint cold-start**: random initialization + MLP-linear protection + epoch resampling.

The warm-start versus cold-start comparison is diagnostically important. If warm-start outperforms cold-start, it suggests that the canonical Ensemble HAT solution provides a favourable basin into which the NL-compensated model can settle. If cold-start matches warm-start, the NL-compensated objective is sufficiently well-conditioned that pre-training on NL = 1.0 is unnecessary. Either outcome would refine the practical recipe for training analog-mapped transformers under severe write asymmetry.

This experiment is the thesis analogue of Route R-A in the post-submission publication plan: a deployment-grade result that couples surrogate-fidelity repair with deployment invariance. Whether it succeeds or fails, it closes the loop opened in Chapter 3.

7.2.2 ImageNet-100 pilot: scaling beyond CIFAR-10

All empirical anchors in this thesis—the $86.37 \pm 1.54\,\%$ Ensemble HAT recovery, the severe-NL ∼80–82% band (revised gradient-scaling recipe), the correlated-D2D bounded degradation ($86.33 \pm 1.61\,\%$ at $\rho = 0$, $84.57 \pm 2.39\,\%$ at $\rho = 0.3$, falling to $82.12 \pm 3.95\,\%$ at $\rho = 0.5$; zone 3A bug-immune canonical protocol), and the 79 % retention plateau—were measured on CIFAR-10 with Tiny-ViT. That choice was deliberate: it isolated the physics–algorithm interaction from the confounding effects of dataset scale and resolution. But it leaves a critical question unanswered. Are the ranking logic and the failure-mode boundaries a small-dataset curiosity, or do they survive the transition to larger vision tasks?

The recommended pilot, specified under the G-AA2 protocol, uses ImageNet-100—a curated 100-class subset of ImageNet-1k—rather than the full 1k dataset. The model remains the Tiny-ViT family, preserving the mixed-signal mapping logic (analog linear layers, digital control-heavy operators) that has been the constant across all preceding chapters. The training protocol begins with the canonical Ensemble HAT cadence: AdamW, cosine annealing, per-epoch D2D resampling, 4-bit differential conductance mapping. The only manipulated variable is dataset scale.

The pilot is designed around two decision points. First, if training instability appears before useful convergence, the resolution or model size should be reduced *before* the hardware protocol is changed. The goal is to test whether the *physics assumptions* scale, not to chase state-of-the-art accuracy. Second, if the 6-bit ADC cliff—a sharp accuracy drop when ADC precision falls below six bits—shifts materially upward or downward relative to the CIFAR-10 observation, that shift is recorded as the main pilot finding rather than forced back to the baseline value.

The success criterion is qualitative, not quantitative: the pilot succeeds if the *ranking* of mitigation strategies survives. Ensemble HAT should still outperform fixed-mask HAT; the severe-NL MLP-linear rescue should still underperform on fresh instances; correlated D2D should still produce bounded rather than catastrophic degradation. If these rankings hold, the deployment envelope derived in this thesis is a *scalable* design principle. If they collapse, the envelope is regime-specific and requires re-derivation at every scale.

This experiment is the natural scaling counterpart to the joint-training experiment of Section 7.2.1. Together, the two thesis-completion experiments answer the questions left open by the manuscript: *does the joint mitigation work?* and *does the ranking survive scale?*

7.3 Language-model CIM: transferring HAT principles beyond vision

Vision transformers were a proof of concept. The dominant transformer workload, measured in inference cost and commercial impact, is the large language model (LLM). Analog CIM offers the same theoretical energy advantage for LLMs—massive matrix–vector products in the MLP and attention layers—but the mapping is complicated by differences in scale, attention geometry, and memory pressure. This section asks how the HAT principles developed in this thesis would transfer to language-model deployment on organic optoelectronic arrays, and identifies the new physics–algorithm boundaries that would need to be mapped.

7.3.1 Scale: parameter count and crossbar tiling

Tiny-ViT has approximately five million parameters. A modest modern LLM has seven to seventy *billion*. Mapping such a model to analog crossbars requires tiling across hundreds or thousands of arrays, each with its own D2D mismatch field, scale-recovery factor, and ADC periphery. The dimensionality of the deployment distribution therefore explodes: instead of resampling a single global mismatch map, HAT would need to resample a *collection* of tile-level maps, possibly with across-tile process gradients that introduce long-range spatial correlation.

Instance-overfitting, the central discovery of Chapter , may behave differently under tiling. On the one hand, the sheer number of independent mismatch realizations could act as an implicit ensemble, averaging out local perturbations and reducing the variance of the fresh-instance accuracy. On the other hand, if each tile presents a distinct spatial signature, the model may overfit to the particular *configuration* of tile mismatches, producing a new form of "tile-configuration overfitting" that is just as catastrophic as the single-array collapse. Which regime dominates is an empirical question that can only be answered once large-scale tiled simulators exist.

A second scale effect concerns weight precision. The canonical experiments use 4-bit differential conductance mapping. For vision transformers on CIFAR-10, this is sufficient. For LLMs, the dynamic range of the MLP weight matrices is substantially larger, and aggressive quantization can degrade perplexity even in the digital baseline. Whether organic devices can support the higher bit-widths required for LLM deployment—or whether alternative mapping strategies such as weight grouping or vector-matrix tiling can compensate—is an open device–architecture co-design question.

7.3.2 Attention geometry: from spatial patches to temporal tokens

In vision transformers, self-attention operates over a fixed grid of image patches; the sequence length is constant at inference time, and the KV states for all patches are computed in parallel. In autoregressive language models, attention operates over a variable-length token sequence. Each new token requires the key and value activations of *all prior tokens* to be available. In a digital accelerator, these KV activations are stored in high-bandwidth SRAM; in an analog CIM system, they must either be stored in analog form on the crossbar itself or converted back to digital after every layer.

Analog KV storage introduces retention drift as a first-class inference constraint. Chapter 3 showed that weight retention drifts to a 79 % plateau under double-exponential decay. If KV activations are stored as conductance states, they would drift on a comparable timescale, distorting the attention scores for tokens generated more than a few hundred milliseconds after the KV write. This makes the retention problem *dynamic* rather than static: the accuracy penalty depends not only on the time since programming but on the temporal position of each token in the sequence.

Digital KV storage avoids drift but reintroduces the ADC/DAC energy and latency that analog CIM is meant to eliminate. The trade-off is stark: analog storage is energy-efficient but retention-limited; digital storage is retention-robust but bandwidth-limited. A hybrid strategy—analog storage for the most recent tokens, digital paging for older ones—may be optimal, but its design requires a joint model of attention sparsity, retention dynamics, and peripheral energy that does not yet exist.

7.3.3 Head parallelism and mismatch correlation

Multi-head attention offers natural parallelism: each head can be mapped to a separate crossbar array, allowing simultaneous computation of multiple attention subspaces. In the vision experiments, the number of heads is small (e.g., four or eight) and the mismatch per head is treated as part of the global D2D field. In LLMs, the number of heads is larger (thirty-two to ninety-six in current architectures), and the mismatch per head may introduce *structural* variation in the multi-head ensemble.

If one head suffers severe D2D mismatch while its neighbours are clean, the learned aggregation of head outputs may amplify the erroneous head. Ensemble HAT training would need to resample mismatch not just per epoch but per *head group*, ensuring that the model learns robust aggregation across head-level noise realizations. The optimal granularity—per head, per head-group, or per layer—is unknown and likely task-dependent. This is a direct analogue of the cadence question studied in Chapter 2: just as per-epoch resampling outperforms per-batch for the global mismatch field, there may be an optimal head-resampling cadence that balances diversity and optimization stability.

7.3.4 Toward a language-model HAT protocol

Despite these differences, the *core principle* of this thesis remains valid for language models: train over the deployment distribution of hardware instances, not a single instance. The practical challenge is cost. Training an LLM is already computationally expensive; adding D2D resampling at tile or head granularity multiplies the forward-pass overhead. Several efficiency strategies suggest themselves.

First, resample mismatch only for the analog-mapped layers, keeping the digital embedding, normalization, and softmax layers deterministic. This reduces the noise dimensionality without sacrificing the key physics. Second, use a *lightweight* surrogate for the mismatch field: instead of drawing full spatial maps, sample low-rank approximations that capture the dominant process gradients. Third, evaluate generalization not by classification accuracy but by *perplexity* on a held-out token sequence, measuring the fresh-instance protocol as a perplexity gap between training-time and unseen tile configurations.

This direction is flagged as Route R-B in the post-thesis publication plan: organic-RRAM language-model CIM scaling. It is not a short-term thesis experiment—the computational budget exceeds what is available—but it is the logical next target for the research instrument built in this thesis.

7.4 Hardware-in-the-loop roadmap: from simulation to silicon

Simulation is a powerful lens, but it is not the destination. The ultimate goal is to deploy HAT-trained models on physical organic optoelectronic arrays. This section outlines a three-stage roadmap—simulation fidelity, field-programmable gate array (FPGA) emulation, and chip prototype—with milestones and blockers at each stage. The roadmap is deliberately staged to de-risk the transition: simulation identifies which physics matter; FPGA validates control logic and digital periphery; the chip validates the device itself.

7.4.1 Stage I: closing the simulation fidelity gap

The current framework uses first-order behavioural surrogates. Write nonlinearity is approximated by scaling the STE gradient (Equations (1.5)–(1.6) of Chapter 1); retention is modelled by a double-exponential fit with literature-derived time constants; correlated D2D uses a separable AR(1) proxy. These approximations were necessary to make the design-space sweeps tractable, but they create an *approximation bound*: every failure mode in Chapter 3 is a boundary of the surrogate, not necessarily of the physical device.

The first milestone is therefore to replace the surrogates with *measured* device profiles. Critical measurements include:

- **Spatial covariance kernels:** real fabricated arrays exhibit anisotropic correlation (stronger along shared gate buses than along bitlines) and long-range wafer-scale gradients. A measured kernel would test whether the bounded-degradation trend observed under the AR(1) proxy ($86.33 \pm 1.61\,\%$ at $\rho = 0$, $84.57 \pm 2.39\,\%$ at $\rho = 0.3$, and $82.12 \pm 3.95\,\%$ at $\rho = 0.5$) holds under realistic spatial structure.
- **Pulse-accurate programming transients:** the dual-attention NL collapse—in which QKV and output-projection linearization drives accuracy below the unmitigated baseline—was discovered under a gradient-scaling surrogate. A pulse-faithful programming model would test whether this collapse is a real physical effect or an artifact of the first-order approximation.
- **Temperature-dependent retention laws:** the $79\,\%$ plateau was computed at room temperature with fixed time constants. Real organic devices are temperature-sensitive; a measured Arrhenius or Williams-Landel-Ferry law would determine whether the plateau shifts upward (better materials) or downward (worse environments).

The blocker is not measurement technique—the required instruments are standard—but *statistical representativeness*. Organic devices often suffer from batch-to-batch variability that exceeds within-batch D2D spread. A profile derived from one fabrication run may not generalise to the next, undermining the very notion of a canonical "device profile." Solving this requires either run-to-run calibration protocols or Bayesian profile updating, neither of which is mature.

7.4.2 Stage II: FPGA emulation of the mixed-signal stack

Before committing to a full custom chip, the analog forward path can be emulated on FPGA to validate digital control logic and evaluate end-to-end inference latency. The

FPGA would implement the complete mixed-signal pipeline: conductance quantization, D2D and cycle-to-cycle (C2C) noise injection, retention drift, inverse-gamma front-end compensation, ADC quantization with DNL, and dynamic scale recovery. The weights are HAT-trained from the simulator; the FPGA merely executes the noisy forward pass in real time.

The milestone is a real-time inference demonstration of Tiny-ViT on FPGA with injected noise that matches the simulator to within 0.1 pp accuracy. Achieving this would confirm two things. First, that the digital peripheral overhead (quantization, noise generation, scale recovery) does not dominate the analog MAC savings. Second, that the control logic—epoch counters, mismatch map buffers, recalibration triggers—can be implemented with finite state machines and on-chip memory.

The blocker is FPGA resource scaling. A single 256×256 crossbar emulated with DSP slices consumes a significant fraction of a mid-range FPGA. Tiling to the size required for even Tiny-ViT forces a hybrid partitioning: some layers run on the FPGA-emulated analog tile, others fall back to digital soft-core processors. The energy and latency numbers from such a hybrid system are not directly comparable to the manuscript's 11.45× energy-reduction estimate, but they are invaluable for identifying which digital periphery blocks are the true bottlenecks.

7.4.3 Stage III: organic RRAM chip prototype

The final stage is fabrication of a small-scale organic RRAM array and on-chip inference demonstration. A realistic first target is a 64×64 or 128×128 array, large enough to map a small MLP (e.g., a two-layer network for MNIST) but small enough to yield with modest lithographic resources. The milestone is not ImageNet-scale performance; it is *functional verification*: a HAT-trained model, transferred from simulation to the physical array without per-device retraining, achieves accuracy within 5 pp of the simulated prediction.

The blockers at this stage are as much ecological as electrical. Organic devices are sensitive to humidity, temperature, and oxygen ingress. A chip that works in a glovebox may fail at ambient conditions. Contact resistance variability—not modelled in the current framework because it is layout-dependent—can dominate the array-level noise budget. And the lack of foundry-grade design kits for organic electronics means that every transistor, via, and passivation layer must be co-designed with the materials scientist.

Nevertheless, the staged roadmap provides a decision framework. If Stage I shows that the surrogates are faithful, the simulation results in this thesis can be treated as predictive. If Stage II shows that digital periphery is the bottleneck, the research priority shifts from device improvement to ADC design and scale-recalibration circuits. If Stage III shows that ambient sensitivity is the dominant failure mode, the deployment envelope must be redefined around encapsulation and thermal management rather than write nonlinearity.

7.4.4 Calibration and test infrastructure

A hidden blocker that cuts across all three stages is calibration infrastructure. Analog arrays require per-column ADC calibration, reference-cell tracking for scale recovery, and temperature compensation. The framework's dynamic scale recalibration (Section 1.2.5 of Chapter 1) assumes perfect knowledge of the drifted conductances. In silicon, the equivalent operation requires dummy devices, intermittent readout, and digital feedback loops whose energy and latency are not included in the manuscript's energy model.

Building this test bench is not a footnote; it is a research contribution in its own right. The thesis framework provides the *algorithmic* specification of what must be calibrated (layer-wise scale factors, retention time constants, correlation lengths). The hardware roadmap provides the *physical* path to implementing that calibration. Closing the gap between the two is the work of the next several years.

7.5 Theory directions: from empirical recipes to principled guarantees

The HAT taxonomy in Chapter 2 and the failure-mode atlas in Chapter 3 are empirical. They tell us *what* happens when the resampling cadence or the noise profile is changed, but they do not derive *why* the observed boundaries are where they are. This section outlines three theory directions that can transform the empirical recipes into principled guarantees: HAT as a regularizer, ensemble-frequency effective width, and probably approximately correct (PAC)–Bayes bounds for device-instance generalisation.

7.5.1 HAT as a data-dependent regulariser

Ensemble HAT minimises an empirical expectation over mismatch maps:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta}\ \mathbb{E}_{M\sim p(M)}\left[\frac{1}{N}\sum_{n=1}^{N}\ell(f(x_n;\theta,M),y_n)\right]. \tag{7.1}$$

This objective is structurally identical to adding a data-dependent noise regulariser to the loss landscape. Drawing a fresh M at each epoch convolves the empirical risk with the mismatch distribution, smoothing sharp minima and favouring flat regions that are compatible with many hardware instances. The connection to randomized smoothing and implicit regularisation is immediate: the D2D resampling acts as a controlled perturbation of the forward path, and the straight-through estimator propagates the smoothed gradient back to the weights.

In the companion theory contribution (K-V5), this intuition is formalised. The key result is that the expected gradient variance under epoch-level resampling bounds the

Hessian trace of the loss landscape at the converged point. Specifically, if σ_M^2 denotes the variance of the mismatch distribution and $H(\theta)$ the Hessian of the clean risk, then

$$\operatorname{Tr} H(\theta^\star_{\mathrm{ens}}) \;\leq\; \frac{C}{\sigma_M^2}\,\mathbb{E}\big[\|\nabla_\theta \ell\|^2\big]\,, \tag{7.2}$$

where C is a constant that depends on the Lipschitz constant of the model and the learning rate. The inequality shows that *larger* mismatch variance forces the optimizer into a *flatter* minimum, provided the resampling cadence is slow enough for the optimizer to settle within each realisation. This is the theoretical counterpart to the empirical observation that Ensemble HAT preserves $86.37 \pm 1.54\,\%$ fresh-instance accuracy: the flat minimum is robust to instance shift because it is not tightly coupled to any single spatial signature.

The regulariser perspective also explains why fixed-mask HAT fails. With $\sigma_M^2 = 0$ after the initial draw, the bound collapses and the optimizer is free to memorise the specific mismatch field, leading to the $10.00\,\%$ collapse observed in Chapter . The theory therefore unifies the two largest empirical findings of the thesis under a single geometric principle.

7.5.2 Ensemble-frequency effective width

Not all resampling schedules are equal. Chapter 2 showed that per-epoch resampling outperforms both fixed-mask and per-batch resampling. The theory direction K-V6 introduces the concept of *ensemble-frequency effective width* to explain why.

Define the effective width as

$$W_{\mathrm{eff}} = \frac{E}{\tau_{\mathrm{mix}}}, \tag{7.3}$$

where E is the number of training epochs and τ_{mix} is the optimizer's mixing time measured in epochs—the number of epochs required for the weight distribution to forget its initial condition. Per-epoch resampling presents one independent mismatch realisation per epoch, so the number of independent samples is E; if $\tau_{\mathrm{mix}} \approx 1$ epoch (a plausible estimate for AdamW with moderate learning rate), then $W_{\mathrm{eff}} \approx E$. Per-batch resampling presents B realisations per epoch, but the mixing time is also shorter because the weights are updated more frequently; if $\tau_{\mathrm{mix}} \approx 1/B$ epochs, then $W_{\mathrm{eff}} \approx BE/B = E$, apparently comparable. However, the shorter mixing time means the optimizer never fully settles into a basin compatible with any single realisation, producing the non-stationary gradient field that empirically underperforms.

The effective-width hypothesis predicts an inverted-U relationship between resampling frequency and fresh-instance accuracy. At very low W_{eff} (fixed mask), the model overfits to one instance. At very high W_{eff} (per-batch), the noise prevents convergence. At moderate W_{eff} (per-epoch), the optimizer enjoys enough stability to descend within each block and enough diversity to generalise across blocks. This prediction is consistent with the exploratory ablation in Chapter 2: per-epoch reaches 88.41 % held-out accuracy, fixed-mask 87.18 %, and per-batch 86.16 %.

Testing the effective-width hypothesis requires a systematic sweep over resampling cadences intermediate between per-epoch and per-batch—for example, resampling every k mini-batches with $k \in \{1, 10, 50, 200\}$. Such a sweep would locate the peak of the inverted-U and determine whether it is task-dependent or universal. It would also provide a principled recipe for choosing the cadence given a new model, dataset, and hardware noise level.

7.5.3 PAC-Bayes bounds for device-instance generalisation

Standard generalisation theory bounds the gap between training and test error under the assumption that both sets are drawn i.i.d. from the same distribution. In the hardware-instance setting, the *data* are i.i.d., but the *deployment environment* shifts: the training environment is a finite sample of mismatch maps, while the test environment is a fresh draw from $p(M)$. PAC-Bayes theory is well suited to this setting because it explicitly bounds the risk of a posterior hypothesis in terms of a prior over environments.

Treat the mismatch distribution $p(M)$ as a prior over "hardware environments." During Ensemble HAT training, the model sees E independent draws $M^{(1)}, \ldots, M^{(E)}$. The learned weights θ define a posterior $q(\theta)$ concentrated around the empirical minimiser. A preliminary PAC-Bayes bound for the fresh-instance risk takes the form

$$R_{\text{fresh}}(q) \;\le\; \hat{R}_{\text{ens}}(q) \;+\; \sqrt{\frac{\mathrm{KL}(q\|p) + \ln\bigl(2\sqrt{E}/\delta\bigr)}{2(E-1)}}, \tag{7.4}$$

where $\hat{R}_{\text{ens}}(q)$ is the empirical Ensemble HAT risk averaged over the E epochs, $\mathrm{KL}(q\|p)$ is the Kullback-Leibler divergence between posterior and prior, and δ is the confidence level.

Several insights follow from Equation (7.4). First, the bound tightens as $O(1/\sqrt{E})$, explaining why longer training (more independent mismatch draws) improves fresh-instance accuracy up to a plateau. Second, the KL term penalises overly complex posteriors; this provides a theoretical justification for weight decay and early stopping, both of which shrink the effective posterior volume. Third, the bound is *distribution-dependent*: if the training mismatch distribution $p_{\text{train}}(M)$ differs from the deployment distribution $p_{\text{deploy}}(M)$—for example, because of correlated D2D or batch-to-batch fabrication drift—the KL term must be replaced by a divergence between the two distributions, and the bound widens proportionally.

Future work can tighten the bound in three directions. First, replace the i.i.d. prior over M with a Gaussian-process prior that captures spatial correlation, reducing the effective complexity penalty for structured mismatch. Second, incorporate the optimizer trajectory into the posterior definition, yielding a data-dependent PAC-Bayes bound that is tighter than the worst-case static bound. Third, extend the bound to the tiled setting of Section 7.3.1, where the environment prior is over collections of tile-level maps rather than single global maps. Each extension would bring the theory closer to the physical reality of large-scale analog accelerators.

Together, the three theory directions—regularisation, effective width, and PAC-Bayes bounds—provide a path from the empirical discoveries of this thesis to a rigorous science of hardware-instance generalisation.

7.6 Conclusion: one story

A thesis is often read as a sequence of chapters, each with its own figures, equations, and contributions. But the true product is not the sequence; it is the *story* that runs through it. This concluding section tells that story in three acts, then gathers its threads into a single claim about what this thesis has built and where it leads.

7.6.1 Act I: the instrument and the discovery

The story opens with a gap. Device scientists publish conductance windows and retention curves; system engineers need deployment accuracy. Chapter 1 closed that gap by building `compute-ViT`, a profile-driven simulation framework in which every physically relevant assumption enters through a structured configuration object. The framework is not merely an implementation detail; it is a *reusable research instrument*. A literature-derived prior and an in-house measured profile populate the same fields and drive the same training scripts. Switching from one device generation to the next requires only replacing the profile instance.

With the instrument in hand, Chapter asked the obvious question: if we train a vision transformer on a noisy analog array, does it survive deployment? The answer was a stark $10.00 \pm 0.00\,\%$ collapse. Standard hardware-aware training, which fixes a single device-to-device mismatch field for the entire training run, overfits that instance as if it were a hidden domain label. The discovery is not that analog noise hurts accuracy—that was known—but that *robustness to analog noise is not the same as robustness to hardware-instance shift*. The remedy, Ensemble HAT, is simple in retrospect: resample the mismatch field at epoch boundaries so that the optimizer learns over a *distribution* of arrays rather than memorising one sample from it. The result—$86.37 \pm 1.54\,\%$ fresh-instance accuracy—is the empirical anchor of the thesis.

Chapter 2 placed that anchor in a broader design space. Ensemble HAT is not an isolated trick; it is one point in a taxonomy of resampling cadences, noise profiles, and transfer guarantees. Per-epoch resampling outperforms per-batch because it creates a block-stationary optimisation landscape: stable enough to descend within each block, diverse enough to generalise across blocks. Uniform and proportional noise profiles define distinct deployment regimes, and cross-regime training collapses catastrophically. These distinctions provide a *language* for discussing HAT that did not exist before: practitioners can now specify not merely “we did HAT” but “we trained with epoch-level resampling, uniform noise, and an implicit expectation over mismatch maps.”

7.6.2 Act II: the atlas

If the first act built the instrument and discovered the central phenomenon, the second act mapped its boundaries. Chapter 3 organised the empirical landscape as a structured catalogue of named, reproducible breakdown patterns. Each entry—hardware-instance overfitting, dual-attention nonlinear-write collapse, MLP-only diagnostic bound, correlated-D2D bounded degradation, retention plateau at 79 %—is annotated with a trigger, a signature, a mitigation, and an open question.

The atlas is not a pessimistic document. It is a *deployment envelope*. It says: if your device operates at NL = 1.0 with i.i.d. mismatch, you can expect $86.37 \pm 1.54\,\%$ on fresh instances. At $\rho = 0.3$, the mean is $84.57 \pm 2.39\,\%$; at $\rho = 0.5$, it drops modestly to $82.12 \pm 3.95\,\%$—bounded degradation, not catastrophic collapse. If retention drift is present but scale recalibration is applied, the accuracy stabilises in the 79.13–79.51 % plateau. These numbers are contracts between physics and algorithm. They tell a device engineer how much nonlinearity must be eliminated to reach a given accuracy target, and they tell an algorithm designer which training interventions are deployment-grade and which are merely diagnostic.

A cross-cutting theme binds the atlas entries together: the primacy of *distributional robustness* over instance-specific rescue. Standard HAT rescues source-domain accuracy; MLP-linearisation rescues source-domain accuracy under severe NL; but neither survives fresh-instance evaluation. Only interventions that explicitly average over the deployment distribution—Ensemble HAT, dynamic scale recalibration—produce results that are stable across multiple hardware realisations. This principle is the methodological legacy of the thesis.

7.6.3 Act III: the path forward

The third act is the outlook developed in the preceding sections of this chapter. It asks: what happens when we push beyond the boundaries of the atlas? The joint MLP-linear + Ensemble HAT experiment tests whether the two deepest failure modes can be simultaneously repaired. The ImageNet-100 pilot tests whether the ranking logic survives dataset scale. The language-model direction tests whether the principles transfer to the dominant AI workload of the coming decade. The hardware roadmap lays out a staged path from simulation to silicon. And the theory directions offer a route from empirical recipes to principled guarantees.

These threads are not disjoint futures; they are convergent. A language-model HAT protocol will need the regularisation theory to choose resampling cadences across thousands of tiles. A chip prototype will need the FPGA stage to validate calibration logic. And every stage will need the profile-driven framework as its common language. The thesis therefore does not end with a single result; it ends with a *platform*.

7.6.4 The core contributions

It is worth stating the contributions plainly, because a thesis that spans simulation, algorithm, and hardware can easily be misread as three small papers bound together. It is not. The contributions are three facets of one story.

Contribution 1: compute-ViT as a reusable research instrument. The framework is the first simulation environment designed specifically for organic optoelectronic CIM that treats device profiles as interchangeable structured inputs. Its pluggable physical interfaces, PyTorch autograd integration, and external validation against CrossSim make it a tool that can outlive the specific experiments reported here.

Contribution 2: the HAT taxonomy and the discovery of hardware-instance overfitting. The taxonomy formalises the design space of resampling cadences and noise profiles, and the instance-overfitting discovery redefines the evaluation criterion for analog-deployment experiments. The fresh-instance protocol (10 arrays $\times$ 5 Monte Carlo evaluations) is not an appendix detail; it is a minimal bar for claiming deployment-grade robustness.

Contribution 3: a quantified deployment envelope for organic CIM. The failure-mode atlas provides the first system-level accuracy contracts for Vision Transformers on organic optoelectronic arrays. It tells the materials scientist which device properties matter most, the circuit designer which peripheral overhead is acceptable, and the algorithm designer which training protocol to choose for a given physical regime.

7.6.5 The final word

This thesis does not claim that organic optoelectronic compute-in-memory is ready for ImageNet-scale deployment today. The severe-NL residual gap, the retention plateau, and the environmental sensitivity of organic devices are real blockers that will require years of joint materials–circuit–algorithm research to overcome. What the thesis claims is something more modest and, perhaps, more durable: that the transition from "device works" to "system works" requires a new kind of experimental science. That science must be *profile-driven*, treating device characterisations as structured inputs rather than narrative context. It must be *distributional*, training over the deployment distribution rather than a single hardware sample. And it must be *transfer-oriented*, evaluating every intervention on fresh instances before declaring it deployment-grade.

The clean room is quiet. The fabrication tools are precise. But the real world is a distribution of arrays, temperatures, and inference latencies that no single measurement can capture. Building a bridge between those two worlds is slow, incremental, and often humbling. Yet every time a trained model survives a fresh hardware instance—every time the accuracy stays above 80 % on an array it has never seen—the bridge lengthens. This thesis has lengthened it a little. The path ahead is to keep building.

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